
Neural Logistic Bandits

Seounghin Bae¹ Dabeen Lee^{2,3}

Abstract

We study the problem of *neural logistic bandits*, where the main task is to learn an unknown reward function within a logistic link function using a neural network. Existing approaches either exhibit unfavorable dependencies on κ , where $1/\kappa$ represents the minimum variance of reward distributions, or suffer from direct dependence on the feature dimension d , which can be huge in neural network-based settings. In this work, we introduce a novel Bernstein-type inequality for self-normalized vector-valued martingales that is designed to bypass a direct dependence on the ambient dimension. This lets us deduce a regret upper bound that grows with the *effective dimension* $\tilde{d}$, not the feature dimension, while keeping a minimal dependence on κ . Based on the concentration inequality, we propose two algorithms, NeuralLog-UCB-1 and NeuralLog-UCB-2, that guarantee regret upper bounds of order $\tilde{O}(\tilde{d}\sqrt{\kappa T})$ and $\tilde{O}(\tilde{d}\sqrt{T/\kappa})$, respectively, improving on the existing results. Lastly, we report numerical results on both synthetic and real datasets to validate our theoretical findings.

1. Introduction

Contextual bandits form the foundation of modern sequential decision-making problems, driving applications such as recommendation systems, advertising, and interactive information retrieval (Li et al., 2010). Although upper confidence bound (UCB)-based linear contextual bandit algorithms achieve near-optimal guarantees when rewards are linear in the feature vector (Abbasi-Yadkori et al., 2011), many real-world scenarios exhibit nonlinear reward structures that demand more expressive models. Motivated by

this, several approaches have been developed to capture complex reward functions that go beyond the linear case, such as those based on generalized linear models (Filippi et al., 2010; Li et al., 2017), reproducing kernel Hilbert space (Srinivas et al., 2009; Valko et al., 2013), and deep neural networks (Riquelme et al., 2018; Zhou et al., 2020).

Among these settings, *logistic bandits* are particularly relevant when the reward is binary (e.g., click vs. no-click); the random reward in each round follows a Bernoulli distribution, whose parameter is determined by the chosen action. Extending logistic bandits via a neural network-based approximation framework, we consider *neural logistic bandits* and address two significant challenges: (i) handling the nonlinearity of the reward function, characterized by the worst-case variance of a reward distribution $1/\kappa$ where κ scales exponentially with the size of the decision set, and (ii) controlling the dependence on the feature dimension d , which can be extremely large due to the substantial number of parameters in deep neural networks.

For logistic bandits, (Fauray et al., 2020) introduced a variance-adaptive analysis by incorporating the true reward variance of each action into the design matrix. This avoids using a uniform worst-case variance bound of $1/\kappa$ for all actions, thus reducing the dependency of the final regret on κ . Building on this, (Abeille et al., 2021) achieved the best-known κ dependence. However, both algorithms explicitly rely on the ambient feature dimension d , so their direct extensions to the neural bandit setting induce poor regret performance. On the other hand, (Verma et al., 2025) derived a regret upper bound for the neural logistic bandit that scales with a data-adaptive effective dimension $\tilde{d}$ rather than the full ambient dimension d . This approach offers an improved performance measure as d increases with the number of parameters in the neural network, often deliberately overparameterized to avoid strong assumptions about the reward function. However, their method still relies on a pessimistic variance estimate, and integrating the variance-aware analysis of (Fauray et al., 2020) into a data-adaptive regret framework remains challenging, resulting in a suboptimal dependence on κ .

Motivated by these limitations, we propose algorithms that do not require worst-case estimates in both the variance of the reward distribution and the feature dimension, thus

¹Department of Industrial & Systems Engineering, KAIST

²Department of Mathematical Sciences & Research Institute of Mathematics, Seoul National University ³ Interdisciplinary Program in Artificial Intelligence, Seoul National University. Correspondence to: Dabeen Lee <dabeenl@snu.ac.kr>.

Table 1. Comparison of algorithms for (neural) logistic bandits. d denotes the dimension of the feature vector, and T represents the total number of rounds. p denotes the total number of parameters of the underlying neural network, and $\tilde{d}$ denotes the effective dimension.

Algorithm	Regret $\tilde{O}(\cdot)$	
	Logistic Bandits	Neural Bandits
NCBF-UCB (Verma et al., 2025)	$\kappa d\sqrt{T}$	$\kappa\tilde{d}\sqrt{T}$
Logistic-UCB-1 (Fauray et al., 2020)	$d\sqrt{\kappa T}$	$p\sqrt{\kappa T}$
NeuralLog-UCB-1 (Algorithm 1)	$d\sqrt{\kappa T}$	$\tilde{d}\sqrt{\kappa T}$
ada-OFU-ECOLOG (Fauray et al., 2022)	$d\sqrt{T/\kappa^*}$	$p\sqrt{T/\kappa^*}$
NeuralLog-UCB-2 (Algorithm 2)	$d\sqrt{T/\kappa^*}$	$\tilde{d}\sqrt{T/\kappa^*}$

achieving the most favorable regret bound for neural logistic bandits. Central to this approach is our new Bernstein-type self-normalized inequality for vector-valued martingales, which allows us to derive a regret upper bound that scales with the effective dimension $\tilde{d}$, and at the same time, matches the best-known dependency on κ . Our main contributions are summarized below:

- We tackle the two main challenges in *neural logistic bandits*: (i) a practical regret upper bound should avoid a direct dependence on d , the ambient dimension of the feature vector, and (ii) it needs to minimize the factor of κ , a problem-dependent constant that increases exponentially with the size of the decision set. To address these challenges, we propose a new Bernstein-type tail inequality for self-normalized vector-valued martingales that yields a bound of order $\tilde{O}(\sqrt{\tilde{d}})$, where $\tilde{d}$ is a data-adaptive effective dimension. This is the first tail inequality that achieves favorable results in both respects, while the previous bound of (Fauray et al., 2020) is $\tilde{O}(\sqrt{d})$ which directly depends on d , and that of (Verma et al., 2025) is $\tilde{O}(\sqrt{\kappa d})$ that includes an additional factor of $\sqrt{\kappa}$.
- Based on our tail inequality, we develop our first algorithm, NeuralLog-UCB-1 which guarantees a regret upper bound of order $\tilde{O}(\tilde{d}\sqrt{\kappa T})$. This improves upon the regret upper bound of order $\tilde{O}(\kappa\tilde{d}\sqrt{T})$ due to (Verma et al., 2025). Furthermore, we provide a fully data-adaptive UCB on $\tilde{d}$ by adaptively controlling the regularization term of our loss function according to previous observations. Our choice of UCB also avoids the projection step required in the previous approach of (Fauray et al., 2020), which was to constrain the parameters to a certain set during training.
- We propose our second algorithm, NeuralLog-UCB-2, as a refined variant of NeuralLog-UCB-1. We show that NeuralLog-UCB-2 achieves a regret upper bound of $\tilde{O}(\tilde{d}\sqrt{T/\kappa^*})$. This result matches the best-known

dependency on κ while avoiding the direct dependence on d seen in $\tilde{O}(d\sqrt{T/\kappa^*})$ given by (Abeille et al., 2021). The improvement comes from the fact that NeuralLog-UCB-2 replaces the true reward variance within the design matrix with a neural network estimated variance, thereby maintaining sufficient statistics for our variance-adaptive UCB in each round and completely removing the worst-case estimate of variance κ . Our numerical results show that NeuralLog-UCB-2 outperforms all baselines, thus validating our theoretical framework.

2. Preliminaries

Logistic bandits. We consider the contextual logistic bandit problem. Let T be the total number of rounds. In each round $t \in [T]$, the agent observes an action set $\mathcal{X}_t$, consisting of K contexts drawn from a feasible set $\mathcal{X} \subset \mathbb{R}^d$. The agent then selects an action $x_t \in \mathcal{X}_t$ and observes a binary (random) reward $r_t \in \{0, 1\}$. This reward is generated by the logistic model governed by the unknown latent reward function $h : \mathbb{R}^d \rightarrow \mathbb{R}$. Specifically, we define a sigmoid function $\mu(x) = (1 + \exp(-x))^{-1}$ and denote its first and second derivatives as $\dot{\mu}$ and $\ddot{\mu}$. Then, the probability distribution of the reward r_t under action x is given by $r_t \sim \text{Bern}(\mu(h(x_t)))$. Let x_t^* be an optimal action in round t , i.e., $x_t^* = \arg \max_{x \in \mathcal{X}_t} \mu(h(x))$. Then the agent’s goal is to minimize the cumulative regret, defined as $\text{Regret}(T) = \sum_{t=1}^T \mu(h(x_t^*)) - \sum_{t=1}^T \mu(h(x_t))$. Finally, we introduce the standard assumption on the problem-dependent parameters κ and R (Fauray et al., 2020; Verma et al., 2025):

Assumption 2.1 (Informal). There exist constants $\kappa, R > 0$, such that $1/\kappa \leq \dot{\mu}(\cdot) \leq R$.

The formal definition of κ and R for the arm set $\mathcal{X}$ and the parameter set Θ is deferred to Assumption 6.3. Notice that for the sigmoid link function, we have $\mu(\cdot)$, $R \leq 1/4$.

Neural bandits. Neural contextual bandit methods address the limitation of traditional (generalized) linear reward models (Filippi et al., 2010; Fauray et al., 2020) by approximating $h(\cdot)$ with a fully connected deep neural network $f(x; \theta)$, which allows them to capture complex, possibly nonlinear, reward structures. In this work, we consider a neural network given by $f(x; \theta) = \sqrt{m}W_L \text{ReLU}(W_{L-1} \text{ReLU}(\dots \text{ReLU}(W_1 x)))$, where $L \geq 2$ is the depth of the neural network, $\text{ReLU}(x) = \max\{x, 0\}$, $W_1 \in \mathbb{R}^{m \times d}$, $W_i \in \mathbb{R}^{m \times m}$ for $2 \leq i \leq L-1$, and $W_L \in \mathbb{R}^{1 \times m}$. The flattened parameter vector is given by $\theta = [\text{vec}(W_1)^\top, \dots, \text{vec}(W_L)^\top]^\top \in \mathbb{R}^p$, where p is the total number of parameters, i.e., $p = m + md + m^2(L-1)$. We denote the gradient of the neural network by $g(x; \theta) = \nabla_\theta f(x; \theta) \in \mathbb{R}^p$.

Notation. For a positive integer n , let $[n] = \{1, \dots, n\}$. For any $x \in \mathbb{R}^d$, $\|x\|_2$ denotes the ℓ_2 norm, and $[x]_i$ denotes its i -th coordinate. Given $x \in \mathbb{R}^d$ and a positive-definite matrix $A \in \mathbb{R}^{d \times d}$, we define $\|x\|_A = \sqrt{x^\top A x}$. We use $\tilde{O}(\cdot)$ to hide the logarithmic factors.

3. Variance- and Data-Adaptive Self-Normalized Martingale Tail Inequality

In this section, we first introduce our new Bernstein-type tail inequality for self-normalized martingales, which leads to a regret analysis that is variance- and data-adaptive. Then we compare it with some existing tail inequalities from prior works.

Theorem 3.1. *Let $\{\mathcal{G}_t\}_{t=1}^\infty$ be a filtration, and $\{x_t, \eta_t\}_{t \geq 1}$ be a stochastic process where $x_t \in \mathbb{R}^d$ is $\mathcal{G}_t$ -measurable and $\eta_t \in \mathbb{R}$ is $\mathcal{G}_{t+1}$ -measurable. Suppose there exist constants $M, R, N, \lambda > 0$ and the parameter $\theta^* \in \mathbb{R}^d$ such that for all $t \geq 1$, $|\eta_t| \leq M$, $\mathbb{E}[\eta_t | \mathcal{G}_t] = 0$, $\mathbb{E}[\eta_t^2 | \mathcal{G}_t] \leq \dot{\mu}(x_t^\top \theta^*)$, and $\|x_t\|_2 \leq N$. Define H_t and s_t as follows:*

$$H_t = \sum_{i=1}^t \dot{\mu}(x_i^\top \theta^*) x_i x_i^\top + \lambda \mathbf{I}, \quad s_t = \sum_{i=1}^t x_i \eta_i.$$

Then, for any $0 < \delta < 1$ and any $t > 0$, with probability at least $1 - \delta$:

$$\begin{aligned} \|s_t\|_{H_t^{-1}} &\leq 8 \sqrt{\log \frac{\det H_t}{\det \lambda \mathbf{I}} \log \left(\frac{4t^2}{\delta} \right)} + \frac{4MN}{\sqrt{\lambda}} \log \frac{4t^2}{\delta} \\ &\leq 8 \sqrt{\log \det \left(\sum_{i=1}^t \frac{R}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log \frac{4t^2}{\delta} + \frac{4MN}{\sqrt{\lambda}} \log \frac{4t^2}{\delta}}. \end{aligned}$$

Our proof of Theorem 3.1 is given in Section D. The second inequality in Theorem 3.1 follows from Assumption 2.1 which states that $\dot{\mu}(\cdot) \leq R \leq 1/4$. Notice that the tail inequality is data-adaptive, as it does not explicitly depend on d . Moreover, the term $\log \frac{\det H_t}{\det \lambda \mathbf{I}}$ can decrease depending on the observed feature vectors (e.g., it becomes 0 if $\{x_i\}_{i=1}^t$ are all $\mathbf{0}$). By incorporating non-uniform variances when defining H_t , our design matrix enables a variance-adaptive analysis and drops the worst-case variance dependency κ .

The seminal work of (Abbasi-Yadkori et al., 2011) provided a variant of the Azuma-Hoeffding tail inequality for vector-valued martingales, under the assumption that the martingale difference η_t is M -sub-Gaussian. Their tail bound shows that $\|s_t\|_{\tilde{V}_t^{-1}} = \tilde{O}(M\sqrt{d})$, where $\tilde{V}_t = \sum_{i=1}^t x_i x_i^\top + \lambda \mathbf{I}$. Extending this result to (neural) logistic bandits, (Verma et al., 2025) incorporated the worst-case variance κ into the design matrix $V_t = \sum_{i=1}^t x_i x_i^\top + \kappa \lambda \mathbf{I}$ to deduce

$$\|s_t\|_{H_t^{-1}} \leq \sqrt{\kappa} \|s_t\|_{V_t^{-1}} \leq M \sqrt{\kappa \log \frac{\det V_t}{\det \kappa \lambda \mathbf{I}}} + 2\kappa \log \frac{1}{\delta}.$$

Here, the first inequality is a consequence of $H_t \succeq (1/\kappa)V_t$, and this step incurs the factor $\sqrt{\kappa}$. Note that this bound is also data-adaptive, yielding an overall order of $\tilde{O}(\sqrt{\kappa d})$.

Another line of work by (Faury et al., 2020) provided a Bernstein-type tail inequality for the same setting considered in Theorem 3.1, using $|\eta_t| \leq M (= 1)$, $\mathbb{E}[\eta_t^2 | \mathcal{G}_t] = \sigma_t^2$, and $\|x_t\|_2 \leq N (= 1)$ for all $t \geq 1$. Their analysis directly takes the design matrix H_t , and they deduce the following inequality avoiding the $\sqrt{\kappa}$ factor:

$$\|s_t\|_{H_t^{-1}} \leq \frac{2MN}{\sqrt{\lambda}} \left(\log \frac{\det H_t}{\delta \det \lambda \mathbf{I}} + d \log(2) \right) + \frac{\sqrt{\lambda}}{2MN}.$$

The inequality requires a specific λ value for the regularization term, given by $\lambda = \tilde{O}(dM^2N^2)$, to achieve the final order of $\tilde{O}(\sqrt{d})$. Although $\log \frac{\det H_t}{\det \lambda \mathbf{I}}$ is data-adaptive, the term $d \log(2)$ introduces an explicit dependence on d that cannot be removed (even with a different choice of λ). The tail bound has been used in subsequent works (Abeille et al., 2021; Faury et al., 2022), making a dependence on d inherent. Hence, we need a new variance-adaptive analysis for neural logistic bandits.

Compared with (Faury et al., 2020), our tail inequality in Theorem 3.1 is derived from a different technique based on Freedman's inequality ((Freedman, 1975), Lemma H.1), which is the key factor behind our improvement. Unlike (Faury et al., 2020), which works with a d -dimensional martingale and thereby incurs an explicit dependence on d , we instead use a one-dimensional martingale to track the growth of the self-normalized error $\|s_t\|_{H_t^{-1}}$, bypassing this vector-level issue. As a result, we obtain a data- and variance-adaptive inequality whose leading term depends on the effective dimension $\tilde{d}$, together with an improved dependence on κ , thanks to the variance sensitivity of Freedman's inequality.

4. Neural Logistic Bandits with Improved UCB

This section introduces our first algorithm, NeuralLog-UCB-1, described in Algorithm 1. In the initialization step, we set the initial parameter θ_0 of the neural network according to the standard initialization process described in (Zhou et al., 2020). For $1 \leq l \leq L - 1$, W_l is set as $\begin{bmatrix} W & \mathbf{0} \\ \mathbf{0} & W \end{bmatrix}$, where each entry of W is independently sampled from $N(0, 4/m)$ while W_L is set to $[w, -w]$, where each entry of w is independently sampled from $N(0, 2/m)$. Next, we set the initial regularization parameter as $\lambda_0 = 8\sqrt{2}C_1 L^{1/2} S^{-1} \log(4/\delta)$ for some absolute constant $C_1 > 0$. The value of λ_0 is chosen so that λ_0 is less than the minimum value among $\lambda_1, \dots, \lambda_T$, where λ_t is updated as in Equation (3). We can verify this by showing that $\lambda_0 \leq \min \lambda_1$ and that $\{\lambda_t\}_{t \geq 1}$ is monotonically non-decreasing in t , which im-

plies $\lambda_0 \leq \min\{\lambda_t\}_{t \geq 1}$.

After the initialization step, in each round t , the agent receives the context set $\mathcal{X}_t \subset X$ and calculates $\text{UCB}_t(x) = \mu(f(x; \theta_{t-1})) + R\sqrt{\kappa}\nu_{t-1}^{(1)}\|g(x; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}$ for every action $x \in \mathcal{X}_t$, where

$$\nu_t^{(1)} = C_6(1 + \sqrt{LS} + LS^2)\iota_t + 1, \quad (1)$$

$$\begin{aligned} \iota_t = 16 \sqrt{\log \det \left(\sum_{i=1}^t \frac{g(x_i; \theta_0)g(x_i; \theta_0)^\top}{4m\lambda_0} + \mathbf{I} \right) \log \frac{4t^2}{\delta}} \\ + 8C_1 \sqrt{\frac{L}{\lambda_0}} \log \frac{4t^2}{\delta}, \end{aligned} \quad (2)$$

where $\delta \in (0, 1)$ is a confidence parameter, and S is a norm parameter of the parameter set defined in Definition 6.2 for some absolute constants $C_1, C_6 > 0$. The first term, $\mu(f(x; \theta_{t-1}))$, estimates the expected value of the reward, and the second term can be viewed as the exploration bonus. Then, we choose our action x_t optimistically by maximizing the UCB value, i.e., $x_t = \arg \max_{x \in \mathcal{X}_t} \text{UCB}_t(x)$, and receive a reward r_t . At the end of each round, we update the parameters based on the observations $\{x_i, r_i\}_{i=1}^t$ collected so far. We set the regularization parameter λ_t , with an absolute constant $C_1 > 0$, as follows:

$$\begin{aligned} \lambda_t \leftarrow \frac{64}{S^2} \log \det \left(\sum_{i=1}^t \frac{g(x_i; \theta_0)g(x_i; \theta_0)^\top}{4m\lambda_0} + \mathbf{I} \right) \log \frac{4t^2}{\delta} \\ + \frac{16C_1^2 L}{S^2 \lambda_0} \log^2 \frac{4t^2}{\delta}. \end{aligned} \quad (3)$$

Then we update ι_t , $\nu_t^{(1)}$, and V_t . Lastly, as described in Subroutine `TrainNN`, we update the parameters of the neural network through gradient descent to minimize the regularized negative log-likelihood loss function $\mathcal{L}_t(\theta)$ and obtain θ_t . In contrast to (Verma et al., 2025), which used a constant regularization parameter λ , we adaptively control the regularization parameter λ_t and employ it in both our design matrix V_t and our loss function $\mathcal{L}_t(\theta)$. This yields a fully data-adaptive concentration inequality between θ_t and the desired parameter, as will be shown in Lemma 4.5.

Now, we present our theoretical results for Algorithm 1. Let $\mathbf{H}$ denote the neural tangent kernel (NTK) matrix computed on all TK context–arm feature vectors over T rounds. Its formal definition is deferred to Definition C.1. Define $\mathbf{h} = [h(x)]_{x \in \mathcal{X}_t, t \in [T]} \in \mathbb{R}^{TK}$. We begin with the following assumptions.

Assumption 4.1. There exists $\lambda_{\mathbf{H}} > 0$ such that $\mathbf{H} \succeq \lambda_{\mathbf{H}} \mathbf{I}$.

Assumption 4.2. For every $x \in \mathcal{X}_t$ and $t \in [T]$, we have $\|x\|_2 = 1$ and $[x]_j = [x]_{j+d/2}$.

Both assumptions are mild and standard in the neural bandit literature (Zhou et al., 2020; Zhang et al., 2021). Assumption 4.1 states that the NTK matrix is nonsingular, which

Algorithm 1 NeuralLog-UCB-1

input Neural network $f(x; \theta)$ with width m and depth L , initialized with parameter θ_0 , step size η , number of gradient descent steps J , norm parameter S , confidence parameter δ

init $\lambda_0 = 8\sqrt{2}C_1 L^{1/2} S^{-1} \log(4/\delta)$, $V_0 = \kappa\lambda_0 \mathbf{I}$

1: **for** $t = 1, \dots, T$ **do**

2: $x_t \leftarrow \arg \max_{x \in \mathcal{X}_t} \mu(f(x; \theta_{t-1})) + R\sqrt{\kappa}\nu_{t-1}^{(1)} \times \|g(x; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}$

3: Select x_t and receive r_t

4: Update λ_t as in Equation (3), ι_t as in Equation (2), $\nu_t^{(1)}$ as in Equation (1)

5: $\theta_t \leftarrow \text{TrainNN}(\lambda_t, \eta, J, m, \{x_i, r_i\}_{i=1}^t, \theta_0)$

6: $V_t \leftarrow \sum_{i=1}^t \frac{1}{m} g(x_i; \theta_0)g(x_i; \theta_0)^\top + \kappa\lambda_t \mathbf{I}$

7: **end for**

Subroutine `TrainNN`

input Regularization parameter λ_t , step size η , number of gradient descent steps J , network width m , observations $\{x_s, r_s\}_{s=1}^t$, initial parameter θ_0

1: Define $\mathcal{L}_t(\theta) = -\sum_{i=1}^t [r_i \log \mu(f(x_i; \theta)) + (1 - r_i) \log(1 - \mu(f(x_i; \theta)))] + \frac{1}{2} m \lambda_t \|\theta - \theta_0\|_2^2$

2: **for** $j = 1, \dots, J - 1$ **do**

3: $\theta^{(j+1)} = \theta^{(j)} - \eta \nabla \mathcal{L}_t(\theta^{(j)})$

4: **end for**

5:

return $\theta^{(J)}$

holds if no two context vectors are parallel. Assumption 4.2 ensures that $f(x^i; \theta_0) = 0$ for all $i \in [TK]$ at initialization. This assumption is made for analytical convenience and can be ensured by building a new context $x' = [x^\top, x^\top]^\top / \sqrt{2}$.

Next, define $\tilde{\mathbf{H}} = \sum_{t=1}^T \sum_{x \in \mathcal{X}_t} \frac{1}{m} g(x; \theta_0)g(x; \theta_0)^\top$, which is the design matrix containing all possible context–arm feature vectors over the T rounds. Then, we can use the following definition:

Definition 4.3. Let $\tilde{d} := \log \det(\frac{R}{\lambda_0} \tilde{\mathbf{H}} + \mathbf{I})$ denote the effective dimension.

We mention that previous works (Zhou et al., 2020; Verma et al., 2025) for neural contextual bandits have defined the effective dimension $\tilde{d}$ in slightly different ways. (Zhou et al., 2020) set $\tilde{d} = \log \det(\frac{1}{\lambda} \mathbf{H} + \mathbf{I}) / \log(1 + TK/\lambda)$ for neural contextual linear bandits, while (Verma et al., 2025) defined $\tilde{d} = \log \det(\frac{1}{\kappa\lambda} \tilde{\mathbf{H}} + \mathbf{I})$. However, these definitions have the same asymptotic order as ours in Definition 4.3 up to logarithmic factors.

Next, to improve readability, we summarize the conditions for the upcoming theorems and lemmas:

Condition 4.4. Suppose Assumptions 2.1, 4.1 and 4.2 hold (a formal definition of Assumption 2.1 is deferred

to Assumption 6.3). The width m is large enough to control the estimation error of the NN (details are deferred to Condition C.2). Set S as a norm parameter satisfying $S \geq \sqrt{2\mathbf{h}^\top \mathbf{H}^{-1} \mathbf{h}}$. The regularization parameter λ_t follows the update rule in Equation (3). For training the NN, set the number of gradient descent iterations as $J = 2 \log(\lambda_t S / (\sqrt{T} \lambda_t + C_4 T^{3/2} L)) TL / \lambda_t$, and the step size as $\eta = C_5 (mTL + m\lambda_t)^{-1}$ for some absolute constants $C_4, C_5 > 0$.

In particular, when m is sufficiently large, we observe that the true reward function $h(x)$ behaves like a linear function (see Lemma 6.1). Then, using the tail inequality given in Theorem 3.1 and the update rule for λ_t , we obtain the following data- and variance-adaptive concentration inequality between θ_t and θ^* :

Lemma 4.5. *Define*

$$H_t := \sum_{i=1}^t \frac{\dot{\mu}(g(x_i; \theta_0)^\top (\theta - \theta_0))}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I}.$$

Under Condition 4.4, there exists an absolute constant $C_1, C_6 > 0$, such that for all $t > 0$ with probability at least $1 - \delta$, $\sqrt{m} \|\theta^ - \theta_t\|_{H_t(\theta^*)} \leq \nu_t^{(1)}$, where $\nu_t^{(1)}$ is defined in Equation (1).*

The proof is deferred to Section E.1. We now present Theorem 4.6, which gives the desired regret upper bound of Algorithm 1.

Theorem 4.6. *Under Condition 4.4, with probability at least $1 - \delta$, the regret of Algorithm 1 satisfies*

$$\text{Regret}(T) = \tilde{O} \left(S^2 \tilde{d} \sqrt{\kappa T} + S^{2.5} \sqrt{\kappa \tilde{d} T} \right).$$

Remark 4.7. Our results, especially Theorem 3.1, extend naturally to the (neural) dueling bandit setting. In this variant, the learner selects a pair of context–arms $\{x_{t,1}, x_{t,2}\}$ in each round t and observes a binary outcome $r_t \in \{0, 1\}$ indicating whether $x_{t,1}$ is preferred over $x_{t,2}$. The preference probability is modeled as $\mathbb{P}(x_{t,1} \succ x_{t,2}) = \mathbb{P}(r_t = 1 | x_{t,1}, x_{t,2}) = \mu(h(x_{t,1}) - h(x_{t,2}))$. The prior work of Verma et al. (2025) (Theorem 3) establishes a regret upper bound of $\tilde{O}(\kappa \tilde{d} \sqrt{T})$, whereas our analysis can improve this to $\tilde{O}(\tilde{d} \sqrt{\kappa T})$.

5. Refined Algorithm with Neural Network-Estimated Variance

In this section, we explain NeuralLog-UCB-2, which guarantees the tightest regret upper bounds. Although Lemma 4.5 establishes a variance-adaptive concentration inequality with $H_t(\theta^*)$, the agent lacks full knowledge of θ^* and must therefore use the crude bound $H_t(\theta^*) \preceq \kappa^{-1} V_t$, which incurs an extra factor of $\sqrt{\kappa}$. In this section, we introduce

Algorithm 2 NeuralLog-UCB-2

input Neural network $f(x; \theta)$ with width m and depth L , initialized with parameter θ_0 , step size η , number of gradient descent steps L , norm parameter S , confidence parameter δ

init $\lambda_0 = 8\sqrt{2}C_1 L^{1/2} S^{-1} \log(4/\delta)$, $W_0 = \lambda_0 \mathbf{I}$

- 1: **for** $t = 1, \dots, T$ **do**
- 2: $x_t \leftarrow \arg \max_{x \in \mathcal{X}_t} g(x; \theta_0)^\top (\theta_{t-1} - \theta_0) + \nu_{t-1}^{(2)} \times \|g(x; \theta_0) / \sqrt{m}\|_{W_{t-1}^{-1}}$
- 3: **Select** x_t and receive r_t
- 4: **Update** λ_t as in Equation (3), ι_t as in Equation (2), $\nu_t^{(2)}$ as in Equation (4)
- 5: $\theta_t \leftarrow \text{TrainNN}(\lambda_t, \eta, J, m, \{x_i, r_i\}_{i=1}^t, \theta_0)$
- 6: $W_t \leftarrow \sum_{i=1}^t \frac{\dot{\mu}(f(x_i; \theta_i))}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I}$
- 7: **end for**

NeuralLog-UCB-2, which replaces $H_t(\theta^*)$ with a neural network–estimated variance-adaptive design matrix. We begin by stating a concentration result for θ^* around θ_t using the new design matrix W_t .

Lemma 5.1. *Define*

$$W_t = \sum_{i=1}^t \frac{\dot{\mu}(f(x_i; \theta_i))}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I},$$

and a confidence set $\mathcal{W}_t$ as

$$\mathcal{W}_t = \{\theta : \sqrt{m} \|\theta - \theta_t\|_{W_t} \leq \nu_t^{(2)}\}, \quad (4)$$

where $\nu_t^{(2)} := \{\theta : \sqrt{m} \|\theta - \theta_t\|_{W_t} \leq C_7(1 + \sqrt{L}S + LS^2)\iota_t + 1$ with an absolute constant $C_7 > 0$, and ι_t defined at Equation (2). Then under Condition 4.4, for all $t > 0$, $\theta^ \in \mathcal{W}_t$ with probability at least $1 - \delta$.*

We give the proof of the lemma in Section F.1. The matrix W_t maintains sufficient statistics via the neural network–estimated variance, and the ellipsoidal confidence set $\mathcal{W}_t$ changes the original problem into a closed-form optimistic formulation. Specifically, after the same initialization step as for Algorithm 1, the agent selects action x_t in each round t according to the following rule:

$$\begin{aligned} x_t &\leftarrow \arg \max_{x \in \mathcal{X}_t, \theta \in \mathcal{W}_{t-1}} \langle g(x; \theta_0), (\theta - \theta_0) \rangle \\ &= \arg \max_{x \in \mathcal{X}_t} g(x; \theta_0)^\top (\theta_{t-1} - \theta_0) + \nu_{t-1}^{(2)} \|x\|_{W_{t-1}^{-1}}. \end{aligned} \quad (5)$$

For the regret upper bound of Algorithm 2, we define another problem-dependent quantity κ^* as $1/\kappa^* = \frac{1}{T} \sum_{t=1}^T \dot{\mu}(h(x_t^*))$, consistent with the definition in (Abeille et al., 2021). Both κ^* and κ scale exponentially with S . We now state our regret upper bound for NeuralLog-UCB-2 and provide a proof outline.

Theorem 5.2. Under Condition 4.4, with probability at least $1 - \delta$, the regret of Algorithm 2 satisfies:

$$\text{Regret}(T) = \tilde{\mathcal{O}}\left(S^2 \tilde{d} \sqrt{T/\kappa^*} + S^{2.5} \tilde{d}^{0.5} \sqrt{T/\kappa^*} + S^4 \kappa \tilde{d}^2 + S^{4.5} \kappa \tilde{d}^{1.5} + S^5 \kappa \tilde{d}\right).$$

Remark 5.3. It is possible to further reduce the regret bound in Theorem 5.2 to $\tilde{\mathcal{O}}(S \tilde{d} \sqrt{T/\kappa^*})$ by combining Theorem 3.1 and the logistic bandit analysis of (Fauray et al., 2022), which achieved $\tilde{\mathcal{O}}(S \tilde{d} \sqrt{T/\kappa^*})$. However, this approach requires a projection step for θ_t , incurring an additional $\mathcal{O}(d^2 \log(1/\epsilon))$ computational cost for ϵ -accuracy. A couple of recent works eliminated the dependence on S in the leading term, achieving $\tilde{\mathcal{O}}(d \sqrt{T/\kappa^*})$. Nonetheless, (Sawarni et al., 2024) relied on a nonconvex optimization subroutine, while the PAC-Bayes analysis in (Lee et al., 2024a) with a uniform prior does not yield data-adaptive regret.

6. Regret Analyses

This section outlines the regret analysis for Algorithms 1 and 2 and provides proof sketch for Theorems 4.6 and 5.2. Let us start by stating some basic results on the NTK analysis and logistic bandits. The following lemma shows that for all $x \in \mathcal{X}_t$ and $t \in [T]$, the true reward function $h(x)$ can be expressed as a linear function.

Lemma 6.1 (Lemma 5.1, (Zhou et al., 2020)). If $m \geq C_0 T^4 K^4 L^6 \log(T^2 K^2 L/\delta)/\lambda_{\mathbf{H}}^4$ for some absolute constant $C_0 > 0$, then with probability at least $1 - \delta$, there exists $\theta^* \in \mathbb{R}^p$ such that

$$h(x) = g(x; \theta_0)^\top (\theta^* - \theta_0),$$

$$\sqrt{m} \|\theta^* - \theta_0\|_2 \leq \sqrt{2 \mathbf{h}^\top \mathbf{H}^{-1} \mathbf{h}} \leq S,$$

for all $x \in \mathcal{X}_t$, $t \in [T]$.

Based on Lemma 6.1, we define the parameter set Θ and the parameters κ and R , which is consistent with the standard logistic bandits literature (Fauray et al., 2020):

Definition 6.2. Let $\Theta := \{\theta \in \mathbb{R}^p : \sqrt{m} \|\theta - \theta_0\|_2 \leq S\}$ denote the parameter set.

Assumption 6.3 (Formal). There exist constants $\kappa, R > 0$ such that for all $x \in \mathcal{X}$, $\theta \in \Theta$,

$$1/\kappa \leq \dot{\mu}(g(x; \theta_0)^\top (\theta - \theta_0)) \leq R.$$

6.1. Proof sketch of Theorem 4.6

Let $|\mu(h(x)) - \mu(f(x; \theta_{t-1}))|$ denote the *prediction error* of x in round t , which is the estimation error between the true reward and our trained neural network. We show that with Lemma 4.5 and large enough m , the prediction error is upper bounded as follows:

Lemma 6.4. Under Condition 4.4, for all $x \in \mathcal{X}_t$, $t \in [T]$, with probability at least $1 - \delta$,

$$|\mu(h(x)) - \mu(f(x; \theta_{t-1}))|$$

$$\leq R \sqrt{\kappa} \nu_t^{(1)} \|g(x; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + \epsilon_{3,t-1},$$

where $\epsilon_{3,t} = C_3 R m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda_0^{-2/3}$ for some absolute constant $C_3 > 0$.

Based on the results so far, we can upper bound the *per-round regret* in round t as follows:

$$\mu(h(x_t^*)) - \mu(h(x_t))$$

$$\leq \mu(f(x_t^*; \theta_{t-1})) + R \sqrt{\kappa} \nu_t^{(1)} \|g(x_t^*; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}$$

$$+ \epsilon_{3,t-1} - \mu(h(x_t))$$

$$\leq \mu(f(x_t; \theta_{t-1})) + R \sqrt{\kappa} \nu_t^{(1)} \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}$$

$$+ \epsilon_{3,t-1} - \mu(h(x_t))$$

$$\leq 2R \sqrt{\kappa} \nu_t^{(1)} \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + 2\epsilon_{3,t-1},$$

where the first and last inequalities follow from Lemma 6.4, and the second inequality holds due to the optimistic rule of Algorithm 1. The cumulative regret can be decomposed as $\text{Regret}(T) = \sum_{t=1}^T \mu(h(x_t^*)) - \mu(h(x_t)) \leq 2R \sqrt{\kappa} \nu_T^{(1)} \sqrt{T \sum_{t=1}^T \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}^2} + 2T \epsilon_{3,T}$, for which we use the Cauchy-Schwarz inequality. We have $\nu_T^{(1)} = \tilde{\mathcal{O}}(\sqrt{\tilde{d}})$, and using the elliptical potential lemma (Lemma H.2) on $\sum_{t=1}^T \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}^2$ gives $\tilde{\mathcal{O}}(\tilde{d})$. Finally, setting m large enough under Condition 4.4, the approximation error term gives $T \epsilon_{3,T} = \mathcal{O}(1)$. See Section G.1 for details.

6.2. Proof sketch of Theorem 5.2

Let $(x_t, \tilde{\theta}_{t-1}) \in \mathcal{X}_t \times \mathcal{W}_{t-1}$ be selected by the optimistic rule at time t . The per-round regret can be decomposed with a second-order Taylor expansion as follows:

$$\mu(h(x_t^*)) - \mu(h(x_t))$$

$$\leq \mu(g(x_t; \theta_0)^\top (\tilde{\theta}_{t-1} - \theta_0)) - \mu(g(x_t; \theta_0)^\top (\theta^* - \theta_0))$$

$$\leq \dot{\mu}(h(x_t)) g(x_t; \theta_0)^\top (\tilde{\theta}_{t-1} - \theta^*)$$

$$+ 1 \cdot [g(x_t; \theta_0)^\top (\tilde{\theta}_{t-1} - \theta^*)]^2,$$

where the first inequality follows from the optimistic rule of Algorithm 2, and we use $\ddot{\mu}(\cdot) \leq 1$ for the second one. To analyze the first term on the right-hand side of the second inequality, we compare $\dot{\mu}(h(x_t))$ and $\dot{\mu}(f(x_t; \theta_t))$ and rewrite the term as $\sqrt{\dot{\mu}(h(x_t))} \sqrt{\dot{\mu}(f(x_t; \theta_t))} g(x_t; \theta_0)^\top \tilde{\theta}_{t-1} - \theta^* \|_{W_{t-1}^{-1}}$. Summing this for $t = 1, \dots, T$, we apply the elliptical potential lemma (Lemma H.2) and Lemma 5.1. For

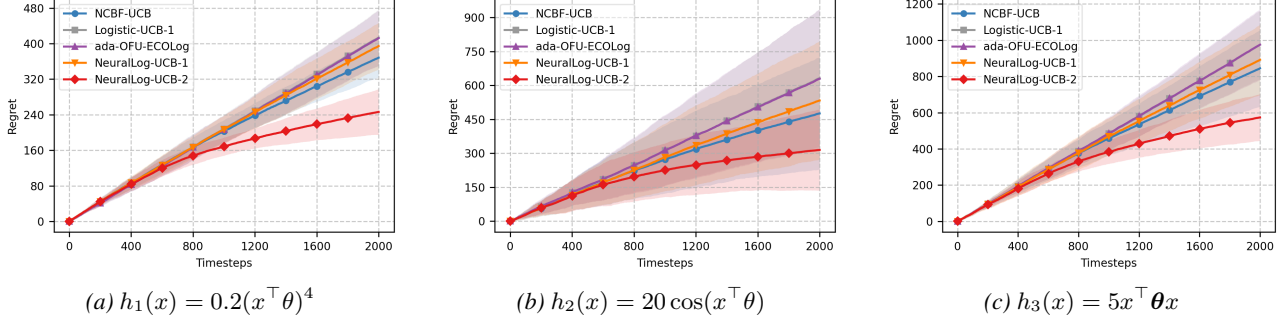


Figure 1. Comparison of cumulative regret of baseline algorithms for nonlinear reward functions.

the second term, since we do not enforce any projection or constraint during training, θ_{t-1} may stay outside Θ . We show that the number of such rounds is $\tilde{O}(\kappa \tilde{d}^2)$. Applying Assumption 6.3 then yields a crude bound of $\kappa \|g(x_t; \theta_0)\|_{V_{t-1}}^2 \|\tilde{\theta}_{t-1} - \theta^*\|_{W_t}^2$. Based on this, the second term can be bounded from above in a similar way. Details are covered in Section G.2.

7. Experiments

In this section, we empirically evaluate the performance of our algorithms. Additional results along with further details are deferred to Section A due to space constraints.

Synthetic dataset. We begin our experiments with a synthetic dataset. We use three nonlinear synthetic latent reward functions: $h_1(x) = 0.2(x^\top \theta)^4$, $h_2(x) = 20 \cos(x^\top \theta)$, $h_3(x) = 5x^\top \theta x$, where x represents the features of a context-arm pair, and $\theta \in \mathbb{R}^d$ and $\theta \in \mathbb{R}^{d \times d}$ are parameters whose elements are independently sampled from $\text{Unif}(-1, 1)$. Subsequently, the agent receives a reward generated by $r_t \sim \text{Bern}(\mu(h_i(x)))$, for $i \in \{1, 2, 3\}$. We set the feature vector dimension to $d = 20$ and the number of arms to $K = 5$. We compare our method against five baseline algorithms in Table 1: (1) NCBF-UCB (Verma et al., 2025); (2) Logistic-UCB-1 (Fauray et al., 2020); (3) ada-OFU-ECOLog (Fauray et al., 2022); (4) NeuralLog-UCB-1; and (5) NeuralLog-UCB-2. For brevity, we will denote algorithms by their number (e.g. algorithm (1)).

Following practical adjustments from previous neural bandits experiments (Zhou et al., 2020; Zhang et al., 2021; Verma et al., 2025), for algorithms (1,4,5), we use the gradient of the current neural network $g(x; \theta_t)$ instead of $g(x; \theta_0)$. We replace $g(x; \theta_t)/\sqrt{m}$ with $g(x; \theta_t)$ and $m\lambda \|\theta - \theta_0\|_2^2/2$ with $\lambda \|\theta\|_2^2$. Previous works simplify the UCB estimation process by fixing parameters for the exploration bonus for practical reasons. In this work, however, we consider the time-varying data-adaptive values of the exploration bonus, characterized by $\text{UCB}_t(x) = \mu(x; \theta_{t-1}) + \sigma(x; \nu, \{x_i, \theta_{i-1}\}_{i=1}^{t-1}, \lambda, S, \kappa)$. Here, μ is the

mean estimate and σ is the exploration bonus, parameterized by an exploration parameter ν , previous observations $\{x_i, \theta_{i-1}\}_{i=1}^{t-1}$, λ , S , and κ . Details of UCB for each algorithm are deferred to Section 5. We use $S = 1$, $\kappa = 10$ and fixed values of ν and λ with the best parameter values using grid search over $\{0.01, 0.1, 1, 10, 100\}$.

We use a two-layer neural network $f(x; \theta)$ with a width of $m = 20$. As in (Zhou et al., 2020), to reduce the computational burden of the high-dimensional design matrices V_t and W_t , we approximated these matrices with diagonal matrices. We update the parameters every 50 rounds, using 100 gradient descent steps per update with a learning rate of 0.01. For each algorithm, we repeat the experiments 10 times over $T = 2000$ timesteps and compare the average cumulative regret with a 96% confidence interval.

Real-world dataset. In the real-world experiments, we use three datasets from K -class classification tasks: `mnist` (LeCun et al., 1998), `mushroom`, and `shuttle` from the UCI Machine Learning Repository (Dua & Graff, 2019). To adapt these datasets to the K -armed logistic bandit setting, we construct K context-arm feature vectors in each round t as follows: given a feature vector $x \in \mathbb{R}^d$, we define $x^{(1)} = [x, \mathbf{0}, \dots, \mathbf{0}]$, $\dots$, $x^{(K)} = [\mathbf{0}, \dots, \mathbf{0}, x] \in \mathbb{R}^{dK}$. The agent receives a reward of 1 if it selects the correct class, and 0 otherwise. All other adjustments for the neural bandit experiments and the neural network training process follow the simulation setup. Details, including data preprocessing, are deferred to Section A.

Regret comparison. Figures 1 and 2 summarize the average cumulative regret for the five baseline algorithms (1–5) tested with the synthetic and real-world datasets, respectively. We observe that the algorithms using linear assumptions on the latent reward function $h(x)$, namely (2) and (3), exhibit the lowest performance, as the true function is nonlinear. Although algorithm (1) can handle nonlinear reward functions and achieves moderate performance, our proposed methods, especially (5), yield the best results by reducing the dependence on κ .

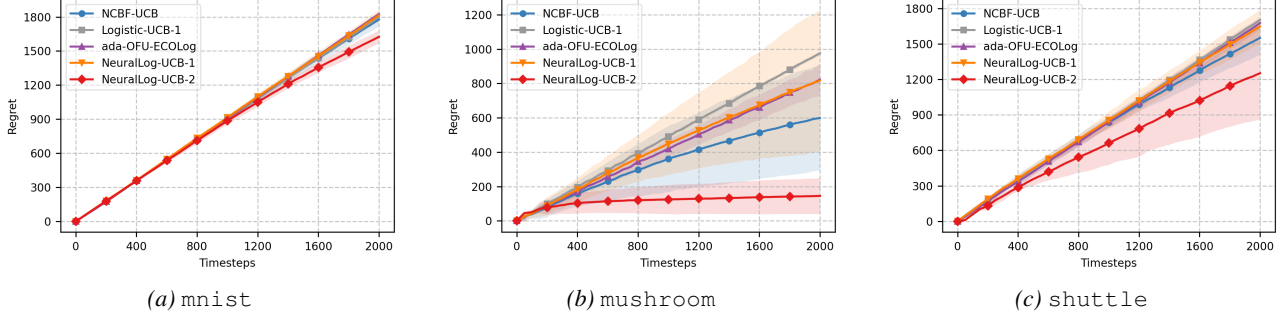


Figure 2. Comparison of cumulative regret of baseline algorithms for real-world dataset.

8. Conclusion and Future Work

In this paper, we study the *neural logistic bandit* problem. We identify the unique challenges of this setting and propose a novel approach based on a new tail inequality for martingales. This inequality enables an analysis that is both variance- and data-adaptive, yielding improved regret bounds for neural logistic bandits. We introduce two algorithms: NeuralLog-UCB-1 that achieves a regret bound of $\tilde{O}(\tilde{d}\sqrt{\kappa T})$ and NeuralLog-UCB-2 that attains a tighter bound of $\tilde{O}(\tilde{d}\sqrt{T/\kappa^*})$ by leveraging the neural network-estimated variance. Our experimental results validate these theoretical findings and demonstrate that our methods outperform the existing approaches.

One potential direction for future work is to improve the dependence on the norm of the unknown parameter S . Although recent frameworks due to (Sawarni et al., 2024; Lee et al., 2024a) have removed the dependence on S from the leading term, they require an additional training step or impose additional constraints. Such requirements are undesirable when trying to integrate neural bandit frameworks. Hence, it is a promising future research direction to eliminate the dependence on S without additional computations. Additional future directions are deferred to Section K.

Acknowledgements

We thank the review team for their careful review and valuable feedback. This work was supported by the National Research Foundation of Korea (NRF) grant (No. RS-2024-00350703) and the Institute of Information & communications Technology Planning & evaluation (IITP) grants (No. IITP-2026-RS-2024-00437268) and (No. RS-2021-II211343, Artificial Intelligence Graduate School Program (Seoul National University)) funded by the Korea government (MSIT).

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Abbasi-Yadkori, Y., Pál, D., and Szepesvári, C. Improved algorithms for linear stochastic bandits. *Advances in neural information processing systems*, 24, 2011.
- Abeille, M., Faury, L., and Calauzènes, C. Instance-wise minimax-optimal algorithms for logistic bandits. In *International Conference on Artificial Intelligence and Statistics*, pp. 3691–3699. PMLR, 2021.
- Ban, Y., Yan, Y., Banerjee, A., and He, J. EE-net: Exploitation-exploration neural networks in contextual bandits. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=X_ch3VrNSRg.
- Dong, S., Ma, T., and Van Roy, B. On the performance of thompson sampling on logistic bandits. In *Conference on Learning Theory*, pp. 1158–1160. PMLR, 2019.
- Dua, D. and Graff, C. Uci machine learning repository. <https://archive.ics.uci.edu/ml>, 2019. Accessed: 2025-03-21.
- Faury, L., Abeille, M., Calauzènes, C., and Fercoq, O. Improved optimistic algorithms for logistic bandits. In *International Conference on Machine Learning*, pp. 3052–3060. PMLR, 2020.
- Faury, L., Abeille, M., Jun, K.-S., and Calauzènes, C. Jointly efficient and optimal algorithms for logistic bandits. In *International Conference on Artificial Intelligence and Statistics*, pp. 546–580. PMLR, 2022.

- Filippi, S., Cappe, O., Garivier, A., and Szepesvári, C. Parametric bandits: The generalized linear case. *Advances in neural information processing systems*, 23, 2010.
- Freedman, D. A. On tail probabilities for martingales. *the Annals of Probability*, pp. 100–118, 1975.
- Gales, S. B., Sethuraman, S., and Jun, K.-S. Norm-agnostic linear bandits. In *International Conference on Artificial Intelligence and Statistics*, pp. 73–91. PMLR, 2022.
- Jacot, A., Gabriel, F., and Hongler, C. Neural tangent kernel: Convergence and generalization in neural networks. *Advances in neural information processing systems*, 31, 2018.
- Jia, Y., ZHANG, W., Zhou, D., Gu, Q., and Wang, H. Learning neural contextual bandits through perturbed rewards. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=7inCJ3MhXt3>.
- Jun, K.-S., Jain, L., Mason, B., and Nassif, H. Improved confidence bounds for the linear logistic model and applications to bandits. In *International Conference on Machine Learning*, pp. 5148–5157. PMLR, 2021.
- Kassraie, P. and Krause, A. Neural contextual bandits without regret. In *International Conference on Artificial Intelligence and Statistics*, pp. 240–278. PMLR, 2022.
- Kim, W., Lee, K., and Paik, M. C. Double doubly robust thompson sampling for generalized linear contextual bandits. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 8300–8307, 2023.
- Kim, Y., Yang, I., and Jun, K.-S. Improved regret analysis for variance-adaptive linear bandits and horizon-free linear mixture mdps. *Advances in Neural Information Processing Systems*, 35:1060–1072, 2022.
- Kveton, B., Zaheer, M., Szepesvari, C., Li, L., Ghavamzadeh, M., and Boutilier, C. Randomized exploration in generalized linear bandits. In *International Conference on Artificial Intelligence and Statistics*, pp. 2066–2076. PMLR, 2020.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Lee, J. and Oh, M.-h. Nearly minimax optimal regret for multinomial logistic bandit. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=Q4NWfStqVf>.
- Lee, J. and Oh, M.-h. Improved online confidence bounds for multinomial logistic bandits. *arXiv preprint arXiv:2502.10020*, 2025.
- Lee, J., Yun, S.-Y., and Jun, K.-S. A unified confidence sequence for generalized linear models, with applications to bandits. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024a. URL <https://openreview.net/forum?id=MDdOQayWTA>.
- Lee, J., Yun, S.-Y., and Jun, K.-S. Improved regret bounds of (multinomial) logistic bandits via regret-to-confidence-set conversion. In *International Conference on Artificial Intelligence and Statistics*, pp. 4474–4482. PMLR, 2024b.
- Li, L., Chu, W., Langford, J., and Schapire, R. E. A contextual-bandit approach to personalized news article recommendation. In *Proceedings of the 19th international conference on World wide web*, pp. 661–670, 2010.
- Li, L., Lu, Y., and Zhou, D. Provably optimal algorithms for generalized linear contextual bandits. In *International Conference on Machine Learning*, pp. 2071–2080. PMLR, 2017.
- Riquelme, C., Tucker, G., and Snoek, J. Deep bayesian bandits showdown: An empirical comparison of bayesian deep networks for thompson sampling. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=SyYe6k-CW>.
- Saito, Y., Aihara, S., Matsutani, M., and Narita, Y. Open bandit dataset and pipeline: Towards realistic and reproducible off-policy evaluation. *arXiv preprint arXiv:2008.07146*, 2020.
- Sawarni, A., Das, N., Barman, S., and Sinha, G. Generalized linear bandits with limited adaptivity. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=FTPDBQuT4G>.
- Srinivas, N., Krause, A., Kakade, S. M., and Seeger, M. Gaussian process optimization in the bandit setting: No regret and experimental design. *arXiv preprint arXiv:0912.3995*, 2009.
- Valko, M., Korda, N., Munos, R., Flouounas, I., and Cristianini, N. Finite-time analysis of kernelised contextual bandits. In *Proceedings of the Twenty-Ninth Conference on Uncertainty in Artificial Intelligence*, UAI’13, pp. 654–663, Arlington, Virginia, USA, 2013. AUAI Press.
- Verma, A., Dai, Z., Lin, X., Jaillet, P., and Low, B. K. H. Neural dueling bandits: Preference-based optimization with human feedback. In *The Thirteenth*

International Conference on Learning Representations, 2025. URL <https://openreview.net/forum?id=VELhv9BBfn>.

Xu, P., Wen, Z., Zhao, H., and Gu, Q. Neural contextual bandits with deep representation and shallow exploration. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=xnYACQquaGV>.

Zahavy, T. and Mannor, S. Deep neural linear bandits: Overcoming catastrophic forgetting through likelihood matching. *arXiv preprint arXiv:1901.08612*, 2019.

Zenati, H., Bietti, A., Diemert, E., Mairal, J., Martin, M., and Gaillard, P. Efficient kernelized ucb for contextual bandits. In *International Conference on Artificial Intelligence and Statistics*, pp. 5689–5720. PMLR, 2022.

Zhang, W., Zhou, D., Li, L., and Gu, Q. Neural thompson sampling. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=tkAtoZkcUnm>.

Zhang, Y.-J. and Sugiyama, M. Online (multinomial) logistic bandit: Improved regret and constant computation cost. *Advances in Neural Information Processing Systems*, 36, 2024.

Zhou, D., Li, L., and Gu, Q. Neural contextual bandits with ucb-based exploration. In *International Conference on Machine Learning*, pp. 11492–11502. PMLR, 2020.

Zhou, D., Gu, Q., and Szepesvari, C. Nearly minimax optimal reinforcement learning for linear mixture markov decision processes. In *Conference on Learning Theory*, pp. 4532–4576. PMLR, 2021.

A. Deferred Experiments from Section 7

Here we introduce the deferred details and experiments from Section 7. All experiments were conducted on a server equipped with an Intel Xeon Gold 6248R 3.00GHz CPU (32 cores), 512GB of RAM, and 4 GeForce RTX 4090 GPUs.

Details of UCB. We define the UCB value as $\mu(x; \theta_{t-1}) + \sigma(x; \nu, \{x_i, \theta_{i-1}\}_{i=1}^{t-1}, \lambda, S, \kappa)$. For the exploration bonus σ , we match the orders of λ , S , κ , and the effective dimension $\tilde{d}$ for each algorithm and then multiply by the exploration parameter ν . Specifically, the effective dimension is defined as follows: for algorithm (1), we use $\log \det(\sum \frac{1}{\kappa} g(x; \theta) g(x; \theta)^\top + \mathbf{I})$; for algorithms (2) and (3), we use $\log \det(\sum R x x^\top + \mathbf{I})$; and for algorithms (4) and (5) we use $\log \det(\sum R g(x; \theta) g(x; \theta)^\top + \mathbf{I})$.

Although algorithms (2) and (3) require an additional step (e.g., nonconvex projection) to ensure that θ_t remains in the desired set, empirical observations from (Fauray et al., 2020; 2022) indicate that θ_t almost always satisfies this condition. Consequently, we streamline all baseline algorithms into two steps: (i) choose the action with the highest UCB and (ii) update the parameters via gradient descent.

Preprocessing for real-world datasets. For consistency with the synthetic environment, we rescale each component of every feature vector $x \in \mathbb{R}^d$ to the range $[-1, 1]$ by applying a normalization of $2 \frac{[x]_j - \min(x)}{\max(x) - \min(x)} - 1$ for all $j \in [d]$. In the `mnist` dataset, we resize each 28×28 image to 7×7 , flatten it, and treat the result as a 196-dimensional feature vector. The `mushroom` dataset provides 22 categorical features. We assign each character a random value in $[-1, 1]$ for normalization and set the label to 1 for edible ('e') and 0 for poisonous ('p') mushrooms. The `shuttle` dataset consists of 7 numerical features, to which we apply the same min-max normalization as used for `mnist`.

Varying effective dimension $\tilde{d}$. To evaluate the influence of $\tilde{d}$ on data-adaptive algorithms, we compare cumulative regret across different values of $\tilde{d}$. We control $\tilde{d}$ by limiting the total number of context-arm feature vectors during training. Allowing redundant vectors reduces $\tilde{d}$. For a low effective dimension (Figures 3a, 3d and 3g), we use only 10 feature vectors randomly placed across the training. For a medium $\tilde{d}$ (Figures 3b, 3e and 3h), we use 50 vectors. For a high $\tilde{d}$ (Figures 3c, 3f and 3i), we use 10000 distinct vectors. Note that Figures 1 and 2 use 2500 vectors. Figure 3 shows that our algorithm, especially algorithm (5), performs best across all those settings and adapts effectively to different environments of the effective dimensions.

B. Related Work

Logistic bandits. (Filippi et al., 2010) introduced the generalized linear bandit framework and derived a regret bound of $\tilde{O}(\kappa d \sqrt{T})$, laying the groundwork for modeling logistic bandits. Subsequent work, starting with (Fauray et al., 2020), has focused on reducing the dependence on κ through variance-aware analyses (see also (Dong et al., 2019; Abeille et al., 2021)). In particular, (Abeille et al., 2021) established a lower bound of $\Omega(d \sqrt{T/\kappa^*})$, statistically closing the gap. However, there is still room for improvement in algorithmic efficiency (Fauray et al., 2022; Zhang & Sugiyama, 2024; Lee & Oh, 2024) and in mitigating the influence of the norm parameter S , with several recent advances addressing this issue (Lee et al., 2024b; Sawarni et al., 2024; Lee et al., 2024a; Lee & Oh, 2025). Another line of research investigates the finite-action setting. When feature vectors are drawn i.i.d. from an unknown distribution whose covariance matrix has a strictly positive minimum eigenvalue, (Li et al., 2017) achieved a regret of $\tilde{O}(\kappa \sqrt{dT})$, while (Kim et al., 2023) and (Jun et al., 2021) further improved it to $\tilde{O}(\sqrt{\kappa d T})$ and $\tilde{O}(\sqrt{dT})$, respectively.

Neural bandits. Advances in deep neural networks have spurred numerous methods that integrate deep learning with contextual bandit algorithms (Riquelme et al., 2018; Zahavy & Mannor, 2019; Kveton et al., 2020). (Zhou et al., 2020) was among the first to formalize neural bandits, proposing the *NeuralUCB* algorithm, which attains a regret bound of $\tilde{O}(\tilde{d} \sqrt{T})$ by leveraging neural tangent kernel theory (Jacot et al., 2018). Building on this foundation, many studies have extended linear contextual bandit algorithms to the neural setting (Zhang et al., 2021; Kassraie & Krause, 2022; Ban et al., 2022; Xu et al., 2022; Jia et al., 2022). The work most closely related to ours is (Verma et al., 2025), which first addressed both logistic and dueling neural bandits and proposed UCB- and Thompson-sampling-based algorithms with a regret bound of $\tilde{O}(\kappa \tilde{d} \sqrt{T})$.

C. Useful Lemmas for Neural Bandits

In this section, we present several lemmas that enable the neural bandit analysis to quantify the approximation error incurred when approximating the unknown reward function $h(x)$ with the neural network $f(x; \theta)$. We begin with the definition of the

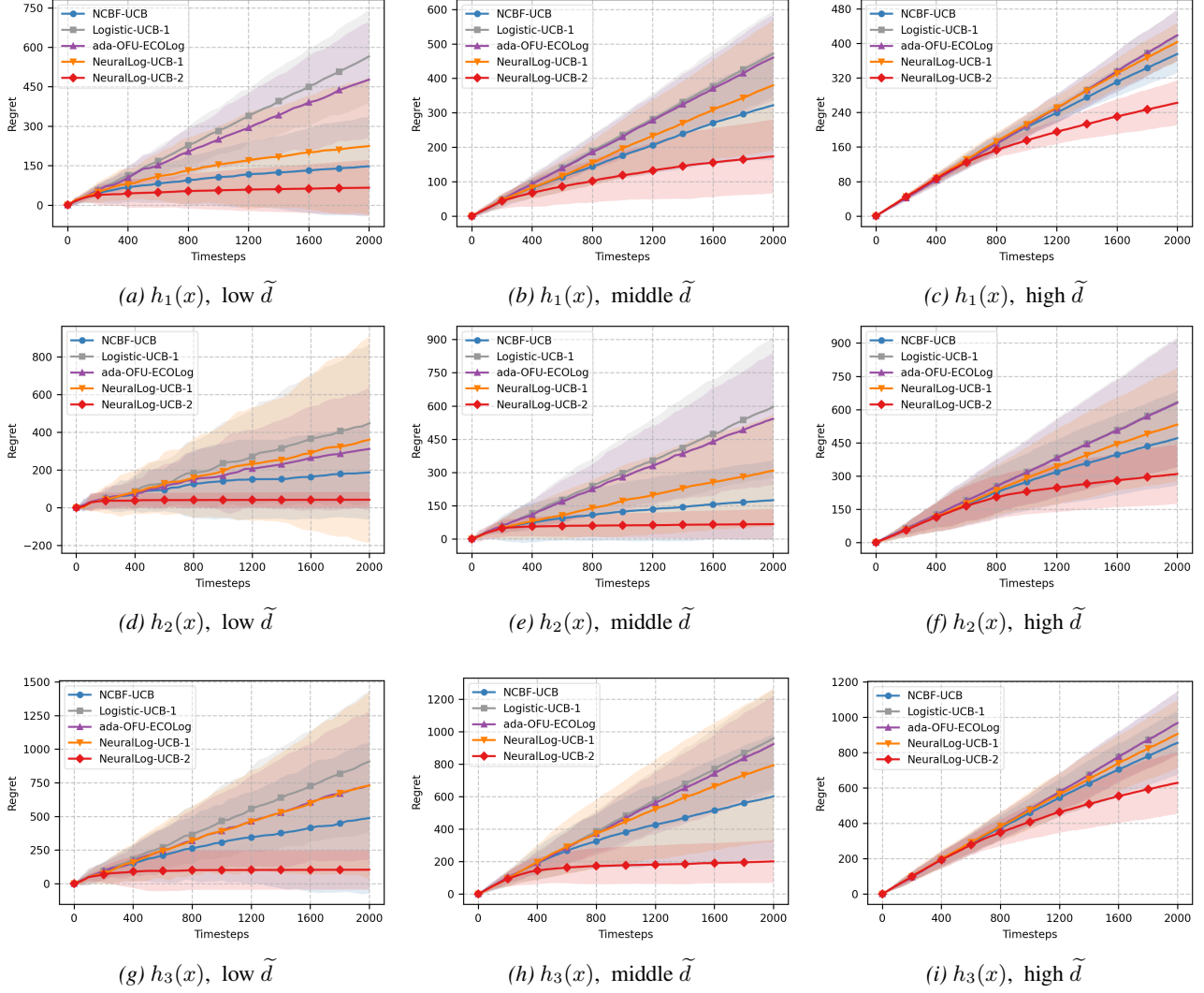


Figure 3. Comparison of cumulative regret of baseline algorithms with varying effective dimension $\tilde{d}$.

neural tangent kernel (NTK) matrix (Jacot et al., 2018):

Definition C.1. Denote all contexts until round T as $\{x^i\}_{i=1}^{TK}$. For $i, j \in [TK]$, define

$$\begin{aligned}\hat{\mathbf{H}}_{i,j}^{(1)} &= \Sigma_{i,j}^{(1)} = \langle x^i, x^j \rangle, \quad \mathbf{A}_{i,j}^{(l)} = \begin{pmatrix} \Sigma_{i,j}^{(l)} & \Sigma_{i,j}^{(l)} \\ \Sigma_{i,j}^{(l)} & \Sigma_{i,j}^{(l)} \end{pmatrix}, \\ \Sigma_{i,j}^{(l+1)} &= 2\mathbb{E}_{(u,v) \sim (\mathbf{0}, \mathbf{A}_{i,j}^{(l)})} \max\{u, 0\} \max\{v, 0\}, \\ \hat{\mathbf{H}}_{i,j}^{(l+1)} &= 2\hat{\mathbf{H}}_{i,j}^{(l)} \mathbb{E}_{(u,v) \sim N(\mathbf{0}, \mathbf{A}_{i,j}^{(l)})} \mathbf{1}(u \geq 0) \mathbf{1}(v \geq 0) + \Sigma_{i,j}^{(l+1)}.\end{aligned}$$

Then, $\mathbf{H} = (\hat{\mathbf{H}}^{(L)} + \Sigma^{(L)})/2$ is called the NTK matrix on the context set.

Next, we introduce a condition on the neural network width m , which is crucial for ensuring that the approximation error remains sufficiently small.

Condition C.2. For an absolute constant $C_0 > 0$, the width of the NN m satisfies:

$$\begin{aligned} m &\geq C_0 \max \left\{ T^4 K^4 L^6 \log(T^2 K^2 L / \delta) \lambda_{\mathbf{H}}^4, L^{-3/2} \lambda_0^{1/2} [\log(T K L^2 / \delta)]^{3/2} \right\} \\ m(\log m)^{-3} &\geq C_0 T^7 L^{21} \lambda_0^{-1} + C_0 T^{16} L^{27} \lambda_0^{-7} R^6 + C_0 T^{10} L^{21} \lambda_0^{-4} R^6 + C_0 T^7 L^{18} \lambda_0^{-4}. \end{aligned}$$

We assume that m satisfies Condition C.2 throughout. For readability, we denote the error probability by δ in all probabilistic statements. We now restate Lemma 6.1, which shows that, for every $x \in \mathcal{X}_t$ and $t \in [T]$, the true reward function $h(x)$ can be represented as a linear function:

Lemma C.3 (Lemma 5.1, (Zhou et al., 2020)). *If $m \geq C_0 T^4 K^4 L^6 \log(T^2 K^2 L / \delta) / \lambda_{\mathbf{H}}^4$ for some absolute constant $C_0 > 0$, then with probability at least $1 - \delta$, there exists $\theta^* \in \mathbb{R}^p$ such that*

$$h(x) = g(x; \theta_0)^\top (\theta^* - \theta_0), \quad \sqrt{m} \|\theta^* - \theta_0\|_2 \leq \sqrt{2\mathbf{h}^\top \mathbf{H}^{-1} \mathbf{h}} \leq S,$$

for all $x \in \mathcal{X}_t$, $t \in [T]$.

Assuming θ_t remains close to its initialization θ_0 , we can apply the following lemmas: Lemma C.4 provides upper bounds on the norms $\|g(x; \theta)\|_2$ and $\|g(x; \theta) - g(x; \theta_0)\|_2$, while Lemma C.5 bounds the approximation error between $f(x; \theta)$ and its linearization $g(x; \theta_0)^\top (\theta - \theta_0)$.

Lemma C.4 (Lemma B.5 and B.6, (Zhou et al., 2020)). *Let $\tau = 3\sqrt{\frac{t}{m\lambda_t}}$. Then there exist absolute constants $C_1, C_2 > 0$, such that for all $x \in \mathcal{X}_t$, $t \in [T]$ and for all $\|\theta - \theta_0\|_2 \leq \tau$, with probability of at least $1 - \delta$,*

$$\begin{aligned} \|g(x; \theta)\|_2 &\leq C_1 \sqrt{mL} \\ \|g(x; \theta) - g(x; \theta_0)\|_2 &\leq C_2 \sqrt{m \log m} \tau^{1/3} L^{7/2} = C_2 m^{1/3} \sqrt{\log m} t^{1/6} \lambda_t^{-1/6} L^{7/2}. \end{aligned}$$

Lemma C.5 (Lemma B.4, (Zhou et al., 2020)). *Let $\tau = 3\sqrt{\frac{t}{m\lambda_t}}$. Then there exists an absolute constant $C_3 > 0$, for all $x \in \mathcal{X}_t$, $t \in [T]$, and for all $\|\theta - \theta_0\|_2 \leq \tau$, with probability of at least $1 - \delta$,*

$$\begin{aligned} \left| f(x; \theta) - g(x; \theta_0)^\top (\theta - \theta_0) \right|_2 \\ \leq C_3 \tau^{4/3} L^3 \sqrt{m \log m} = C_3 m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda_t^{-2/3}. \end{aligned}$$

Finally, we state a lemma that establishes an upper bound on the distance between θ_t and θ_0 . It also shows that, although the loss function $\mathcal{L}_t(\theta)$ is non-convex and hence the iterate θ_t obtained after J steps of gradient descent may differ from the ideal maximum likelihood estimator, this discrepancy remains sufficiently small. The proof is deferred to Section C.1:

Lemma C.6. *Define the auxiliary loss function $\tilde{\mathcal{L}}(\theta)$ as*

$$\begin{aligned} \tilde{\mathcal{L}}(\theta) &= - \sum_{i=1}^t r_i \log \mu(g(x_i; \theta_0)^\top (\theta - \theta_0)) + (1 - r_i) \log(1 - \mu(g(x_i; \theta_0)^\top (\theta - \theta_0))) \\ &\quad + \frac{m\lambda_t}{2} \|\theta - \theta_0\|_2^2, \end{aligned}$$

and the auxiliary sequence $\{\tilde{\theta}^{(j)}\}_{j=1}^J$ associated with the auxiliary loss $\tilde{\mathcal{L}}(\theta)$. Let the MLE estimator as $\hat{\theta}_t = \arg \min_{\theta} \tilde{\mathcal{L}}(\theta)$. Then there exist absolute constants $\{C_i\}_{i=1}^5 > 0$ such that if $J = 2 \log(\lambda_t S / (\sqrt{T} \lambda_t + C_4 T^{3/2} L)) T L / \lambda_t$ and $\eta = C_5 (m T L + m \lambda_t)^{-1}$, then with probability at least $1 - \delta$,

$$\begin{aligned} \|\theta_t - \tilde{\theta}_t\|_2 &\leq \sqrt{\frac{t}{m\lambda_t}}, \quad \|\theta_t - \theta_0\|_2 \leq 3\sqrt{\frac{t}{m\lambda_t}} = \tau, \\ \|\theta_t - \hat{\theta}_t\|_2 &\leq 2(1 - \eta m \lambda_t)^{J/2} t^{1/2} m^{-1/2} \lambda_t^{-1/2} \\ &\quad + C_2 m^{-2/3} \sqrt{\log m} t^{7/6} \lambda_t^{-7/6} L^{7/2} + C_1 C_3 R m^{-2/3} \sqrt{\log m} L^{7/2} t^{5/3} \lambda_t^{-5/3}. \end{aligned}$$

C.1. Proof of Lemma C.6

For simplicity, we omit the subscript t by default. First, recall the definition of the auxiliary sequence $\{\tilde{\theta}^{(j)}\}_{j=1}^J$ associated with the auxiliary loss $\tilde{L}(\theta)$; its update rule is given by:

$$\begin{aligned}\tilde{\theta}^{(j+1)} &= \tilde{\theta}^{(j)} - \eta \nabla \tilde{L}(\tilde{\theta}^{(j)}) \\ &= \tilde{\theta}^{(j)} - \eta \left[m\lambda \tilde{\theta}^{(j)} - \sum_{i=1}^t (\mu(g(x_i; \theta_0)^\top (\tilde{\theta}^{(j)} - \theta_0)) - r_i) g(x_i; \theta_0) \right].\end{aligned}$$

Similarly, the update rule for $\theta^{(j)}$ is given by:

$$\theta^{(j+1)} = \theta^{(j)} - \eta \left[m\lambda \theta^{(j)} - \sum_{i=1}^t (\mu(f(x; \theta^{(j)})) - r_i) g(x_i; \theta^{(j)}) \right].$$

Also notice that

$$\nabla^2 \tilde{L}(\theta) = \sum_{i=1}^t \mu(g(x_i; \theta_0)^\top (\theta - \theta_0)) (1 - \mu(g(x_i; \theta_0)^\top (\theta - \theta_0))) g(x_i; \theta_0) g(x_i; \theta_0)^\top + m\lambda \mathbf{I}. \quad (6)$$

Now, we start the proof with

$$\|\theta^{(j+1)} - \hat{\theta}\|_2 \leq \underbrace{\|\tilde{\theta}^{(j+1)} - \hat{\theta}\|_2}_{(\text{term 1})} + \underbrace{\|\theta^{(j+1)} - \tilde{\theta}^{(j+1)}\|_2}_{(\text{term 2})} \quad (7)$$

For **(term 1)**, observe from Equation (6) that $\tilde{\mathcal{L}}$ is $(m\lambda)$ -strongly convex, since $(m\lambda)\mathbf{I} \preceq \nabla^2 \tilde{L}(\theta)$. Moreover, $\tilde{\mathcal{L}}$ is a $C_5(tmL + m\lambda)$ -smooth function for some absolute constant $C_5 > 0$, because

$$\begin{aligned}\nabla^2 \tilde{L}(\theta) &\preceq \sum_{i=1}^t \frac{1}{2} \cdot \frac{1}{2} \cdot g(x_i; \theta_0) g(x_i; \theta_0)^\top + m\lambda \mathbf{I} \\ &\preceq \left(\sum_{i=1}^t \frac{1}{4} \|g(x_i; \theta_0)\|_2^2 + m\lambda \right) \mathbf{I} \\ &\preceq C_5(tmL + m\lambda) \mathbf{I},\end{aligned}$$

where the first inequality follows from $\mu(\cdot)(1 - \mu(\cdot)) \leq 1/4$, the second follows because for any $u, x \in \mathbb{R}^d$, $u^\top x x^\top u \leq \|u\|_2^2 \|x\|_2^2 \leq u^\top (\|x\|_2^2 \mathbf{I}) u$, and the last inequality follows from Lemma C.4.

Then, with our choice of $\eta = C_5(tmL + m\lambda)$, standard results for gradient descent on $\tilde{L}$ imply that $\tilde{\theta}^{(j)}$ converges to $\hat{\theta}_t$ at the rate as

$$\begin{aligned}\|\tilde{\theta}^{(j)} - \hat{\theta}\|_2^2 &\leq \frac{2}{m\lambda} \cdot (\tilde{\mathcal{L}}(\tilde{\theta}^{(j)}) - \tilde{\mathcal{L}}(\hat{\theta})) \\ &\leq (1 - \eta m\lambda)^j \cdot \frac{2}{m\lambda} \cdot (\tilde{\mathcal{L}}(\theta_0) - \tilde{\mathcal{L}}(\hat{\theta})) \\ &\leq (1 - \eta m\lambda)^j \cdot \frac{2}{m\lambda} \cdot \tilde{\mathcal{L}}(\theta_0),\end{aligned}$$

where the first and the second inequalities follow from the strong convexity and the smoothness of $\tilde{L}$. Furthermore, we have

$$\tilde{\mathcal{L}}(\theta_0) = - \sum_{i=1}^t r_i \log \mu(0) + (1 - r_i) \log(1 - \mu(0)) + \frac{m\lambda_t}{2} \|\theta_0 - \theta_0\|_2^2 = - \sum_{i=1}^t \log 0.5 \leq t,$$

Plugging this back gives

$$\|\tilde{\theta}^{(j)} - \hat{\theta}\|_2 \leq (1 - \eta m\lambda)^{j/2} \sqrt{2t/(m\lambda)}. \quad (8)$$

Next, consider **(term 2)**. From the definition of the update rule, it follows that:

$$\begin{aligned}
 \text{(term 2)} &\leq (1 - \eta m \lambda) \|\theta^{(j)} - \tilde{\theta}^{(j)}\|_2 \\
 &\quad + \eta \left\| \sum_{i=1}^t (\mu(f(x; \theta^{(j)}) - r_i) g(x_i; \theta^{(j)}) - \sum_{i=1}^t (\mu(g(x_i; \theta_0)^\top (\tilde{\theta}^{(j)} - \theta_0)) - r_i) g(x_i; \theta_0)) \right\|_2 \\
 &\leq (1 - \eta m \lambda) \|\theta^{(j)} - \tilde{\theta}^{(j)}\|_2 + \underbrace{\eta \sum_{i=1}^t \left\| (\mu(f(x; \theta^{(j)}) - r_i) [g(x_i; \theta^{(j)}) - g(x_i; \theta_0)]) \right\|_2}_{\text{(term 3)}} \\
 &\quad + \underbrace{\eta \sum_{i=1}^t \left\| [\mu(f(x; \theta^{(j)})) - \mu(g(x; \theta_0)^\top (\tilde{\theta}^{(j)} - \theta_0))] g(x_i; \theta_0) \right\|_2}_{\text{(term 4)}}. \tag{9}
 \end{aligned}$$

Considering each term of Equation (9), there exist absolute constants $C_1, C_2, C_3 > 0$ such that

$$\text{(term 3)} \leq \eta \sum_{i=1}^t \left\| 1 \cdot [g(x_i; \theta^{(j)}) - g(x_i; \theta_0)] \right\|_2 \leq C_2 \eta m^{1/3} \sqrt{\log m} t^{7/6} \lambda^{-1/6} L^{7/2} \tag{10}$$

$$\begin{aligned}
 \text{(term 4)} &\leq \eta \sum_{i=1}^t \left\| R[f(x; \theta^{(j)}) - g(x; \theta_0)^\top (\tilde{\theta}^{(j)} - \theta_0)] g(x_i; \theta_0) \right\|_2 \\
 &\leq \eta R \sum_{i=1}^t \left\| f(x; \theta^{(j)}) - g(x; \theta_0)^\top (\tilde{\theta}^{(j)} - \theta_0) \right\|_2 \cdot \|g(x_i; \theta_0)\|_2 \\
 &\leq C_3 \eta R \sum_{i=1}^t m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda^{-2/3} \|g(x_i; \theta_0)\|_2 \\
 &\leq C_1 C_3 \eta R m^{1/3} \sqrt{\log m} L^{7/2} t^{5/3} \lambda^{-2/3}. \tag{11}
 \end{aligned}$$

For **(term 3)**, we apply Lemma C.4. For **(term 4)**, the first inequality follows from the R -Lipschitz continuity of $\mu(\cdot)$, the second follows from the Cauchy–Schwarz inequality, the third follows from Lemma C.5, and the final inequality follows from Lemma C.4 after summing over t .

Substituting Equations (10) and (11) into Equation (9) yields

$$\begin{aligned}
 \|\theta^{(j+1)} - \tilde{\theta}^{(j+1)}\|_2 &\leq (1 - \eta m \lambda) \|\theta^{(j)} - \tilde{\theta}^{(j)}\|_2 \\
 &\quad + C_2 \eta m^{1/3} \sqrt{\log m} t^{7/6} \lambda^{-1/6} L^{7/2} + C_1 C_3 \eta R m^{1/3} \sqrt{\log m} L^{7/2} t^{5/3} \lambda^{-2/3} \tag{12}
 \end{aligned}$$

By iteratively applying Equation (12) from 0 to j , we obtain

$$\begin{aligned}
 \|\theta^{(j+1)} - \tilde{\theta}^{(j+1)}\|_2 &\leq \frac{C_2 \eta m^{1/3} \sqrt{\log m} t^{7/6} \lambda^{-1/6} L^{7/2} + C_1 C_3 \eta R m^{1/3} \sqrt{\log m} L^{7/2} t^{5/3} \lambda^{-2/3}}{\eta m \lambda} \\
 &\leq C_2 m^{-2/3} \sqrt{\log m} t^{7/6} \lambda^{-7/6} L^{7/2} + C_1 C_3 R m^{-2/3} \sqrt{\log m} L^{7/2} t^{5/3} \lambda^{-5/3}. \tag{13}
 \end{aligned}$$

By substituting Equations (8) and (13) into Equation (7) and setting $j = J - 1$, we complete the proof of the upper bound for $\|\theta_t - \hat{\theta}_2\|_2$. Likewise, from Equation (13), setting $j = J - 1$ and following the width condition in Condition 4.4 yields

$$\begin{aligned}
 \|\theta_t - \tilde{\theta}_t\|_2 &\leq \sqrt{\frac{t}{m \lambda_t}} \left(C_2 m^{-1/6} \sqrt{\log m} t^{2/3} \lambda_t^{-2/3} L^{7/2} + C_1 C_3 R m^{-1/6} \sqrt{\log m} L^{7/2} t^{7/6} \lambda_t^{-7/6} \right) \\
 &\leq \sqrt{\frac{t}{m \lambda_t}} \left(C_2 m^{-1/6} \sqrt{\log m} T^{2/3} \lambda_0^{-2/3} L^{7/2} + C_1 C_3 R m^{-1/6} \sqrt{\log m} L^{7/2} T^{7/6} \lambda_0^{-7/6} \right) \\
 &\leq \sqrt{\frac{t}{m \lambda}}, \tag{14}
 \end{aligned}$$

which completes the bound on $\|\theta_t - \tilde{\theta}_t\|_2$. Finally, observe that

$$\|\theta_t - \theta_0\|_2 \leq \|\theta_t - \tilde{\theta}_t\|_2 + \|\tilde{\theta}_t - \theta_0\|_2,$$

where Equation (14) gives $\|\theta_t - \tilde{\theta}_t\|_2 \leq \tau/3$, and for the second term

$$\frac{m\lambda_t}{2} \|\theta_t - \theta_0\|_2^2 \leq \tilde{L}(\tilde{\theta}_t) \leq \tilde{L}(\theta_0) = \sum_{i=1}^t r_i \log \mu(0) + (1 - r_i) \log(1 - \mu(0)) \leq t \log 2,$$

which implies $\|\tilde{\theta}_t - \theta_0\|_2 \leq 2\sqrt{t/(m\lambda_t)} = 2\tau/3$. Combining these results completes the proof of the bound on $\|\theta_t - \theta_0\|_2$.

D. Proof of Theorem 3.1

Our proof technique is primarily inspired by the recent work of (Zhou et al., 2021), which integrates non-uniform variance into the analysis of linear bandits. For brevity, let $\sigma_t^2 = \dot{\mu}(x_t^\top \theta^*)$, which yields

$$H_t = \sum_{i=1}^t \dot{\mu}(x_i^\top \theta^*) x_i x_i^\top + \lambda \mathbf{I} = \sum_{i=1}^t \sigma_i^2 x_i x_i^\top + \lambda \mathbf{I}.$$

We introduce the following additional definitions:

$$\begin{aligned} \beta_t &= 8\sqrt{\log \frac{\det H_t}{\det \lambda \mathbf{I}} \log(4t^2/\delta)} + \frac{4MN}{\sqrt{\lambda}} \log(4t^2/\delta) \\ s_t &= \sum_{i=1}^t x_i \eta_i, \quad Z_t = \|s_t\|_{H_t^{-1}}, \quad w_t = \|x_t\|_{H_{t-1}^{-1}}, \quad \mathcal{E}_t = \mathbb{1}\{0 \leq s \leq t, Z_s \leq \beta_s\} \end{aligned} \quad (15)$$

for $t \geq 1$, where we set $s_0 = 0, Z_0 = 0, \beta_0 = 0$.

Since $H_t = H_{t-1} + \sigma_t^2 x_t x_t^\top$, by the matrix inversion lemma,

$$\begin{aligned} H_t^{-1} &= H_{t-1}^{-1} - \frac{H_{t-1}^{-1}(\sigma_t x_t)(\sigma_t x_t)^\top H_{t-1}^{-1}}{1 + (\sigma_t x_t)^\top H_{t-1}^{-1}(\sigma_t x_t)} \\ &= H_{t-1}^{-1} - \frac{\sigma_t^2 H_{t-1}^{-1} x_t x_t^\top H_{t-1}^{-1}}{1 + \sigma_t^2 w_t^2}. \end{aligned} \quad (16)$$

We begin by establishing a crude upper bound on Z_t . In particular, we have

$$\begin{aligned} Z_t^2 &= \|s_t\|_{H_t^{-1}}^2 = (s_{t-1} + x_t \eta_t)^\top H_t^{-1} (s_{t-1} + x_t \eta_t) \\ &= s_{t-1}^\top H_t^{-1} s_{t-1} + 2\eta_t x_t^\top H_t^{-1} s_{t-1} + \eta_t^2 x_t^\top H_t^{-1} x_t \\ &\leq Z_{t-1}^2 + \underbrace{2\eta_t x_t^\top H_t^{-1} s_{t-1}}_{(\text{term 1})} + \underbrace{\eta_t^2 x_t^\top H_t^{-1} x_t}_{(\text{term 2})}, \end{aligned}$$

where the inequality follows from the fact that $H_t \succeq H_{t-1}$. For **(term 1)**, from the matrix inversion lemma Equation (16), we have

$$\begin{aligned} (\text{term 1}) &= 2\eta_t \left(x_t^\top H_{t-1}^{-1} s_{t-1} - \frac{\sigma_t^2 x_t^\top H_{t-1}^{-1} x_t x_t^\top H_{t-1}^{-1} s_{t-1}}{1 + \sigma_t^2 w_t^2} \right) \\ &= 2\eta_t \left(x_t^\top H_{t-1}^{-1} s_{t-1} - \frac{\sigma_t^2 w_t^2 x_t^\top H_{t-1}^{-1} s_{t-1}}{1 + \sigma_t^2 w_t^2} \right) \\ &= \frac{2\eta_t x_t^\top H_{t-1}^{-1} s_{t-1}}{1 + \sigma_t^2 w_t^2}. \end{aligned}$$

For **(term 2)**, again from the matrix inversion lemma Equation (16), we have

$$\text{(term 2)} = \eta_t^2 \left(x_t^\top H_{t-1}^{-1} x_t - \frac{\sigma_t^2 x_t^\top H_{t-1}^{-1} x_t x_t^\top H_{t-1}^{-1} x_t}{1 + \sigma_t^2 w_t^2} \right) = \eta_t^2 \left(w_t^2 - \frac{\sigma_t^2 w_t^4}{1 + \sigma_t^2 w_t^2} \right) = \frac{\eta_t^2 w_t^2}{1 + \sigma_t^2 w_t^2}. \quad (17)$$

Therefore, we have

$$Z_t^2 \leq Z_{t-1}^2 + \frac{2\eta_t x_t^\top H_{t-1}^{-1} s_{t-1}}{1 + \sigma_t^2 w_t^2} + \frac{\eta_t^2 w_t^2}{1 + \sigma_t^2 w_t^2},$$

and by summing this up from $i = 1$ to t gives,

$$Z_t^2 \leq \sum_{i=1}^t \frac{2\eta_i x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} + \sum_{i=1}^t \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2}. \quad (18)$$

We give two lemmas to upper bound each term.

Lemma D.1. *Let $s_i, w_i, \mathcal{E}_i$ be as defined in Equation (15). Then, with probability at least $1 - \delta/2$, simultaneously for all $t \geq 1$ it holds that*

$$\sum_{i=1}^t \frac{2\eta_i x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} \mathcal{E}_{i-1} \leq \frac{3}{4} \beta_t^2.$$

Lemma D.2. *Let w_i be as defined in Equation (15). Then, with probability at least $1 - \delta/2$, simultaneously for all $t \geq 1$ it holds that*

$$\sum_{i=1}^t \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \leq \frac{1}{4} \beta_t^2.$$

Now consider the event $\mathcal{E}$ in which the conclusions of Lemma D.1 and Lemma D.2 hold. We claim that, on this event, for any $i \geq 0$, $Z_i \leq \beta_i$. We prove this by induction on i . For the base case $i = 0$, the claim holds by definition, since $\beta_0 = 0 = Z_0$. Now fix any $t \geq 1$ and assume that for all $0 \leq i < t$ we have $Z_i \leq \beta_i$. Under this induction hypothesis, it follows that $\mathcal{E}_1 = \mathcal{E}_2 = \dots = \mathcal{E}_{t-1} = 1$. Then by Equation (18), we have

$$Z_t^2 \leq \sum_{i=1}^t \frac{2\eta_i x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} + \sum_{i=1}^t \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} = \sum_{i=1}^t \frac{2\eta_i x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} \mathcal{E}_{i-1} + \sum_{i=1}^t \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2}. \quad (19)$$

Since on the event $\mathcal{E}$ the conclusion of Lemma D.1 and Lemma D.2 hold, we have

$$\sum_{i=1}^t \frac{2\eta_i x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} \mathcal{E}_{i-1} \leq \frac{3}{4} \beta_t^2, \quad \sum_{i=1}^t \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \leq \frac{1}{4} \beta_t^2. \quad (20)$$

Therefore, substituting Equation (20) into Equation (19) yields $Z_t \leq \beta_t(\delta)$, which completes the induction. By a union bound, the events in Lemma D.1 and Lemma D.2 both hold with probability at least $1 - \delta$. Hence, with probability at least $1 - \delta$, for all t , $Z_t \leq \beta_t$.

D.1. Proof of Lemma D.1

We now proceed to apply Freedman's inequality, as stated in Lemma H.1. We have

$$\left| \frac{2x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} \mathcal{E}_{i-1} \right| \leq \frac{2\|x_i\|_{H_{i-1}^{-1}} [\|s_{i-1}\|_{H_{i-1}^{-1}} \mathcal{E}_{i-1}]}{1 + \sigma_i^2 w_i^2} \leq \frac{2w_i \beta_{i-1}}{1 + \sigma_i^2 w_i^2} \leq \min\{1/\sigma_i, 2w_i\} \beta_{i-1}. \quad (21)$$

Here, the first inequality follows from the Cauchy-Schwarz inequality, the second follows from the definition of $\mathcal{E}_{i-1}$, and the final inequality follows by simple algebra. For simplicity, let ℓ_i denote

$$\ell_i = \frac{2\eta_i x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} \mathcal{E}_{i-1}.$$

We now apply Freedman's inequality from Lemma H.1 to the sequences $(\ell_i)_i$ and $(\mathcal{G}_i)_i$. First, note that $\mathbb{E}[\ell_i|\mathcal{G}_i] = 0$. Moreover, by Equation (21), the following inequalities hold almost surely:

$$|\ell_i| \leq M\beta_{i-1} \min\{1/\sigma_i, 2w_i\} \leq \frac{2MN}{\sqrt{\lambda}}\beta_t, \quad (22)$$

where the last inequality follows since $(\beta_i)_i$ is non-decreasing in i and by the fact that

$$w_i = \|x_i\|_{H_{i-1}^{-1}} \leq \|x_i\|_2/\sqrt{\lambda} \leq N/\sqrt{\lambda}. \quad (23)$$

We also have

$$\begin{aligned} \sum_{i=1}^t \mathbb{E}[\ell_i^2|\mathcal{G}_i] &\leq \sum_{i=1}^t \sigma_i^2 \left(\frac{2x_i^\top H_{i-1}^{-1} s_{i-1}}{1 + \sigma_i^2 w_i^2} \mathcal{E}_{i-1} \right)^2 \\ &\leq \sum_{i=1}^t \sigma_i^2 (\min\{1/\sigma_i, 2w_i\} \beta_{i-1})^2 \\ &= \sum_{i=1}^t (\min\{1, 2w_i \sigma_i\} \beta_{i-1})^2 \\ &\leq 4\beta_t^2 \sum_{i=1}^t \min\{1, (w_i \sigma_i)^2\}, \end{aligned}$$

where the first inequality holds by the definition of σ_i , the second inequality follows from Equation (21), the third inequality holds since $(\beta_i)_i$ is non-decreasing. Since

$$\sum_{i=1}^t \min\{1, (w_i \sigma_i)^2\} = \sum_{i=1}^t \min\{1, \|\sigma_i x_i\|_{H_{i-1}^{-1}}^2\} \leq 2 \log \frac{\det H_t}{\det \lambda \mathbf{I}}, \quad (24)$$

where the last inequality follows from Lemma H.2. Substituting this back yields,

$$\sum_{i=1}^t \mathbb{E}[\ell_i^2|\mathcal{G}_i] \leq 8\beta_t^2 \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right). \quad (25)$$

Therefore, by Equations (22) and (25), using Lemma H.1, we know that for any t , with probability at least $1 - \delta/(4t^2)$, we have

$$\begin{aligned} \sum_{i=1}^t \ell_i &\leq \sqrt{16\beta_t^2 \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta)} + \frac{2}{3} \cdot \frac{2MN}{\sqrt{\lambda}} \beta_t \log(4t^2/\delta) \\ &\leq \frac{\beta_t^2}{4} + 16 \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta) + \frac{\beta_t^2}{4} + \frac{4M^2 N^2}{\lambda} \log^2(4t^2/\delta) \\ &\leq \frac{\beta_t^2}{2} + \frac{1}{4} \left[8 \sqrt{\log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta)} + \frac{4MN}{\sqrt{\lambda}} \log(4t^2/\delta) \right]^2 \\ &= \frac{3}{4} \beta_t^2, \end{aligned} \quad (26)$$

where the second inequality follows from the fact that $2\sqrt{|ab|} \leq |a| + |b|$, and the final equality follows from the definition of β_t . Applying a union bound to Equation (26) from $t = 1$ to ∞ and using the fact that $\sum_{t=1}^{\infty} t^{-2} < 2$ completes the proof.

D.2. Proof of Lemma D.2

Similarly to Lemma D.1, we apply Freedman's inequality from Lemma H.1 to the sequences $(\ell_i)_i$ and $(\mathcal{G}_i)_i$, where now

$$\ell_i = \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} - \mathbb{E} \left[\frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \middle| \mathcal{G}_i \right].$$

First, with Equation (23), we derive a crude upper bound for the following term:

$$\left| \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \right| \leq |\eta_i^2 w_i^2| \leq \frac{M^2 N^2}{\lambda}. \quad (27)$$

Now, for any i , we have $\mathbb{E}[\ell_i | \mathcal{G}_i] = 0$ almost surely. Furthermore, we can see that

$$\begin{aligned} \sum_{i=1}^t \mathbb{E}[\ell_i^2 | \mathcal{G}_i] &\leq \sum_{i=1}^t \mathbb{E} \left[\frac{\eta_i^4 w_i^4}{(1 + \sigma_i^2 w_i^2)^2} \middle| \mathcal{G}_i \right] \\ &\leq \frac{M^2 N^2}{\lambda} \sum_{i=1}^t \mathbb{E} \left[\frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \middle| \mathcal{G}_i \right] \\ &\leq \frac{M^2 N^2}{\lambda} \sum_{i=1}^t \frac{\sigma_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \\ &\leq \frac{2M^2 N^2}{\lambda} \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right), \end{aligned} \quad (28)$$

where the first inequality follows from the fact that $\mathbb{E}(X - \mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$, the second follows from Equation (27), the third follows from the definition of η_i , and the fourth follows from the bound $\sigma_i^2 w_i^2 / (1 + \sigma_i^2 w_i^2) \leq \min\{1, \sigma_i^2 w_i^2\}$ together with the result in Equation (24). Furthermore, applying Equation (27) again gives

$$|\ell_i| \leq \left| \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \right| + \left| \mathbb{E} \left[\frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \middle| \mathcal{G}_i \right] \right| \leq \frac{2M^2 N^2}{\lambda}, \quad (29)$$

almost surely. Therefore by Equation (28) and Equation (29), using Lemma H.1, we know that for any t , with probability at

least $1 - \delta/(4t^2)$, we have

$$\begin{aligned}
 & \sum_{i=1}^t \frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \\
 & \leq \sum_{i=1}^t \mathbb{E} \left[\frac{\eta_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} \middle| \mathcal{G}_i \right] + \sqrt{\frac{4M^2 N^2}{\lambda} \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta)} \\
 & \quad + \frac{2}{3} \cdot \frac{2M^2 N^2}{\lambda} \log(t^2/\delta) \\
 & \leq \sum_{i=1}^t \frac{\sigma_i^2 w_i^2}{1 + \sigma_i^2 w_i^2} + \sqrt{\frac{4M^2 N^2}{\lambda} \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta)} \\
 & \quad + \frac{4M^2 N^2}{\lambda} \log(4t^2/\delta) \\
 & \leq 2 \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) + \sqrt{\frac{4M^2 N^2}{\lambda} \log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta)} \\
 & \quad + \frac{4M^2 N^2}{\lambda} \log(4t^2/\delta) \\
 & \leq \frac{1}{4} \cdot \left[8 \sqrt{\log \det \left(\sum_{i=1}^t \frac{\sigma_i^2}{\lambda} x_i x_i^\top + \mathbf{I} \right) \log(4t^2/\delta)} + \frac{4MN}{\sqrt{\lambda}} \log(4t^2/\delta) \right]^2 \\
 & = \frac{1}{4} \beta_t^2,
 \end{aligned} \tag{30}$$

where the second inequality follows from the definition of σ_i^2 , the third follows from the bound $\sigma_i^2 w_i^2 / (1 + \sigma_i^2 w_i^2) \leq \min\{1, \sigma_i^2 w_i^2\}$ together with the result in Equation (24), and the final inequality follows from the definition of β_t . Applying a union bound to Equation (30) for $t = 1$ to ∞ and using the fact that $\sum_{t=1}^{\infty} t^{-2} < 2$ completes the proof.

E. Proof of Lemmas in Section 6

For clarity, we assume that Condition 4.4 always holds. We then define the quantity $\alpha(z', z'')$ via the mean-value theorem, and introduce two additional analogous definitions for brevity as follows:

$$\begin{aligned}
 \alpha(z', z'') &= \frac{\mu(z') - \mu(z'')}{z' - z''} = \int_{v=0}^1 \dot{\mu}(z' + v(z'' - z')) dv, \\
 \alpha(x, \theta', \theta'') &= \alpha\left(g(x; \theta_0)^\top (\theta' - \theta_0), g(x; \theta_0)^\top (\theta'' - \theta_0)\right), \\
 \alpha(x', x'', \theta) &= \alpha\left(g(x'; \theta_0)^\top (\theta - \theta_0), g(x''; \theta_0)^\top (\theta - \theta_0)\right).
 \end{aligned} \tag{31}$$

For the design matrix X_t associated with the time-varying regularization parameter λ_t , we denote by $\tilde{X}_t$ the corresponding matrix formed using the initial regularization parameter λ_0 . For example,

$$\begin{aligned}
 \tilde{V}_t &= \sum_{i=1}^t \frac{1}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \kappa \lambda_0 \mathbf{I}, \\
 \tilde{H}_t(\theta) &= \sum_{i=1}^t \frac{1}{m} \dot{\mu}(g(x_i; \theta_0)^\top (\theta_i - \theta_0)) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_0 \mathbf{I} \\
 \tilde{W}_t &= \sum_{i=1}^t \frac{1}{m} \dot{\mu}(f(x_i; \theta_i)) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_0 \mathbf{I}.
 \end{aligned} \tag{32}$$

E.1. Proof of Lemma 4.5

First, we define the auxiliary loss $\tilde{L}_t(\theta)$

$$\begin{aligned}\tilde{\mathcal{L}}_t(\theta) &= -\sum_{i=1}^t r_i \log \mu(g(x_i; \theta_0)^\top (\theta - \theta_0)) + (1 - r_i) \log(1 - \mu(g(x_i; \theta_0)^\top (\theta - \theta_0))) \\ &\quad + \frac{m\lambda_t}{2} \|\theta - \theta_0\|_2^2,\end{aligned}$$

and its maximum likelihood estimator $\hat{\theta}_t = \arg \min_{\theta} \tilde{\mathcal{L}}_t(\theta)$. Then, we use the following definitions:

$$\begin{aligned}\gamma_t(\theta) &= \sum_{i=1}^t \frac{1}{m} \mu(g(x_i; \theta_0)^\top (\theta - \theta_0)) g(x_i; \theta_0) + \lambda_t (\theta - \theta_0) \\ \Gamma_t(\theta', \theta'') &= \sum_{i=1}^t \frac{1}{m} \alpha(x_i, \theta', \theta'') g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I},\end{aligned}$$

where $\alpha(x_i, \theta', \theta'')$ is defined at Equation (31). We can see that

$$\begin{aligned}\gamma_t(\theta) - \gamma_t(\theta^*) &= \sum_{i=1}^t \frac{1}{m} \left(\mu(g(x_i; \theta_0)^\top (\theta - \theta_0)) - \mu(g(x_i; \theta_0)^\top (\theta^* - \theta_0)) \right) g(x_i; \theta_0) + \lambda_t (\theta - \theta^*) \\ &= \sum_{i=1}^t \frac{1}{m} \alpha(x_i, \theta, \theta^*) g(x_i; \theta_0) g(x_i; \theta_0)^\top (\theta - \theta^*) + \lambda_t (\theta - \theta^*) \\ &= \Gamma_t(\theta, \theta^*) (\theta - \theta^*),\end{aligned}$$

which implies that

$$\|\theta - \theta^*\|_{\Gamma_t(\theta, \theta^*)} = \|\gamma_t(\theta) - \gamma_t(\theta^*)\|_{\Gamma_t^{-1}(\theta, \theta^*)}. \quad (33)$$

Now we provide the following two lemmas:

Lemma E.1. For $\delta \in (0, 1]$, define

$$\mathcal{C}_t = \left\{ \theta : \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{H_t^{-1}(\theta)} \leq \iota_t \right\}, \quad (34)$$

where ι_t is defined at Equation (2). Then for all $t \geq 0$, $\theta^* \in \mathcal{C}_t$ with probability at least $1 - \delta$

Lemma E.2. Let $\delta \in (0, 1]$. Define $\mathcal{C}_t$ as in Equation (34). There exists an absolute constant $C_1 > 0$ such that for all $\theta \in \mathcal{C}_t$,

$$H_t(\theta) \preceq \left(1 + C_1^2 \frac{L}{\lambda_t} \iota_t^2 + C_1 \sqrt{\frac{L}{\lambda_t}} \iota_t \right) \Gamma_t(\theta, \hat{\theta}_t), \quad H_t(\hat{\theta}_t) \preceq \left(1 + C_1^2 \frac{L}{\lambda_t} \iota_t^2 + C_1 \sqrt{\frac{L}{\lambda_t}} \iota_t \right) \Gamma_t(\theta, \hat{\theta}_t).$$

Now we are ready to start the proof.

Proof of Lemma 4.5. For the absolute constants $\{C_i\}_{i=1}^3$, we can start with

$$\sqrt{m} \|\theta_t - \theta^*\|_{H_t(\theta^*)} \quad (35)$$

$$\begin{aligned}&\leq \sqrt{m} \|\hat{\theta}_t - \theta^*\|_{H_t(\theta^*)} + \sqrt{m} \|\theta_t - \hat{\theta}_t\|_{H_t(\theta^*)} \\ &\leq \sqrt{m} \|\hat{\theta}_t - \theta^*\|_{H_t(\theta^*)} + \sqrt{m} \|\theta_t - \hat{\theta}_t\|_2 \cdot (\lambda_t + C_1 t L) \\ &\leq \underbrace{\sqrt{m} \|\hat{\theta}_t - \theta^*\|_{H_t(\theta^*)}}_{\text{(term 1)}} + \underbrace{2(\lambda_t + C_1 t L)(1 - \eta m \lambda_t)^{J/2} t^{1/2} \lambda_t^{-1/2}}_{\text{(term 2)}} \\ &\quad + \underbrace{(\lambda_t + C_1 t L) \left[C_2 m^{-1/6} \sqrt{\log m} t^{7/6} \lambda_t^{-7/6} L^{7/2} + C_1 C_3 R m^{-1/6} \sqrt{\log m} L^{7/2} t^{5/3} \lambda_t^{-5/3} \right]}_{\text{(term 3)}}.\end{aligned} \quad (36)$$

The first inequality is due to triangle inequality. The second inequality is due to $\lambda_{\max}(H_t(\theta^*)) \leq \lambda_t + t \times \|\sqrt{\dot{\mu}(\cdot)}/m \cdot g(\cdot)\|_2^2 \leq \lambda_t + C_1 t L$ where we used Lemma C.4. Finally, the last inequality follows from Lemma C.6.

For **(term 1)**, we rewrite the definition of ι_t and λ_t :

$$\begin{aligned}\iota_t &= 16 \sqrt{\log \det \left(\sum_{i=1}^t \frac{1}{4m^2 \lambda_0} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \mathbf{I} \right) \log \frac{4t^2}{\delta} + 8C_1 \sqrt{\frac{L}{\lambda_0}} \log \frac{4t^2}{\delta}} \\ \lambda_t &= \frac{64}{S^2} \log \det \left(\sum_{i=1}^t \frac{1}{4m \lambda_0} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \mathbf{I} \right) \log \frac{4t^2}{\delta} + \frac{16C_1^2 L}{S^2 \lambda_0} \log^2 \frac{4t^2}{\delta}.\end{aligned}$$

We can see that $\iota_t^2/\lambda_t \leq 8S^2$ (and $\iota_t/\sqrt{\lambda_t} \leq 2\sqrt{2}S$) by the fact that $(a+b)^2 \leq 2a^2 + 2b^2$. Therefore, applying these with Lemmas E.1 and E.2 gives

$$H_t(\theta^*) \preceq (1 + 2\sqrt{2}C_1\sqrt{L}S + 8C_1^2LS^2)\Gamma_t(\theta^*, \hat{\theta}_t), \quad (37)$$

for some absolute constant $C_1 > 0$. Now, back to **(term 1)**, we have

$$\begin{aligned}\sqrt{m} \|\hat{\theta}_t - \theta^*\|_{H_t(\theta^*)} &\leq \sqrt{m(1 + 2\sqrt{2}C_1\sqrt{L}S + 8C_1^2LS^2)} \|\hat{\theta}_t - \theta^*\|_{\Gamma_t(\hat{\theta}_t, \theta^*)} \\ &= \sqrt{m(1 + 2\sqrt{2}C_1\sqrt{L}S + 8C_1^2LS^2)} \|\gamma(\hat{\theta}_t) - \gamma(\theta^*)\|_{\Gamma_t^{-1}(\hat{\theta}_t, \theta^*)} \\ &\leq (1 + 2\sqrt{2}C_1\sqrt{L}S + 8C_1^2LS^2) \sqrt{m} \|\gamma(\hat{\theta}_t) - \gamma(\theta^*)\|_{H_t^{-1}(\theta^*)} \\ &\leq (1 + 2\sqrt{2}C_1\sqrt{L}S + 8C_1^2LS^2) \iota_t.\end{aligned}$$

where the first and the second inequalities follow from Equation (37), the equality is due to Equation (33), and the last inequality follows from Lemma E.1.

For **(term 2)**, plugging in $J = 2 \log(\lambda_t S / (T^{1/2} \lambda_t + C_4 T^{3/2} L)) TL / \lambda_t$, $\eta = C_5 (mTL + m\lambda_t)^{-1}$ gives

$$\begin{aligned}2(\lambda_t + C_1 t L)(1 - \eta m \lambda_t)^{J/2} t^{1/2} \lambda_t^{-1/2} \\ \leq 2(\lambda_t + C_1 t L)(1 - \lambda_t / (TL))^{J/2} T^{1/2} \lambda_t^{-1/2} \\ \leq 2S \sqrt{\lambda_t} \\ \leq \iota_t,\end{aligned}$$

where the last inequality follows from the definition of λ_t and the fact that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$. For **(term 3)**, recall that $\lambda_0 \leq \min\{\lambda_t\}_{t \geq 1}$, then we have

$$\begin{aligned}&C_2 m^{-1/6} \sqrt{\log m} t^{7/6} \lambda_t^{-1/6} L^{7/2} + C_1 C_3 R m^{-1/6} \sqrt{\log m} L^{7/2} t^{5/3} \lambda_t^{-2/3} \\ &\quad + C_1 C_2 m^{-1/6} \sqrt{\log m} t^{13/6} \lambda_t^{-7/6} L^{9/2} + C_1^2 C_3 R m^{-1/6} \sqrt{\log m} L^{9/2} t^{8/3} \lambda_t^{-5/3} \\ &\leq C_2 m^{-1/6} \sqrt{\log m} T^{7/6} \lambda_0^{-1/6} L^{7/2} + C_1 C_3 R m^{-1/6} \sqrt{\log m} L^{7/2} T^{5/3} \lambda_0^{-2/3} \\ &\quad + C_1 C_2 m^{-1/6} \sqrt{\log m} T^{13/6} \lambda_0^{-7/6} L^{9/2} + C_1^2 C_3 R m^{-1/6} \sqrt{\log m} L^{9/2} T^{8/3} \lambda_0^{-5/3} \\ &\leq 1,\end{aligned}$$

where the last inequality can be verified that if the width of the NN m is large enough, satisfying the condition on Condition C.2, **(term 3)** ≤ 1 .

Substituting **(term 1)**, **(term 2)**, and **(term 3)** back to Equation (36) gives

$$\begin{aligned}\sqrt{m} \|\theta_t - \theta^*\|_{H_t(\theta^*)} &\leq (2 + 2\sqrt{2}C_1\sqrt{L}S + 8C_1^2LS^2) \iota_t + 1 \\ &\leq C_6 (1 + \sqrt{L}S + LS^2) \iota_t + 1,\end{aligned}$$

for some absolute constant $C_6 > 0$, concludes the proof. $\square$

E.2. Proof of Lemma E.1

Recall the definition of $\tilde{\mathcal{L}}_t(\theta)$, $\hat{\theta}_t$, $\gamma_t(\theta)$, and $\Gamma_t(\theta', \theta'')$ from Section E.1. Since $\hat{\theta}_t$ is a maximum likelihood estimator, $\tilde{\mathcal{L}}_t(\hat{\theta}_t) = 0$, which gives

$$\sum_{i=1}^t \frac{1}{m} \mu(g(x_i; \theta_0)^\top (\hat{\theta}_t - \theta_0)) g(x_i; \theta_0) + \lambda_t (\hat{\theta}_t - \theta_0) = \sum_{i=1}^t \frac{1}{m} r_i g(x_i; \theta_0). \quad (38)$$

Therefore, we can see that

$$\begin{aligned} & \sqrt{m} \|\gamma(\hat{\theta}_t) - \gamma(\theta^*)\|_{H_t^{-1}(\theta^*)} \\ &= \sqrt{m} \left\| \sum_{i=1}^t \frac{1}{m} [\mu(g(x_i; \theta_0)^\top (\hat{\theta}_t - \theta_0)) - \mu(g(x_i; \theta)^\top (\theta^* - \theta_0))] g(x_i; \theta_0) + \lambda_t \hat{\theta}_t - \lambda_t \theta^* \right\|_{H_t^{-1}(\theta^*)} \\ &= \sqrt{m} \left\| \sum_{i=1}^t \frac{1}{m} [r_i - \mu(g(x_i; \theta_0)^\top (\theta^* - \theta_0))] g(x_i; \theta_0) - \lambda_t (\theta^* - \theta_0) \right\|_{H_t^{-1}(\theta^*)} \\ &\leq \underbrace{\left\| \sum_{i=1}^t \frac{1}{\sqrt{m}} [r_i - \mu(g(x_i; \theta_0)^\top (\theta^* - \theta_0))] g(x_i; \theta_0) \right\|_{H_t^{-1}(\theta^*)}}_{\text{(term 1)}} + \underbrace{\sqrt{\lambda_t m} \|\theta^* - \theta_0\|_2}_{\text{(term 2)}}, \end{aligned} \quad (39)$$

where the first equality follows from the definition, the second equality is due to Equation (38), and the first inequality follows from triangle inequality, and the fact that $\lambda_{\max}(H_t^{-1}(\theta^*)) \leq 1/\sqrt{\lambda_t}$.

For **(term 1)**, we are going to use our new tail inequality for martingales in Theorem 3.1. Define $\eta_i = r_i - \mu(g(x_i; \theta_0)^\top (\theta^* - \theta_0)) = r_i - \mu(h(x_i))$. Then, we can see the following conditions are satisfied:

$$|\eta_i| \leq 1, \quad \mathbb{E}[\eta_i | \mathcal{G}_i] = 0, \quad \mathbb{E}[\eta_i^2 | \mathcal{G}_i] = \dot{\mu}(g(x_i; \theta_0)^\top (\theta^* - \theta_0)).$$

By Lemma C.4 we have $\|g(x_i; \theta_0)/\sqrt{m}\|_2 \leq C_1 \sqrt{L}$ for some absolute constant $C_1 > 0$. Therefore, applying Theorem 3.1 gives

$$\begin{aligned} & \left\| \sum_{i=1}^t \frac{1}{\sqrt{m}} \eta_i g(x_i; \theta_0) \right\|_{H_t^{-1}(\theta^*)} \\ &\leq \left\| \sum_{i=1}^t \frac{1}{\sqrt{m}} \eta_i g(x_i; \theta_0) \right\|_{\tilde{H}_t^{-1}(\theta^*)} \\ &\leq 8 \sqrt{\log \det \left(\sum_{i=1}^t \frac{1}{4m\lambda_0} g(x_i; \theta_0) g(x_i; \theta_0)^\top + I \right) \log \frac{4t^2}{\delta} + 4C_1 \sqrt{\frac{L}{\lambda_0}} \log \frac{4t^2}{\delta}}, \end{aligned} \quad (40)$$

with probability at least $1 - \delta$. Substituting Equation (40) into Equation (39) gives

$$\begin{aligned} & \sqrt{m} \|\gamma(\hat{\theta}_t) - \gamma(\theta^*)\|_{H_t^{-1}(\theta^*)} \\ &\leq 8 \sqrt{\log \det \left(\sum_{i=1}^t \frac{1}{4m\lambda_0} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \mathbf{I} \right) \log \frac{4t^2}{\delta} + 4C_1 \sqrt{\frac{L}{\lambda_0}} \log \frac{4t^2}{\delta} + S \sqrt{\lambda_t}} \\ &\leq 16 \sqrt{\log \det \left(\sum_{i=1}^t \frac{1}{4m\lambda_0} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \mathbf{I} \right) \log \frac{4t^2}{\delta} + 8C_1 \sqrt{\frac{L}{\lambda_0}} \log \frac{4t^2}{\delta}} \\ &= \iota_t. \end{aligned} \quad (41)$$

where the last inequality is due to the update rule of λ_t and the fact that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$. We finish the proof.

E.3. Proof of Lemma E.2

We modified the previous results of (Abeille et al., 2021) (Lemma 2), and (Faury et al., 2022) (proof of Lemma 1), proper to our settings.

Lemma E.3. *Let $\delta \in (0, 1]$. Define $\mathcal{C}_t$ as in Equation (34). There exists an absolute constant $C_1 > 0$ such that for all $\theta \in \mathcal{C}_t$:*

$$\sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)} \leq C_1 \sqrt{\frac{L}{\lambda_t}} \iota_t^2 + \iota_t.$$

The proof is deferred to Section E.4. Following the proof of Lemma E.3, from Equation (42), we have

$$\begin{aligned} \Gamma_t(\theta, \hat{\theta}_t) &\geq \left(1 + C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}\right)^{-1} H_t(\theta) \\ &\geq \left(1 + C_1^2 \frac{L}{\lambda_t} \iota_t^2 + C_1 \sqrt{\frac{L}{\lambda_t}} \iota_t\right)^{-1} H_t(\theta) \end{aligned}$$

where the last inequality follows from applying Lemma E.3 again.

One can achieve the same result for $H_t(\hat{\theta}_t)$ in a similarly way by starting the proof of Lemma E.3 with

$$\begin{aligned} \Gamma_t(\theta, \hat{\theta}_t) &= \sum_{i=1}^t \alpha(x_i, \theta, \hat{\theta}_t) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \\ &\geq \sum_{i=1}^t \left(1 + |g(x_i; \theta_0)^\top (\theta - \hat{\theta}_t)|\right)^{-1} \dot{\mu}(g(x_i; \theta_0)^\top (\hat{\theta}_t - \theta_0)) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \\ &\geq \left(1 + C_1 \sqrt{\frac{L}{\lambda_t}} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}\right)^{-1} H_t(\hat{\theta}_t) \\ &\geq \left(1 + C_1^2 \frac{L}{\lambda_t} \iota_t^2 + C_1 \sqrt{\frac{L}{\lambda_t}} \iota_t\right)^{-1} H_t(\hat{\theta}_t), \end{aligned}$$

where the first inequality follows from Lemma H.3, the second inequality follows the same process of Equation (42), and the last inequality follows from Lemma E.3, finishing the proof.

E.4. Proof of Lemma E.3

Recall the definition of Γ_t and $\alpha(x, \theta', \theta'')$ from Equation (31). We start with

$$\begin{aligned} \Gamma_t(\theta, \hat{\theta}_t) &= \sum_{i=1}^t \alpha(x_i, \theta, \hat{\theta}_t) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \\ &\geq \sum_{i=1}^t \underbrace{\left(1 + |g(x_i; \theta_0)^\top (\theta - \hat{\theta}_t)|\right)^{-1}}_{(\text{term 1})} \dot{\mu}(g(x_i; \theta_0)^\top (\theta - \theta_0)) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I}, \end{aligned}$$

where the inequality follows from Lemma H.3. For (term 1), we have,

$$\begin{aligned} (\text{term 1}) &\leq \|g(x_i; \theta_0)/\sqrt{m}\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)} \cdot \sqrt{m} \|\theta - \hat{\theta}_t\|_{\Gamma_t(\theta, \hat{\theta}_t)} \\ &\leq C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\theta - \hat{\theta}_t\|_{\Gamma_t(\theta, \hat{\theta}_t)} \\ &\leq C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}, \end{aligned}$$

where the first inequality follows from the Cauchy-Schwarz inequality, the second inequality follows from the fact that $\lambda_{\max}(\Gamma(\cdot)^{-1}) \leq \lambda_t^{-1}$ and Lemma C.4, and the last inequality follows from Equation (33). Substituting **(term 1)** back gives,

$$\begin{aligned}
 \Gamma_t(\theta, \hat{\theta}_t) &\geq \left(1 + C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}\right)^{-1} \\
 &\quad \times \sum_{i=1}^t \dot{\mu}(g(x_i; \theta_0)^\top (\theta - \theta_0)) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \\
 &\geq \left(1 + C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}\right)^{-1} \\
 &\quad \times \left(\sum_{i=1}^t \dot{\mu}(g(x_i; \theta_0)^\top (\theta - \theta_0)) g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I}\right) \\
 &= \left(1 + C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}\right)^{-1} H_t(\theta)
 \end{aligned} \tag{42}$$

Using this results, we can further obtain

$$\begin{aligned}
 &\sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}^2 \\
 &\leq \left(1 + C_1 L^{1/2} \lambda_t^{-1/2} \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}\right) \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{H_t^{-1}(\theta)}^2 \\
 &\leq \iota_t^2 + C_1 L^{1/2} \lambda_t^{-1/2} \iota_t^2 \cdot \sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)},
 \end{aligned}$$

where the last inequality follows from Lemma E.1. We solve the polynomial inequality in $\sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)}$ using a fact that for $b, c > 0$ and $x \in \mathbb{R}$, following implication holds: $x^2 \leq bx + c \implies x \leq b + \sqrt{c}$, which finally gives

$$\sqrt{m} \|\gamma_t(\theta) - \gamma_t(\hat{\theta}_t)\|_{\Gamma_t^{-1}(\theta, \hat{\theta}_t)} \leq C_1 \sqrt{\frac{L}{\lambda_t}} \iota_t^2 + \iota_t$$

E.5. Proof of Lemma 6.4

First, we show that we can upper bound on the prediction error for all $x \in \mathcal{X}_t$, $t \in [T]$, which is the difference between the true reward $\mu(h(x))$ with our prediction with the neural network $\mu(f(x; \theta_t))$. For $x \in \mathcal{X}_{t+1}$ and the absolute constant $C_3 > 0$, the prediction error is defined as

$$\begin{aligned}
 &|\mu(h(x)) - \mu(f(x; \theta_t))| \\
 &\leq R[h(x) - f(x; \theta_t)] \\
 &= R[g(x; \theta_0)^\top (\theta^* - \theta_0) - f(x; \theta_t)] \\
 &\leq R[\underbrace{g(x; \theta_0)^\top (\theta^* - \theta_0) - g(x; \theta_0)^\top (\theta_t - \theta_0)}_{\text{(term 1)}} + C_3 m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda_t^{-2/3}],
 \end{aligned} \tag{43}$$

where the first inequality is due to the fact that $\mu(\cdot)$ is R -Lipschitz function, the equality follows from Lemma 6.1, and the last inequality follows from Lemma C.5. For **(term 1)**, we have

$$\begin{aligned}
 \text{(term 1)} &= g(x; \theta_0)^\top (\theta^* - \theta_t) \\
 &= \frac{1}{\sqrt{m}} g(x; \theta_0)^\top \cdot H_t^{-1/2}(\theta^*) \cdot H_t^{1/2}(\theta^*) \cdot \sqrt{m}(\theta^* - \theta_t) \\
 &\leq \|g(x; \theta_0)/\sqrt{m}\|_{H_t^{-1}(\theta^*)} \cdot \sqrt{m} \|\theta^* - \theta_t\|_{H_t(\theta^*)} \\
 &\leq \sqrt{\kappa} \|g(x; \theta_0)/\sqrt{m}\|_{V_t^{-1}} \cdot \sqrt{m} \|\theta^* - \theta_t\|_{H_t(\theta^*)} \\
 &\leq \sqrt{\kappa} \|g(x; \theta_0)/\sqrt{m}\|_{V_t^{-1}} \cdot (C_6(1 + \sqrt{LS} + LS^2) \iota_t + 1),
 \end{aligned} \tag{44}$$

where the first inequality follows from the Cauchy-Schwarz inequality, the second inequality is due to the Assumption 6.3 that $\frac{1}{\kappa} V_t \preceq H_t^{-1}(\theta^*)$, and the last inequality follows from Lemma 4.5. Plugging Equation (44) into Equation (43) gives

$$\begin{aligned} & |\mu(h(x)) - \mu(f(x; \theta_t))| \\ & \leq R\sqrt{\kappa}(C_6(1 + \sqrt{L}S + LS^2)\iota_t + 1)\|g(x; \theta_0)/\sqrt{m}\|_{V_t^{-1}} + C_3 R m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda_t^{-2/3} \\ & \leq R\sqrt{\kappa}(C_6(1 + \sqrt{L}S + LS^2)\iota_t + 1)\|g(x; \theta_0)/\sqrt{m}\|_{V_t^{-1}} + \epsilon_{3,t}, \end{aligned} \quad (45)$$

where the second inequality follows from the fact that $\lambda_0 \leq \min\{\lambda_t\}_{t \geq 1}$ and the definition of $\epsilon_{3,t}$.

F. Proof of lemmas in Section 5

F.1. Proof of Lemma 5.1

Recall the definition of $\tilde{L}_t(\theta)$, $\hat{\theta}_t$, $\gamma_t(\theta)$, $\Gamma_t(\theta', \theta'')$, ι_t , and λ_t from Section E.1. We also use

$$\begin{aligned} W_t &= \sum_{i=1}^t \frac{\dot{\mu}(f(x_i; \theta_i))}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \\ H_t(\hat{\theta}_t) &= \sum_{i=1}^t \frac{\dot{\mu}(g(x_i; \theta_0)^\top (\hat{\theta}_t - \theta_0))}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \\ Z_t &= \sum_{i=1}^t \frac{|\dot{\mu}(f(x_i; \theta_i)) - \dot{\mu}(g(x_i; \theta_0)^\top (\hat{\theta}_t - \theta_0))|}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} \end{aligned}$$

By the definition of Z_t , for any $x \in \mathbb{R}^p$, we have

$$\|x\|_{W_t} \leq \|x\|_{H_t(\hat{\theta}_t) + Z_t} \leq \|x\|_{H_t(\hat{\theta}_t)} + \|x\|_{Z_t}.$$

Now with the above inequality, we can start with

$$\sqrt{m}\|\theta_t - \theta^*\|_{W_t} \leq \underbrace{\sqrt{m}\|\theta_t - \theta^*\|_{H_t(\hat{\theta}_t)}}_{\text{(term 1)}} + \underbrace{\sqrt{m}\|\theta_t - \theta^*\|_{Z_t}}_{\text{(term 2)}}. \quad (46)$$

For **(term 1)**, we directly follow the proof of Lemma 4.5 in Section E.1. Therefore, for the absolute constants $\{C_i\}_{i=1}^3$, we have

$$\begin{aligned} & \sqrt{m}\|\theta_t - \theta^*\|_{H_t(\hat{\theta}_t)} \\ & \leq \sqrt{m}\|\hat{\theta}_t - \theta^*\|_{H_t(\hat{\theta}_t)} + \sqrt{m}\|\theta_t - \hat{\theta}_t\|_{H_t(\hat{\theta}_t)} \\ & \leq \underbrace{\sqrt{m}\|\hat{\theta}_t - \theta^*\|_{H_t(\hat{\theta}_t)}}_{\text{(term 3)}} + \underbrace{2(\lambda_t + C_1 t L)(1 - \eta m \lambda_t)^{J/2} t^{1/2} \lambda_t^{-1/2}}_{\text{(term 4)}} \\ & \quad + \underbrace{(\lambda_t + C_1 t L) \left[C_2 m^{-1/6} \sqrt{\log m} t^{7/6} \lambda_t^{-7/6} L^{7/2} + C_1 C_3 R m^{-1/6} \sqrt{\log m} L^{7/2} t^{5/3} \lambda_t^{-5/3} \right]}_{\text{(term 5)}}. \end{aligned}$$

Using the same argument as in Section E.1, we can see that

$$\text{(term 4)} \leq 2S\sqrt{\lambda_t} \leq \iota_t, \quad \text{(term 5)} \leq 1/2.$$

Note that the upper bound for **(term 5)** has been changed from 1 to $1/2$ solely to unify the constant in the concentration

inequalities of Lemma 4.5 and Lemma 5.1. For **(term 3)**, we have

$$\begin{aligned}
 \sqrt{m} \left\| \hat{\theta}_t - \theta^* \right\|_{H_t(\hat{\theta}_t)} &\leq \sqrt{m(1 + 2\sqrt{2}C_1\sqrt{LS} + 8C_1^2LS^2)} \left\| \hat{\theta}_t - \theta^* \right\|_{\Gamma_t(\hat{\theta}_t, \theta^*)} \\
 &= \sqrt{m(1 + 2\sqrt{2}C_1\sqrt{LS} + 8C_1^2LS^2)} \left\| \gamma(\hat{\theta}_t) - \gamma(\theta^*) \right\|_{\Gamma_t^{-1}(\hat{\theta}_t, \theta^*)} \\
 &\leq (1 + 2\sqrt{2}C_1\sqrt{LS} + 8C_1^2LS^2) \sqrt{m} \left\| \gamma(\hat{\theta}_t) - \gamma(\theta^*) \right\|_{H_t^{-1}(\theta^*)} \\
 &\leq (1 + 2\sqrt{2}C_1\sqrt{LS} + 8C_1^2LS^2) \iota_t,
 \end{aligned}$$

where the first and the second inequalities follow from Lemma E.2, the equality follows from Equation (37), and the last inequality follows from Lemma E.1. Plugging **(term 3-5)** into **(term 1)** gives

$$\textbf{(term 1)} \leq (2 + C_1\sqrt{LS} + C_1LS^2) \iota_t + 1/2.$$

Now, moving on to **(term 2)**, we have

$$\sqrt{m} \|\theta_t - \theta^*\|_{Z_t} \leq \underbrace{\sqrt{m} \|\theta_t - \theta^*\|_2}_{\textbf{(term 6)}} \times \underbrace{\lambda_{\max}^{1/2}(Z_t)}_{\textbf{(term 7)}}.$$

For **(term 7)**, we have

$$\begin{aligned}
 \lambda_{\max}^{1/2}(Z_t) &= \lambda_{\max}^{1/2} \left(\sum_{i=1}^t \frac{|\dot{\mu}(f(x_i; \theta_i)) - \dot{\mu}(g(x_i; \theta_0)^\top (\hat{\theta}_t - \theta_0))|}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top \right) \\
 &\leq \lambda_{\max}^{1/2} \left(\sum_{i=1}^t \frac{C_3 R m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda_t^{-2/3}}{m} g(x_i; \theta_0) g(x_i; \theta_0)^\top \right) \\
 &\leq C_3 R^{1/2} m^{-1/12} (\log m)^{1/4} t^{5/6} L^2 \lambda_t^{-1/3}.
 \end{aligned}$$

Here, the first inequality follows from the Lipschitz continuity of $\dot{\mu}$, the bounds $|\ddot{\mu}| \leq \dot{\mu} \leq R$, and Lemma C.5, while the final inequality follows from $\lambda_{\max}(\sum_{i=1}^t x_i x_i^\top) \leq \sum_{i=1}^t \|x_i\|_2^2$ and used Lemma C.4. For **(term 6)** we have

$$\sqrt{m} \|\theta_t - \theta^*\|_2 \leq \sqrt{m} \|\theta_t - \theta_0\|_2 + \sqrt{m} \|\theta^* - \theta_0\|_2 \leq 3t^{1/2} \lambda_t^{-1/2} + S,$$

where the last inequality follows from Lemmas C.6 and 6.1. Plugging **(term 6-7)** back to **(term 2)** gives,

$$\begin{aligned}
 \textbf{(term 2)} &\leq C_3 R^{1/2} m^{-1/12} (\log m)^{1/4} t^{4/3} L^2 \lambda_t^{-5/6} + C_3 S R^{1/2} m^{-1/12} (\log m)^{1/4} t^{5/6} L^2 \lambda_t^{-1/3} \\
 &\leq C_3 R^{1/2} m^{-1/12} (\log m)^{1/4} T^{4/3} L^2 \lambda_0^{-5/6} + C_3 S R^{1/2} m^{-1/12} (\log m)^{1/4} T^{5/6} L^2 \lambda_0^{-1/3} \\
 &\leq 1/2 + 2S \lambda_t^{1/2} \\
 &\leq 1/2 + \iota_t,
 \end{aligned}$$

where the third inequality is followed by the condition on m in Condition C.2, and the last inequality is due to the update rule of λ_t . Finally, substituting **(term 1-2)** into Equation (46) gives

$$\begin{aligned}
 \sqrt{m} \|\theta_t - \theta^*\|_{W_t} &\leq (3 + 2\sqrt{2}C_1\sqrt{LS} + 8C_1^2LS^2) \iota_t + 1 \\
 &\leq C_7 (1 + \sqrt{LS} + LS^2) \iota_t + 1,
 \end{aligned}$$

for some absolute constant $C_7 > 0$, finishing the proof.

G. Regret Analyses

G.1. Proof of Theorem 4.6

We start with a proposition for the per-round regret:

Proposition G.1. Under Condition 4.4, for all $x \in \mathcal{X}_t$, $t \in [T]$, with probability at least $1 - \delta$,

$$\mu(h(x_t^*)) - \mu(h(x_t)) \leq 2R\sqrt{\kappa}((CLS^2 + 2)\iota_{t-1} + 1)\|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + 2\epsilon_{3,t-1}.$$

Proof. We follow the standard procedure to upper bound the per-round regret with the prediction error under the optimistic rule. For all $t \in [T]$ we have

$$\begin{aligned} \mu(h(x_t^*)) - \mu(h(x_t)) &\leq \mu(f(x_t^*; \theta_{t-1})) + R\sqrt{\kappa}(C_6(1 + \sqrt{LS} + LS^2)\iota_t + 1)\|g(x_t^*; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + \epsilon_{3,t-1} - \mu(h(x_t)) \\ &\leq \mu(f(x_t; \theta_{t-1})) + R\sqrt{\kappa}(C_6(1 + \sqrt{LS} + LS^2)\iota_t + 1)\|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + \epsilon_{3,t-1} - \mu(h(x_t)) \\ &\leq 2R\sqrt{\kappa}((CLS^2 + 2)\iota_{t-1} + 1)\|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + 2\epsilon_{3,t-1}, \end{aligned}$$

where the first and the last inequalities follow from Lemma 6.4, the second inequality comes from the optimistic rule of Algorithm 1, finishing the proof. $\square$

With Proposition G.1, we have

$$\begin{aligned} \mu(h(x_t^*)) - \mu(h(x_t)) &\leq \min \left\{ 2R\sqrt{\kappa}\nu_{t-1}^{(1)}\|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}} + 2\epsilon_{3,t-1}, 1 \right\} \\ &\leq \min \left\{ 2R\sqrt{\kappa}\nu_{t-1}^{(1)}\|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}, 1 \right\} + 2\epsilon_{3,t-1} \\ &\leq 2R\sqrt{\kappa}\nu_{t-1}^{(1)} \min \left\{ \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}, 1 \right\} + 2\epsilon_{3,t-1} \\ &\leq 2R\sqrt{\kappa}\nu_T^{(1)} \min \left\{ \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}, 1 \right\} + 2\epsilon_{3,T}. \end{aligned}$$

Here, the first inequality follows from $0 \leq |\mu(\cdot) - \mu(\cdot)| \leq 1$, the second from the bound $\min\{a + b, 1\} \leq \min\{a, 1\} + b$ for $b > 0$, the third from the facts that $2R\sqrt{\kappa} \geq 1$ and $\nu_t^{(1)} \geq 1$ for all t , thereby using $\min\{ab, 1\} \leq a \min\{b, 1\}$ if $a \geq 1$, and the last inequality follows from the fact that both ν_t and $\epsilon_{3,t}$ are monotonically non-decreasing in t .

Now, we can proceed as

$$\begin{aligned} \text{Regret}(T) &= \sum_{t=1}^T \mu(h(x_t^*)) - \mu(h(x_t)) \\ &\leq 2R\sqrt{\kappa}\nu_T^{(1)} \sum_{t=1}^T \min \left\{ \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}, 1 \right\} + 2T\epsilon_{3,T}, \end{aligned}$$

where we can see that by the condition of m in Condition C.2,

$$T\epsilon_{3,T} = C_3 R m^{-1/6} \sqrt{\log m} L^3 T^{5/3} \lambda_0^{-2/3} \leq 1,$$

plugging this back gives,

$$\begin{aligned} \text{Regret}(T) &\leq 2R\sqrt{\kappa}\nu_T^{(1)} \sum_{t=1}^T \min \left\{ \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}, 1 \right\} + 1 \\ &\leq 2R\sqrt{\kappa}\nu_T^{(1)} \sqrt{T \sum_{i=1}^T \min \left\{ \|g(x_i; \theta_0)/\sqrt{m}\|_{V_{i-1}^{-1}}^2, 1 \right\}} + 1 \\ &\leq 2R\sqrt{\kappa}\nu_T^{(1)} \sqrt{2T \log \det \left(\sum_{t=1}^T \frac{1}{\kappa m \lambda_0} g(x_t; \theta_0) g(x_t; \theta_0)^\top + \mathbf{I} \right)} + 1 \\ &\leq 2R\sqrt{\kappa}\nu_T^{(1)} \sqrt{2Td} + 1, \end{aligned}$$

where the second inequality follows from the Cauchy–Schwarz inequality and the relation $V_{t-1} \succeq \tilde{V}_{t-1}$, the third follows from Lemma H.2, and the final inequality follows from the definition of $\tilde{d}$. Notice that

$$\begin{aligned} \nu_T &= 16 \sqrt{\log \det \left(\sum_{t=1}^T \frac{1}{4m\lambda_0} g(x_t; \theta_0) g(x_t; \theta_0)^\top + \mathbf{I} \right) \log \frac{4T^2}{\delta} + 8C_1 \sqrt{\frac{L}{\lambda_0}} \log \frac{4T^2}{\delta}} \\ &\leq 16 \sqrt{\tilde{d} \log(4T^2/\delta)} + \sqrt{4C_1(2L)^{1/2} S \log^{-1}(4/\delta) \log(4T^2/\delta)}, \end{aligned}$$

where the last inequality follows from the definition of $\tilde{d}$ and the initialization rule of λ_0 , which gives $\nu_T^{(1)} = \tilde{O}(S^2 \sqrt{\tilde{d}} + S^{2.5})$. Finally, plugging $\nu_T^{(1)}$ in gives,

$$\text{Regret}(T) = \tilde{O}\left(S^2 \tilde{d} \sqrt{\kappa T} + S^{2.5} \sqrt{\kappa \tilde{d} T}\right),$$

finishing the proof.

G.2. Proof of Theorem 5.2

First, for each $t \in \mathbb{N}$, define the set of timesteps

$$\mathcal{T}_1(t) = \left\{ t' \in [t] : |f(x_{t'}; \theta_{t'}) - g(x_{t'}; \theta_0)^\top (\theta^* - \theta_0)| \geq 1 \right\}. \quad (47)$$

This set contains exactly those timesteps where $\theta_{t'}$ lies outside the parameter set (when $\|\theta_{t'} - \theta_0\|_2 > S$). Based on this, we form a pruned design matrix by removing the corresponding feature vectors while preserving their original order. In particular, for the regularized covariance matrix V_t , we obtain

$$\underline{V}_t = \sum_{i=1}^t \frac{1}{m} \mathbb{1}\{i \notin \mathcal{T}_1\} g(x_i; \theta_0) g(x_i; \theta_0)^\top + \lambda_t \mathbf{I} = \sum_{i=1}^{t-|\mathcal{T}_1(t)|} \frac{1}{m} g(x_{\tau(i)}; \theta_0) g(x_{\tau(i)}; \theta_0)^\top + \lambda_t \mathbf{I}.$$

Here, $\tau : \{1, \dots, t - |\mathcal{T}_1(t)|\} \rightarrow \{1, \dots, t\}$ maps each j to the j -th smallest element of $[t] \setminus \mathcal{T}_1(t)$. Similarly, we define $\underline{H}_t(\theta)$ and $\underline{W}_t$ as:

$$\begin{aligned} \underline{H}_t(\theta) &= \sum_{i=1}^{t-|\mathcal{T}_1(t)|} \frac{\dot{\mu}(g(x_{\tau(i)}; \theta_0)^\top (\theta - \theta_0))}{m} g(x_{\tau(i)}; \theta_0) g(x_{\tau(i)}; \theta_0)^\top + \lambda_t \mathbf{I}, \\ \underline{W}_t &= \sum_{i=1}^{t-|\mathcal{T}_1(t)|} \frac{\dot{\mu}(f(x_{\tau(i)}; \theta_{\tau(i)}))}{m} g(x_{\tau(i)}; \theta_0) g(x_{\tau(i)}; \theta_0)^\top + \lambda_t \mathbf{I}. \end{aligned}$$

Same way as before, we will denote $\tilde{\underline{V}}_t, \tilde{\underline{H}}_t(\theta), \tilde{\underline{W}}_t$ as the design matrix where the regularization parameter λ_t is replaced to λ_0 .

Using our new design matrices and the self-concordant property of the logistic function (see Lemma H.3; cf. Lemma 9 of (Fauray et al., 2020), Lemma 7 of (Abeille et al., 2021), and Lemma 5 of (Jun et al., 2021)), we can show that the true-variance design matrix $\underline{H}(\theta^*)$ is bounded by the empirical-variance design matrix $\underline{W}_t$.

Proposition G.2. *We have $3\underline{H}_t(\theta^*) \succeq \underline{W}_t \succeq \frac{1}{3}\underline{H}_t(\theta^*)$.*

Next, we define three additional sets of timesteps derived from $\mathcal{T}_1$:

$$\begin{aligned} \mathcal{T}_2 &= \left\{ t \in [T - |\mathcal{T}_1(T)|] : |g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)| \geq 1 \right\}, \\ \mathcal{T}_3 &= \left\{ t \in [T - |\mathcal{T}_1(T)|] : \left\| g(x_{\tau(t)}; \theta_0) / \sqrt{m} \right\|_{\tilde{\underline{V}}_{\tau(t)-1}^{-1}} \geq 1 \right\}, \\ \mathcal{T}_4 &= \left\{ t \in [T - |\mathcal{T}_1(T)|] : \left\| \sqrt{\dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))} g(x_{\tau(t)}; \theta_0) / \sqrt{m} \right\|_{\tilde{\underline{W}}_{\tau(t)-1}^{-1}} \geq 1 \right\}. \end{aligned} \quad (48)$$

We define $\mathcal{T}_2$ to measure the distance between $g(x_{\tau(t)}; \theta_0)^\top \tilde{\theta}_{\tau(t)-1}$ and $h(x_{\tau(t)})$ and control the estimation error of the neural network. We introduce $\mathcal{T}_3, \mathcal{T}_4$ to control the value of $\|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\tilde{V}_{\tau(t)-1}^{-1}}$ and $\|\sqrt{\dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))}g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\tilde{W}_{\tau(t)-1}^{-1}}$ in order to apply the elliptical potential lemma (Lemma H.2).

Next, we introduce two propositions to bound the cardinality of $\mathcal{T}_1(T)$, $\mathcal{T}_2$, $\mathcal{T}_3$ and $\mathcal{T}_4$:

Proposition G.3. *We have $|\mathcal{T}_1(T)| \leq 4\kappa\tilde{d}\nu_T^{(1)2} + 1$ and $|\mathcal{T}_2| \leq 24\kappa\tilde{d}\nu_T^{(2)2}$, where $\nu_t^{(1)}$ and $\nu_t^{(2)}$ are defined at Equations (1) and (4), respectively.*

Proposition G.4. *We have $|\mathcal{T}_3|, |\mathcal{T}_4| \leq 2\tilde{d}$.*

For Proposition G.3, we use the concentration inequalities between $\theta_{\tau(t)}$ and θ^* , and $\tilde{\theta}_{\tau(t)-1}$ and θ^* using Lemmas 4.5 and 5.1. For Proposition G.4 we modified previous results appropriate to our setting called the elliptical potential count lemma (Lemma 7 of (Gales et al., 2022), Lemma 4 of (Kim et al., 2022)).

Now we can start the proof of Theorem 5.2.

Proof of Theorem 5.2. At time t , from the optimistic rule in Equation (5), denote

$$(x_t, \tilde{\theta}_{t-1}) \leftarrow \arg \max_{x \in \mathcal{X}_t, \theta \in \mathcal{W}_{t-1}} \langle g(x; \theta_0), \theta - \theta_0 \rangle \quad (49)$$

We use $\mathcal{T}_1(T) = \mathcal{T}_1$ for brevity. From Equations (47) and (48), we define the combined set of timesteps as

$$\mathcal{T} = \{\mathcal{T}_2 \cup \mathcal{T}_3 \cup \mathcal{T}_4\}.$$

Then we have,

$$\begin{aligned} \text{Regret}(T) &\leq |\mathcal{T}_1| + \sum_{t=1}^{T-|\mathcal{T}_1|} \mu(h(x_{\tau(t)}^*)) - \mu(h(x_{\tau(t)})) \\ &\leq |\mathcal{T}_1| + |\mathcal{T}_2| + |\mathcal{T}_3| + |\mathcal{T}_4| + \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\mu(h(x_{\tau(t)}^*)) - \mu(h(x_{\tau(t)}))] \\ &\leq 4\kappa\tilde{d}\nu_T^{(1)2} + 24\kappa\tilde{d}\nu_T^{(2)2} + 4\tilde{d} + 1 + \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\mu(h(x_{\tau(t)}^*)) - \mu(h(x_{\tau(t)}))]}_{=: \text{Regret}^c(T)}, \end{aligned} \quad (50)$$

where the second inequality follows from the definition of $\mathcal{T}$, and the last inequality follows from Propositions G.3 and G.4. For $\text{Regret}^c(T)$, we have

$$\begin{aligned} \text{Regret}^c(T) &= \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\mu(g(x_{\tau(t)}^*; \theta_0)^\top (\theta^* - \theta_0)) - \mu(g(x_{\tau(t)}; \theta_0)^\top (\theta^* - \theta_0))] \\ &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\mu(g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta_0)) - \mu(g(x_{\tau(t)}; \theta_0)^\top (\theta^* - \theta_0))] \end{aligned}$$

where the equality follows from Lemma 6.1, and the inequality follows from the optimistic rule in Equation (49) since $\tilde{\theta}_{\tau(t)-1}, \theta^* \in \mathcal{W}_{\tau(t)-1}$. With the definition of $\alpha(x, \theta', \theta'')$ at Equation (31), we can continue with

$$\begin{aligned} \text{Regret}^c(T) &\leq \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta^* - \theta_0))g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)]}_{(\text{term 1})} \\ &\quad + \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\alpha(x_{\tau(t)}, \tilde{\theta}_{\tau(t)-1}, \theta^*)[g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)]^2]}_{(\text{term 2})}, \end{aligned} \quad (51)$$

where we used a second-order Taylor expansion and the fact that $|\ddot{\mu}| \leq \dot{\mu}$.

For **(term 2)** we have

$$\begin{aligned}
 \text{(term 2)} &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [1 \cdot [g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)]^2] \\
 &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t)-1}^{-1}}^2 \cdot m \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t)-1}}^2.
 \end{aligned} \tag{52}$$

For $\|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t)-1}^{-1}}$, we have

$$\begin{aligned}
 \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t)-1}^{-1}} &\leq \sqrt{3} \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{H}_{\tau(t)-1}^{-1}(\theta^*)} \\
 &\leq \sqrt{3\kappa} \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}},
 \end{aligned} \tag{53}$$

where the first inequality follows from Proposition G.2 and the second inequality follows from Assumption 6.3. For $\sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t)-1}}$, we have

$$\begin{aligned}
 &\sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t)-1}} \\
 &\leq \sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t)-1}} \\
 &\leq \sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{W_{\tau(t)-1}} \\
 &\leq \left(\sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta_{\tau(t)-1}\|_{W_{\tau(t)-1}} + \sqrt{m} \|\theta_{\tau(t)-1} - \theta^*\|_{W_{\tau(t)-1}} \right) \\
 &\leq 2\nu_{\tau(t)-1}^{(2)},
 \end{aligned} \tag{54}$$

where the first and the second inequality are due to the fact that $\tau(t-1) \leq \tau(t)-1$ and $\underline{W}_t \preceq W_t$ for all t , respectively. The third inequality follows from the triangle inequality, and the last inequality follows from Lemma 5.1 since $\tilde{\theta}_{\tau(t)-1}, \theta^* \in \mathcal{W}_{\tau(t)-1}$. Plugging Equations (53) and (54) back to **(term 2)** gives

$$\begin{aligned}
 \text{(term 2)} &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot 3\kappa \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2 \cdot 4(\nu_{\tau(t)-1}^{(2)})^2 \\
 &\leq 12\kappa (\nu_T^{(2)})^2 \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2,
 \end{aligned}$$

where the inequality holds since $\nu_t^{(2)}$ is monotonically non-decreasing in t . By the definition of $\mathcal{T}_3$, we have $\|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}} < 1$ for all $t \in [T - |\mathcal{T}_1|]$. Therefore,

$$\begin{aligned}
 \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2 &= \min \left\{ 1, \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2 \right\} \\
 &\leq \min \left\{ 1, \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2 \right\} \\
 &\leq \min \left\{ 1, \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2 \right\},
 \end{aligned} \tag{55}$$

where the last inequality follows from the fact that $\lambda_0 \leq \lambda_t$. Substituting Equation (55) gives

$$\begin{aligned}
 (\text{term 2}) &\leq 12\kappa(\nu_T^{(2)})^2 \sum_{t=1}^{T-|\mathcal{T}_1|} \min \left\{ 1, \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t-1)}^{-1}}^2 \right\} \\
 &\leq 24\kappa(\nu_T^{(2)})^2 \log \frac{\det \tilde{V}_{\tau(T-|\mathcal{T}_1|)}}{\det \kappa \lambda_0 \mathbf{I}} \\
 &\leq 24\kappa(\nu_T^{(2)})^2 \log \det \left(\sum_{t=1}^T \frac{1}{\kappa m \lambda_0} g(x_t; \theta_0) g(x_t; \theta_0)^\top + \mathbf{I} \right) \\
 &\leq 24\kappa(\nu_T^{(2)})^2 \tilde{d},
 \end{aligned} \tag{56}$$

where the second inequality follows from Lemma H.2, and the last inequality follows from the definition of $\tilde{d}$.

For **(term 1)**, we consider 2 cases where:

(case 1). if $\dot{\mu}(h(x_{\tau(t)})) \leq \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))$

(case 2). if $\dot{\mu}(h(x_{\tau(t)})) > \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))$

In (case 1), for **(term 1)**, we continue with

$$\begin{aligned}
 &\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta^* - \theta_0)) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)] \\
 &= \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \sqrt{\dot{\mu}(h(x_{\tau(t)}))} \sqrt{\dot{\mu}(h(x_{\tau(t)}))} g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*) \\
 &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \sqrt{\dot{\mu}(h(x_{\tau(t)}))} \sqrt{\dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))} g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*),
 \end{aligned} \tag{57}$$

where the last inequality follows from the assumption of (case 1). For brevity, we denote $\dot{g}(x_t; \theta_0) = \sqrt{\dot{\mu}(f(x_t; \theta_t))} g(x_t; \theta_0)$. Notice that we can represent W_t as $W_t = \sum_{i=1}^t \frac{1}{m} \dot{g}(x_i; \theta_0) \dot{g}(x_i; \theta_0)^\top + \lambda_t \mathbf{I}$. Then we can continue as

$$\begin{aligned}
 (\text{term 1}) &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \sqrt{\dot{\mu}(h(x_{\tau(t)}))} \cdot \dot{g}(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*) \\
 &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \sqrt{\dot{\mu}(h(x_{\tau(t)}))} \cdot \|\dot{g}(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t-1)}^{-1}} \cdot \sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t-1)}}
 \end{aligned}$$

For $\mathbb{1}\{t \notin \mathcal{T}\} \|\dot{g}(x_{\tau(t)}; \theta_0)\|_{\underline{W}_{\tau(t-1)}^{-1}}$, we have

$$\begin{aligned}
 \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|\dot{g}(x_{\tau(t)}; \theta_0)\|_{\underline{W}_{\tau(t-1)}^{-1}} &= \min \left\{ 1, \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|\dot{g}(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t-1)}^{-1}} \right\} \\
 &\leq \min \left\{ 1, \|\dot{g}(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t-1)}^{-1}} \right\},
 \end{aligned} \tag{58}$$

where the inequality follows from the definition of $\mathcal{T}_4$ and the fact that $\lambda_0 \leq \lambda_t$ for all t . Also, using the previous results of

Equation (54), we have $\sqrt{m}\|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\tilde{W}_{\tau(t)-1}} \leq 2\nu_{\tau(t)-1}^{(2)}$. Substituting these back gives

$$\begin{aligned} \text{(term 1)} &\leq 2\nu_T^{(2)} \sum_{t=1}^{T-|\mathcal{T}_1|} \sqrt{\dot{\mu}(h(x_{\tau(t)}))} \cdot \left\{1, \|\dot{g}(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\tilde{W}_{\tau(t)-1}^{-1}}\right\} \\ &\leq 2\nu_T^{(2)} \underbrace{\sqrt{\sum_{t=1}^{T-|\mathcal{T}_1|} \dot{\mu}(h(x_{\tau(t)}))}}_{\text{(term 3)}} \cdot \underbrace{\sqrt{\sum_{t=1}^{T-|\mathcal{T}_1|} \left\{1, \|\dot{g}(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\tilde{W}_{\tau(t)-1}^{-1}}^2\right\}}}_{\text{(term 4)}} \end{aligned}$$

where the first inequality is by the monotonicity of $\mu_t^{(2)}$ in t , and the second inequality follows from the Cauchy-Schwarz inequality. For (term 4), we have

$$\begin{aligned} \sqrt{\sum_{t=1}^{T-|\mathcal{T}_1|} \left\{1, \|\dot{g}(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\tilde{W}_{\tau(t)-1}^{-1}}^2\right\}} &\leq \sqrt{2 \log \frac{\det \tilde{W}_{\tau(T-|\mathcal{T}_1|)}}{\det \lambda_0 \mathbf{I}}} \\ &\leq \sqrt{2 \log \det \left(\sum_{t=1}^T \frac{\dot{\mu}(f(x_t; \theta_t))}{m\lambda_0} g(x_t; \theta_0) g(x_t; \theta_0)^\top + \mathbf{I} \right)} \\ &\leq \sqrt{2\tilde{d}}, \end{aligned}$$

where the first inequality follows from Lemma H.2, and the last inequality follows from the definition of $\tilde{d}$.

For (term 3), we have

$$\begin{aligned} \text{(term 3)}^2 &\leq \sum_{t=1}^T \dot{\mu}(g(x_t; \theta_0)^\top (\theta^* - \theta_0)) \\ &\leq \sum_{t=1}^T \dot{\mu}(g(x_t^*; \theta_0)^\top (\theta^* - \theta_0)) + \sum_{t=1}^T \alpha(x_t, x_t^*, \theta^*) (g(x_t; \theta_0) - g(x_t^*; \theta_0))^\top (\theta^* - \theta_0) \\ &= \frac{T}{\kappa^*} + \sum_{t=1}^T \alpha(x_t, x_t^*, \theta^*) (g(x_t; \theta_0) - g(x_t^*; \theta_0))^\top (\theta^* - \theta_0) \\ &\leq \frac{T}{\kappa^*} + \sum_{t=1}^T \alpha(x_t, x_t^*, \theta^*) (g(x_t^*; \theta_0) - g(x_t; \theta_0))^\top (\theta^* - \theta_0) \\ &= \frac{T}{\kappa^*} + \sum_{t=1}^T \mu(g(x_t^*; \theta_0)^\top (\theta^* - \theta_0)) - \mu(g(x_t; \theta_0)^\top (\theta^* - \theta_0)) \\ &= \frac{T}{\kappa^*} + \sum_{t=1}^T \mu(h(x_t^*)) - \mu(h(x_t)) \\ &= \frac{T}{\kappa^*} + \text{Regret}(T). \end{aligned} \tag{59}$$

Here, the second inequality follows from a first-order Taylor expansion together with the bound $|\dot{\mu}| \leq \dot{\mu}$ and the definition of $\alpha(x', x'', \theta)$ in Equation (31), the first equality follows from the definition of κ^* , namely $1/\kappa^* = \frac{1}{T} \sum_{t=1}^T \dot{\mu}(h(x_t^*))$, the third inequality uses the fact that $h(x_t^*) \geq h(x_t)$, the second equality follows from the mean-value theorem, and the final equality follows from the definition of regret.

Finally, substituting (term 3) and (term 4) back gives

$$\text{(term 1)} \leq 2\nu_T^{(2)} \sqrt{\text{Regret}(T) + T/\kappa^*} \cdot \sqrt{2\tilde{d}}. \tag{60}$$

Now we consider about (case 2), where $\dot{\mu}(h(x_{\tau(t)})) > \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))$. For **(term 1)**, we have

$$\begin{aligned}
 & \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} [\dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta^* - \theta_0)) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)] \\
 & \leq \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta_0)) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)}_{\text{(term 4)}} \\
 & \quad + \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot 1 \cdot [g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta^*) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)]}_{\text{(term 5)}},
 \end{aligned}$$

where the inequality follows from the Taylor expansion, and by the fact that $|\dot{\mu}| \leq \dot{\mu} \leq 1$. For **(term 5)** we have,

$$\begin{aligned}
 & \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot 1 \cdot [g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta^*) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)] \\
 & \leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \|g(x_{\tau(t)}; \theta_0) / \sqrt{m}\|_{\underline{W}_{\tau(t-1)}^{-1}}^2 \\
 & \quad \times \sqrt{m} \|\theta_{\tau(t)} - \theta^*\|_{\underline{W}_{\tau(t-1)}} \times \sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t-1)}}.
 \end{aligned}$$

We have $\mathbb{1}\{t \notin \mathcal{T}\} \cdot \|g(x_{\tau(t)}; \theta_0) / \sqrt{m}\|_{\underline{W}_{\tau(t-1)}^{-1}} \leq 3\kappa \min\{1, \|g(x_{\tau(t)}; \theta_0) / \sqrt{m}\|_{\tilde{\underline{V}}_{\tau(t-1)}^{-1}}^2\}$ using Equations (53) and (55).

Also we have $\sqrt{m} \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t-1)}} \leq 2\nu_{\tau(t)-1}^{(2)}$ using Equation (54). For $\sqrt{m} \|\theta_{\tau(t)} - \theta^*\|_{\underline{W}_{\tau(t-1)}}$, we have

$$\sqrt{m} \|\theta_{\tau(t)} - \theta^*\|_{\underline{W}_{\tau(t-1)}} \leq \sqrt{m} \|\theta_{\tau(t)} - \theta^*\|_{\underline{W}_{\tau(t)}} \leq \sqrt{m} \|\theta_{\tau(t)} - \theta^*\|_{\underline{W}_{\tau(t)}} \leq \nu_{\tau(t)}^{(2)}.$$

Plugging results back gives

$$\begin{aligned}
 \text{(term 5)} & \leq \sum_{t=1}^{T-|\mathcal{T}_1|} 3\kappa \min\left\{1, \|g(x_{\tau(t)}; \theta_0) / \sqrt{m}\|_{\tilde{\underline{V}}_{\tau(t-1)}^{-1}}^2\right\} \cdot \nu_{\tau(t)}^{(2)} \cdot 2\nu_{\tau(t)-1}^{(2)} \\
 & \leq 6\kappa (\nu_T^{(2)})^2 \sum_{t=1}^{T-|\mathcal{T}_1|} \min\left\{1, \|g(x_{\tau(t)}; \theta_0) / \sqrt{m}\|_{\tilde{\underline{V}}_{\tau(t-1)}^{-1}}^2\right\} \\
 & \leq 12\kappa (\nu_T^{(2)})^2 \log \frac{\det \tilde{\underline{V}}_{\tau(T-|\mathcal{T}_1|)}}{\det \kappa \lambda_0 \mathbf{I}} \\
 & \leq 12\kappa \tilde{d} (\nu_T^{(2)})^2
 \end{aligned}$$

where the second inequality is because $\nu_{\tau(t)}^{(2)}$ is non-decreasing in t , the third inequality follows from Lemma H.2, and the last inequality follows from the definition of $\tilde{d}$.

Now, for **(term 4)**, we have

$$\begin{aligned}
 & \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta_0)) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*) \\
 &= \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)})) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)}_{\text{(term 6)}} \\
 &+ \underbrace{\sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \left(\dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta_0)) - \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)})) \right) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)}_{\text{(term 7)}}.
 \end{aligned}$$

For **(term 7)**, recall the definition of $\mathcal{T}_2$. Then for some absolute constant $C_3 > 0$, we have

$$\begin{aligned}
 \text{(term 7)} &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \left| \dot{\mu}(g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta_0)) - \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)})) \right| \cdot |g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)| \\
 &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} R \left| g(x_{\tau(t)}; \theta_0)^\top (\theta_{\tau(t)} - \theta_0) - f(x_{\tau(t)}; \theta_{\tau(t)}) \right| \cdot 1 \\
 &\leq T \cdot C_3 R m^{-1/6} \sqrt{\log mL}^3 T^{2/3} \lambda_0^{-2/3} \\
 &\leq 1,
 \end{aligned}$$

where the second inequality follows from the definition of $\mathcal{T}_2$, the third inequality is due to the fact that $\mu(\cdot)$ is a R -Lipschitz function, the third inequality follows from Lemma C.5, and the last inequality follows from the condition of m in Condition C.2.

For **(term 6)**, we have

$$\begin{aligned}
 & \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)})) g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*) \\
 &\leq \sum_{t=1}^{T-|\mathcal{T}_1|} \mathbb{1}\{t \notin \mathcal{T}\} \cdot \sqrt{\dot{\mu}(h(x_{\tau(t)}))} \sqrt{\dot{\mu}(f(x_{\tau(t)}; \theta_{\tau(t)}))} g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*),
 \end{aligned}$$

where the inequality follows from the assumption of (case 2). Notice that expression is same as the **(term 1)** of (case 1) at Equation (57). Therefore, using the result of Equation (60), we have **(term 6)** $\leq 2\nu_T^{(2)} \sqrt{\text{Regret}(T) + T/\kappa^*} \cdot \sqrt{2\tilde{d}}$.

Finally, plugging **(term 4-7)** into **(term 1)** gives,

$$\text{(term 1)} \leq 2\nu_T^{(2)} \sqrt{\text{Regret}(T) + T/\kappa^*} \cdot \sqrt{2\tilde{d}} + 12\kappa\tilde{d}(\nu_T^{(2)})^2 + 1.$$

Recall the upper bound of **(term 1)** in (case 1) at Equation (60), which is **(term 1)** $\leq 2\nu_T^{(2)} \sqrt{\text{Regret}(T) + T/\kappa^*} \cdot \sqrt{2\tilde{d}}$. Since the upper bound value in (case 2) is strictly larger than that of (case 1), we give a naive bound of **(term 1)** by using the result of (case 2).

Now, substituting **(term 1-2)** into Equation (51) gives

$$\text{Regret}^c(T) \leq 2\nu_T^{(2)} \sqrt{\text{Regret}(T) + T/\kappa^*} \cdot \sqrt{2\tilde{d}} + 36\kappa\tilde{d}(\nu_T^{(2)})^2 + 1.$$

Substituting $\text{Regret}^c(T)$ into Equation (50) gives

$$\text{Regret}(T) \leq 2\nu_T^{(2)} \sqrt{\text{Regret}(T) + T/\kappa^*} \cdot \sqrt{2\tilde{d}} + 4\tilde{d} + 4\kappa\tilde{d}(\nu_T^{(1)})^2 + 60\kappa\tilde{d}(\nu_T^{(2)})^2 + 2.$$

Finally, using the fact that for $b, c > 0$ and $x \in \mathbb{R}$, $x^2 - bx - c \leq 0 \implies x^2 \leq 2b^2 + 2c$, and substituting $\nu_T^{(1)}, \nu_T^{(2)} = \tilde{\mathcal{O}}(S^2 \sqrt{\tilde{d}} + S^{2.5})$, we have

$$\begin{aligned} \text{Regret}(T) &\leq 16(\nu_T^{(2)})^2 + 4\nu_T^{(2)} \sqrt{2\tilde{d}T/\kappa^*} + 8\tilde{d} + 8\kappa\tilde{d}(\nu_T^{(1)})^2 + 120\kappa\tilde{d}(\nu_T^{(2)})^2 + 4 \\ &\leq \tilde{\mathcal{O}}\left(S^2\tilde{d}\sqrt{T/\kappa^*} + S^{2.5}\tilde{d}^{0.5}\sqrt{T/\kappa^*} + S^4\kappa\tilde{d}^2 + S^{4.5}\kappa\tilde{d}^{1.5} + S^5\kappa\tilde{d}\right), \end{aligned}$$

finishing the proof. $\square$

G.3. Proof of Proposition G.2

We suitably modify Lemma 5 of (Jun et al., 2021) for our setting. Define $d(t) = |f(x_t; \theta_t) - g(x_t; \theta_0)^\top(\theta^* - \theta_0)|$. By the definition of $\mathcal{T}_1$, for all $t \notin \mathcal{T}_1(T)$, $d(t) \leq 1$. Recall the definition of $\alpha(z', z'')$ at Equation (31). Then for all $t \notin \mathcal{T}_1(T)$, we have

$$\begin{aligned} \dot{\mu}(f(x_t; \theta_t)) &\geq \frac{d(t)}{\exp(d(t)) - 1} \cdot \alpha\left(f(x_t; \theta_t), g(x_t; \theta_0)^\top(\theta^* - \theta_0)\right) \\ &\geq \frac{d(t)}{\exp(d(t)) - 1} \cdot \frac{1 - \exp(-d(t))}{d(t)} \dot{\mu}(g(x_t; \theta_0)^\top(\theta^* - \theta_0)) \\ &= \frac{1}{\exp(d(t))} \cdot \mu(g(x_t; \theta_0)^\top(\theta^* - \theta_0)) \\ &\geq \frac{1}{d(t)^2 + d(t) + 1} \cdot \mu(g(x_t; \theta_0)^\top(\theta^* - \theta_0)) \\ &\geq \frac{1}{2d(t) + 1} \cdot \mu(g(x_t; \theta_0)^\top(\theta^* - \theta_0)), \end{aligned}$$

where the first and the second inequalities follow from the self-concordant property in Lemma H.3, the third and the fourth inequalities hold since $d(t) \leq 1$. This implies that

$$\underline{W}_t \succeq \frac{1}{2 \max\{d(t')\}_{(t' \in [t]) \cap (t' \notin \mathcal{T}_1(t))} + 1} \underline{H}_t(\theta^*) \succeq \frac{1}{3} \underline{H}_t(\theta^*),$$

In a similar way, we can have

$$\underline{H}_t(\theta^*) \succeq \frac{1}{2 \max\{d(t')\}_{(t' \in [t]) \cap (t' \notin \mathcal{T}_1(t))} + 1} \underline{W}_t \succeq \frac{1}{3} \underline{W}_t.$$

Combining these results, we finish the proof.

G.4. Proof of Proposition G.3

We start with the upper bound of $|\mathcal{T}_1|$. For an absolute constant $C_3 > 0$, we have:

$$\begin{aligned} |\mathcal{T}_1| \cdot \min\{1, 1^2\} &\leq \sum_{t=1}^T \min\left\{1, |f(x_t; \theta_t) - g(x_t; \theta_0)^\top(\theta^* - \theta_0)|^2\right\} \\ &\leq \sum_{t=1}^T \min\left\{1, 2|f(x_t; \theta_t) - g(x_t; \theta_0)^\top(\theta_t - \theta_0)|^2 + 2|g(x_t; \theta_0)^\top(\theta_t - \theta^*)|^2\right\}, \end{aligned}$$

For $2|f(x_t; \theta_t) - g(x_t; \theta_0)^\top(\theta_t - \theta_0)|^2$, we have $|f(x_t; \theta_t) - g(x_t; \theta_0)^\top(\theta_t - \theta_0)| \leq C_3 m^{-1/6} \sqrt{\log m} L^3 t^{2/3} \lambda_t^{-2/3}$ using Lemma C.5. Since the error term is positive, we can take it out of the $\min\{1, \cdot\}$ term, which gives

$$\begin{aligned} |\mathcal{T}_1| &\leq \sum_{t=1}^T \min\left\{1, 2|g(x_t; \theta_0)^\top(\theta_t - \theta^*)|^2\right\} + C_3^2 m^{-1/3} (\log m) L^6 T^{7/3} \lambda_0^{-4/3} \\ &\leq 2 \sum_{t=1}^T \min\left\{1, |g(x_t; \theta_0)^\top(\theta_t - \theta^*)|^2\right\} + 1, \end{aligned}$$

where the last inequality is due to the condition of m at Condition 4.4, and the fact that $\min\{1, ab\} \leq a \min\{1, b\}$ if $a \geq 1$. We further proceed as

$$|\mathcal{T}_1| \leq 2 \sum_{t=1}^T \min \left\{ 1, \|g(x_t; \theta_0)/\sqrt{m}\|_{H_{t-1}^{-1}(\theta^*)}^2 \cdot m \|\theta_t - \theta^*\|_{H_{t-1}(\theta^*)}^2 \right\} + 1.$$

For $m \|\theta_t - \theta^*\|_{H_{t-1}(\theta^*)}^2$, we have

$$m \|\theta_t - \theta^*\|_{H_{t-1}(\theta^*)}^2 \leq m \|\theta_t - \theta^*\|_{H_t(\theta^*)}^2 \leq (\nu_t^{(1)})^2.$$

Since $\nu_t^{(1)} \geq 1$ we can take out of the $\min\{1, \cdot\}$ term, and by the monotonicity of $\nu_t^{(1)}$ in t , we have

$$\begin{aligned} |\mathcal{T}_1| &\leq 2(\nu_T^{(1)})^2 \sum_{t=1}^T \min \left\{ 1, \|g(x_t; \theta_0)/\sqrt{m}\|_{H_{t-1}^{-1}(\theta^*)}^2 \right\} + 1 \\ &\leq 2\kappa(\nu_T^{(1)})^2 \sum_{t=1}^T \min \left\{ 1, \|g(x_t; \theta_0)/\sqrt{m}\|_{\tilde{V}_{t-1}^{-1}}^2 \right\} + 1 \\ &\leq 4\kappa(\nu_T^{(1)})^2 \log \frac{\det \tilde{V}_T}{\det \kappa \lambda_0 \mathbf{I}} + 1 \\ &\leq 4\kappa \tilde{d} (\nu_T^{(1)})^2 + 1, \end{aligned}$$

where the second inequality follows from $\kappa \geq 1$, and $H_t(\theta^*) \succeq (1/\kappa)V_t \succeq (1/\kappa)\tilde{V}_t$, the third inequality follows from Lemma H.2, and the last inequality follows from the definition of $\tilde{d}$.

Next we can show the upper bound of $|\mathcal{T}_2|$ in a similar way:

$$\begin{aligned} |\mathcal{T}_2| \cdot \min\{1, 1^2\} &\leq \sum_{t=1}^{T-|\mathcal{T}_1(T)|} \min \left\{ 1, |g(x_{\tau(t)}; \theta_0)^\top (\tilde{\theta}_{\tau(t)-1} - \theta^*)|^2 \right\} \\ &\leq \sum_{t=1}^{T-|\mathcal{T}_1(T)|} \min \left\{ 1, \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t)-1}^{-1}}^2 \cdot m \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t)-1}}^2 \right\} \end{aligned}$$

We have $\|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{W}_{\tau(t)-1}^{-1}}^2 \leq \sqrt{3\kappa} \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2$ using the result of Equation (53). Also, we have $m \|\tilde{\theta}_{\tau(t)-1} - \theta^*\|_{\underline{W}_{\tau(t)-1}}^2 \leq 4(\nu_{\tau(t)-1}^{(2)})^2$ using the result of Equation (54). Since $\nu_t^{(2)}$ is non-decreasing in t , substituting results back gives

$$\begin{aligned} |\mathcal{T}_2| &\leq 12\kappa(\nu_T^{(2)})^2 \sum_{t=1}^{T-|\mathcal{T}_1(T)|} \min \left\{ 1, \|g(x_{\tau(t)}; \theta_0)/\sqrt{m}\|_{\underline{V}_{\tau(t)-1}^{-1}}^2 \right\} \\ &\leq 24\kappa(\nu_T^{(2)})^2 \log \frac{\det \tilde{V}_{\tau(T-|\mathcal{T}_1|)}}{\det \kappa \lambda_0 \mathbf{I}} \\ &\leq 24\kappa \tilde{d} (\nu_T^{(2)})^2, \end{aligned}$$

where the second inequality follows from Lemma H.2, and the last inequality follows from the definition of $\tilde{d}$, finishing the proof.

G.5. Proof of Proposition G.4

We begin with the case of $\mathcal{T}_3$. We define a new design matrix that consists of all feature vectors of $\underline{V}_t$ up to time t , in their original order, including only those corresponding to timesteps in $\mathcal{T}_3(t)$:

$$\tilde{\mathcal{V}}_t = \sum_{i=1}^{t-|\mathcal{T}_1(t)|} \frac{1}{m} \mathbb{1}\{i \in \mathcal{T}_3\} g(x_{\tau(i)}; \theta_0) g(x_{\tau(i)}; \theta_0)^\top + \lambda_0 \mathbf{I}$$

For brevity we define $j(t) = \tau(t - |\mathcal{T}_1(t)|)$. Then we have

$$\begin{aligned}
 \det(\tilde{V}_T) &= \det\left(\sum_{i=1}^{j(T)} \frac{1}{m} \mathbb{1}\{i \in \mathcal{T}_3\} g(x_{\tau(i)}; \theta_0) g(x_{\tau(i)}; \theta_0)^\top + \lambda_0 \mathbf{I}\right) \\
 &= \det\left(\tilde{V}_{\tau(j(T)-1)} + \frac{1}{m} \mathbb{1}\{\tau(j(T)) \in \mathcal{T}_3\} g(x_{\tau(j(T))}; \theta_0) g(x_{\tau(j(T))}; \theta_0)^\top\right) \\
 &= \det\left(\tilde{V}_{\tau(j(T)-1)}\right) \left(1 + \mathbb{1}\{\tau(j(T)) \in \mathcal{T}_3\} \|g(x_{\tau(j(T))}; \theta_0) / \sqrt{m}\|_{\tilde{V}_{\tau(j(T)-1)}^{-1}}^2\right) \\
 &\geq \det\left(\tilde{V}_{\tau(j(T)-1)}\right) \left(1 + \mathbb{1}\{\tau(j(T)) \in \mathcal{T}_3\}\right),
 \end{aligned}$$

where the third equality follows from the matrix determinant lemma, and the inequality follows from the definition of $\mathcal{T}_3$. Repeating inequalities to $\tilde{V}_{\tau}(0)$ gives

$$\det(\tilde{V}_T) \geq \det\left(\tilde{V}_{\tau(0)}\right) \cdot \left(1 + \mathbb{1}\{\tau(j(T)) \in \mathcal{T}_3\}\right)^{T - |\mathcal{T}_1(T)|} = \det(\kappa \lambda_0 \mathbf{I}) \cdot (1 + 1)^{|\mathcal{T}_3|}.$$

Therefore, we can rewrite as

$$|\mathcal{T}_3| \leq \frac{1}{\log 2} \cdot \log \frac{\det \tilde{V}_T}{\det \kappa \lambda_0 \mathbf{I}} \leq \frac{1}{\log 2} \cdot \log \frac{\det \tilde{V}_T}{\det \kappa \lambda_0 \mathbf{I}} \leq 2\tilde{d},$$

where the last inequality follows from the definition of $\tilde{d}$. We can prove $|\mathcal{T}_4| \leq 2\tilde{d}$ in a similar way, starting by defining $\tilde{W}_t = \sum_{i=1}^{t - |\mathcal{T}_1(t)|} \frac{\dot{\mu}(f(x_{\tau(i)}; \theta_0))}{m} \mathbb{1}\{i \in \mathcal{T}_4\} g(x_{\tau(i)}; \theta_0) g(x_{\tau(i)}; \theta_0)^\top + \lambda_0 \mathbf{I}$ and following the above process.

H. Auxiliary Lemmas

Lemma H.1 ((Freedman, 1975)). *Let $M, v > 0$ be fixed constants. Let $\{x_i\}_{i=1}^n$ be a stochastic process, $\{\mathcal{G}_i\}_i$ be a filtration so that for all $i \in [n]$, x_i is $\mathcal{G}_i$ -measurable, while almost surely $\mathbb{E}[x_i | \mathcal{G}_{i-1}] = 0$, $|x_i| \leq M$ and*

$$\sum_{i=1}^n \mathbb{E}[x_i^2 | \mathcal{G}_{i-1}] \leq v.$$

Then, for any $\delta > 0$, with probability at least $1 - \delta$,

$$\sum_{i=1}^n x_i \leq \sqrt{2v \log(1/\delta)} + 2/3 \cdot M \log(1/\delta).$$

Lemma H.2 (Lemma 11 (Abbasi-Yadkori et al., 2011)). *For any $\lambda > 0$ and sequence $\{x_t\}_{t=1}^T \in \mathbb{R}^d$, define $Z_t = \lambda \mathbf{I} + \sum_{i=1}^t x_i x_i^\top$. Then, provided that $\|x_t\|_2 \leq L$ holds for all $t \in [T]$, we have*

$$\sum_{t=1}^T \min\{1, \|x_t\|_{Z_{t-1}^{-1}}^2\} \leq 2 \log \frac{\det Z_T}{\det \lambda \mathbf{I}} \leq 2d \log \frac{d\lambda + TL^2}{d\lambda}$$

Lemma H.3 (Lemma 7 (Abeille et al., 2021)). *For any $z', z'' \in \mathbb{R}$, we have,*

$$\dot{\mu}(z') \frac{1 - \exp(1 - |z' - z''|)}{|z' - z''|} \leq \int_0^1 \dot{\mu}(z' + v(z'' - z')) dv \leq \dot{\mu}(z') \frac{\exp(|z' - z''|) - 1}{|z' - z''|},$$

Also, we have,

$$\int_0^1 \dot{\mu}(z' + v(z'' - z')) dv \geq \frac{\dot{\mu}(z')}{1 + |z' - z''|}, \quad \int_0^1 \dot{\mu}(z' + v(z'' - z')) dv \geq \frac{\dot{\mu}(z'')}{1 + |z' - z''|}.$$

I. Thompson Sampling-based Variants

In this section, we introduce the Thompson sampling-based variants of Algorithm 1, which we call NeuralLog-TS-1. The proof for the regret of Neural-TS-1 can be obtained by exactly following the proof of Theorem 3.5 of Zhang et al. (2021). To reuse the result of previous work, we match the notations by using the following definitions:

$$\begin{aligned}\sigma_t(x)^2 &:= \kappa \lambda_t \|g(x; \theta_0) / \sqrt{m}\|_{V_t^{-1}}^2 \\ \nu_t &:= \nu_t^{(1)} \lambda_t^{-1/2} = C_6 \lambda_t^{-1/2} (1 + \sqrt{LS} + LS^2) \iota_t + \lambda_t^{-1/2} \\ c_t &:= \nu_T (1 + \sqrt{2 \log(Kt^2)})\end{aligned}$$

and denote $\mathcal{F}_t$ as a filtration containing the history of observations up to iteration t . Also define the set of saturated points as

$$\mathcal{S}_t = \{x \in \mathcal{X}_t : \Delta_t(x) > c_{t-1} \sigma_{t-1}(x) + 2\epsilon'_{t-1}\}, \quad (61)$$

where $\Delta_t(x) = h(x_t^*) - h(x)$ and $\epsilon'_t = R^{-1} \epsilon_{3,t}$. Note that $x_t^* \notin \mathcal{S}_t$.

In round t , for each $x \in \mathcal{X}_t$, we sample a latent reward $\tilde{r}_t(x)$ from the normal distribution

$$\forall x \in \mathcal{X}_t, \quad \tilde{r}_t(x) \sim \mathcal{N}(f(x; \theta_{t-1}), \nu_T^2 \sigma_{t-1}^2(x)),$$

and choose an arm following

$$x_t = \arg \max_{x \in \mathcal{X}_t} \tilde{r}_t(x).$$

Now we introduce two good events: First, define the event $\mathcal{E}_1(t)$ when the following inequality holds for all $x \in \mathcal{X}_t$:

$$|h(x) - f(x; \theta_{t-1})| \leq \nu_T \sigma_{t-1}(x) + \epsilon'_{t-1}. \quad (62)$$

Then, by the direct result of Lemma 6.4, $\mathbb{P}(\mathcal{E}_1(t)) \geq 1 - \delta$. Next, define the event $\mathcal{E}_2(t)$ when the following inequality holds for all $x \in \mathcal{X}_t$:

$$|\tilde{r}_t(x) - f(x; \theta_{t-1})| \leq \nu_T \sqrt{2 \log(Kt^2)} \sigma_{t-1}(x). \quad (63)$$

Since $\tilde{r}_t(x)$ is sampled from $\mathcal{N}(f(x; \theta_{t-1}), \nu_T^2 \sigma_{t-1}^2(x))$, we can use the concentration inequality on Gaussian distributions to obtain $\mathbb{P}(\mathcal{E}_2(t) | \mathcal{F}_{t-1}) \geq 1 - 1/t^2$ for any possible filtration $\mathcal{F}_{t-1}$.

Next, recall the definition of the set of saturated points in Equation (61). We reuse the result of Lemma 4.5 of Zhang et al. (2021) as follows

$$\mathbb{P}(x_t \in \mathcal{X}_t \setminus \mathcal{S}_t \mid \mathcal{F}_{t-1}, \mathcal{E}_1(t)) \geq (4e\sqrt{\pi})^{-1} - 1/t^2. \quad (64)$$

We skip the proof as the same argument can be found in Section B.4 of Zhang et al. (2021). Instead, we give a high-level intuition. By construction, saturated arms are those whose posterior mean reward is significantly worse than that of the optimal arm. Under the good events $\mathcal{E}_1(t)$ and $\mathcal{E}_2(t)$, this gap is reflected both in their true means and in their posterior samples, so with high probability a saturated arm cannot catch up to the optimal arm in terms of the sampled reward.

On the other hand, the posterior for the optimal arm enjoys an anti-concentration property, which is, with constant probability, its sample exceeds its mean by a suitable margin. This is where the factor $(4e\sqrt{\pi})^{-1}$ comes from. Combining these facts, with constant probability the sampled reward of the optimal arm is larger than the samples of all saturated arms, so the arm selected by Thompson sampling must be unsaturated. The $1/t^2$ term accounts for the small probability that one of the good events $\mathcal{E}_1(t)$ or $\mathcal{E}_2(t)$ fails.

Now, with the previous results in place, we derive an upper bound on the expected instantaneous regret. Define $d_t = h(x_t^*) - h(x_t)$. Again, we reuse the result of Lemma 4.6 of Zhang et al. (2021) as follows:

$$\mathbb{E}[d_t \mid \mathcal{F}_{t-1}, \mathcal{E}_1(t)] \leq 44e\sqrt{\pi} C_1 c_t \sqrt{L} \mathbb{E}[\min\{1, \sigma_t(x_t)\} \mid \mathcal{F}_{t-1}, \mathcal{E}_1(t)] + 4\epsilon'_{t-1} + 2/t^2,$$

where $C_1 > 0$ is the same absolute constant that appears in Lemma C.4. By Equation (64), Neural-TS-1 selects an unsaturated arm with constant probability, so in expectation the posterior standard deviation of the played arm is comparable to that of the best unsaturated arm. Under the good events, the posterior means stay close to the true means and saturated arms have very small gaps, which allows us to bound the instantaneous regret d_t by a constant multiple of $\min\{1, \sigma_t(x_t)\}$ plus the approximation terms $4\epsilon'_{t-1} + 2/t^2$. Taking the conditional expectation and using a global control on the posterior variances over time then yields the stated bound.

Now we are ready to start the proof for the regret. Define a stochastic process $(Y_t)_{t=0}^T$ where

$$\begin{aligned}\bar{d}_t &= d_t \mathbf{1}\{\mathcal{E}_1(t)\} \\ X_t &= \bar{d}_t - 44e\sqrt{\pi}C_1c_t\sqrt{L}\min\{1, \sigma_t(x_t)\} - 4\epsilon'_{t-1} - 2/t^2 \\ Y_t &= \sum_{i=1}^t X_i, \quad Y_0 = 0\end{aligned}$$

We can see that (Y_t) is a supermartingale with respect to $\mathcal{F}_t$ since $\mathbb{E}[Y_t - Y_{t-1} \mid \mathcal{F}_{t-1}] = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \leq 0$. Now we prepare to apply the Azuma-Hoeffding inequality for a supermartingale:

Lemma I.1 (Azuma-Hoeffding inequality for supermartingale). *If a supermartingale Y_t , corresponding to a filtration $\mathcal{F}_t$ satisfies $|Y_t - Y_{t-1}| \leq B_t$, then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$Y_t - Y_0 \leq \sqrt{2 \log(1/\delta) \sum_{i=1}^t B_i^2}.$$

To derive an upper bound on $|Y_t - Y_{t-1}|$, we have

$$\begin{aligned}|Y_t - Y_{t-1}| &= |X_t| \leq |\bar{d}_t| + 44e\sqrt{\pi}C_1c_t\sqrt{L}\min\{1, \sigma_t(x_t)\} + 4\epsilon'_{t-1} + 2/t^2 \\ &\leq 4 + 44e\sqrt{\pi}C_1^2c_tL + 4\epsilon'_{t-1}.\end{aligned}$$

where the last inequality follows from Lemma C.4, and $1/t^2 \leq 1$. Now, applying Lemma I.1 with $B_t = 4 + 44e\sqrt{\pi}C_1^2c_tL + 4\epsilon'_{t-1}$ to (Y_t) , with probability at least $1 - \delta$, we have

$$\begin{aligned}\sum_{t=1}^T \bar{d}_t &\leq \underbrace{\sum_{t=1}^T 44e\sqrt{\pi}C_1c_t\sqrt{L}\min\{1, \sigma_t(x_t)\}}_{\text{(term 1)}} + \underbrace{\sum_{t=1}^T 4\epsilon'_{t-1} + \sum_{t=1}^T 2/t^2}_{\text{(term 2)}} \\ &\quad + \underbrace{\sqrt{2 \log(1/\delta) \sum_{t=1}^T (4 + 44e\sqrt{\pi}C_1^2c_tL + 4\epsilon'_{t-1})^2}}_{\text{(term 3)}}.\end{aligned}\tag{65}$$

For **(term 1)**, applying Cauchy-Schwarz inequality,

$$\begin{aligned}\text{(term 1)} &\leq 44e\sqrt{\pi}C_1\nu_T^{(1)}(1 + \sqrt{2 \log(KT^2)})\sqrt{\kappa L} \sqrt{T \sum_{t=1}^T \min\{1, \|g(x_t; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}^2\}} \\ &= \tilde{\mathcal{O}}\left(S^2\tilde{d}\sqrt{\kappa T} + S^{2.5}\sqrt{\kappa\tilde{d}T}\right)\end{aligned}$$

For **(term 2)**, by the condition of m in Condition 4.4, $\sum_{t=1}^T 4\epsilon'_{t-1} \leq 1$, and $\sum_{t=1}^T 2/t^2 \leq \pi^2/3$. For **(term 3)**, since $\nu_T^{(1)} = \tilde{\mathcal{O}}(S^2\sqrt{\tilde{d}} + S^{2.5})$, and $\lambda_0^{-1/2} = \mathcal{O}(S^{0.5})$

$$\begin{aligned}\text{(term 3)} &\leq (4 + 44e\sqrt{\pi}C_1^2\nu_T^{(1)}\lambda_0^{-1/2}(1 + \sqrt{2 \log(KT^2)}) + 4\epsilon'_T)\sqrt{2 \log(1/\delta)T} \\ &= \tilde{\mathcal{O}}\left(S^{2.5}\sqrt{\tilde{d}T} + S^3\sqrt{T}\right)\end{aligned}$$

Combining results, we have

$$\sum_{t=1}^T \tilde{d}_t \leq \tilde{\mathcal{O}} \left(S^2 \tilde{d} \sqrt{\kappa T} + S^{2.5} \sqrt{\kappa \tilde{d} T} + S^3 \sqrt{T} \right)$$

with probability at least $1 - \delta$. Notice that $\text{Regret}(T) \leq \sum_{t=1}^T R |h(x_t^*) - h(x_t)|$. Therefore $R \sum_{t=1}^T \tilde{d}_t$ upper bounds the regret with probability at least $1 - \delta$. Finally, replacing δ by $\delta/2$ for both cases and applying the union bound finishes the proof.

Remark I.2 (Discussion on Thompson sampling-based variants of NeuralLog-UCB-2). In analogy with the Thompson sampling extension of NeuralLog-UCB-1, one can also consider a Thompson sampling-based variant of NeuralLog-UCB-2 as follows. Define $\sigma'_t(x)^2 := \lambda_t \|g(x; \theta_0) / \sqrt{m}\|_{W_{t-1}^{-1}}$ and $\nu'_t := \nu_{t-1}^{(2)} \lambda_t^{-1/2}$, and for all $x \in \mathcal{X}_t$ sample $\tilde{r}'_t(x) \sim \mathcal{N}(g(x; \theta_0)^\top (\theta_{t-1} - \theta_0), \nu'^2_t \sigma'^2_{t-1}(x))$, then choose $x_t = \arg \max_{x \in \mathcal{X}_t} \tilde{r}'_t(x)$. However, our current regret analysis for Thompson sampling-based algorithms proceeds by defining a stochastic process associated with the per-round regret and then applying a concentration inequality for this process to obtain an upper bound on the per-round regret. In order to fully exploit W_t from Algorithm 2 within this framework, a much more delicate analysis of the second-order Taylor expansion of the per-round regret would be required.

More concretely, if we proceed the analysis in a naive way and consider the regret bound obtained for such a NeuralLog-TS-2 algorithm, then, denoting by **(term 1')** the counterpart of **(term 1)** in Equation (65), and focusing only on the dependence on κ , we obtain

$$\text{(term 1')} \lesssim \sum_{t=1}^T \min\{1, \|g(x; \theta_0) / \sqrt{m}\|_{W_{t-1}^{-1}}\} \lesssim \sqrt{\kappa T \sum_{t=1}^T \min\{1, \|g(x; \theta_0) / \sqrt{m}\|_{V_{t-1}^{-1}}^2\}},$$

where we see that the additional $\sqrt{\kappa}$ factor is reintroduced. Treating this issue within our current proof technique therefore appears to be a non-trivial problem, and we leave a sharper analysis of such Thompson sampling-based variants of NeuralLog-UCB-2 for future work.

As we have seen in Remark I.2, although NeuralLog-TS-2 does not attain a regret bound with the same dependence on κ as NeuralLog-UCB-2, the algorithm itself is well defined, just like NeuralLog-TS-1. In Section J, we present additional experiments including these two algorithms and demonstrate their practical performance.

J. Additional Experiments

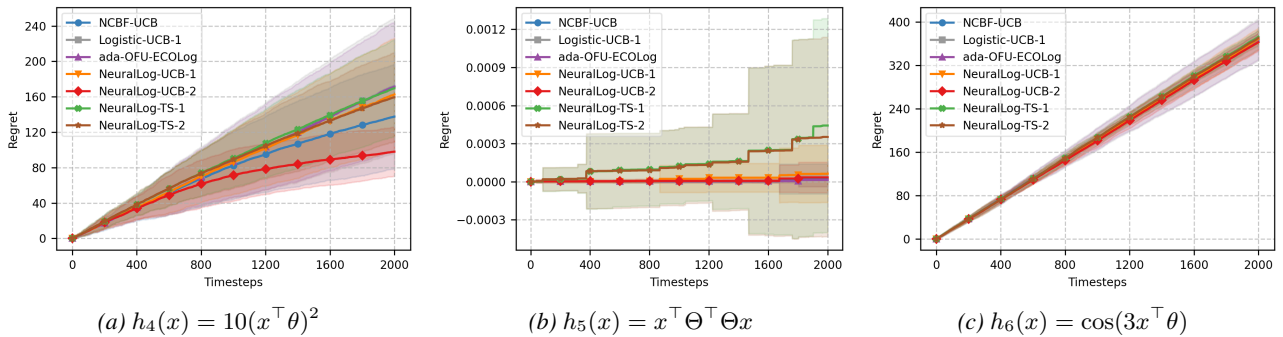


Figure 4. Comparison of cumulative regret of baseline algorithms for nonlinear reward functions.

We compare five baseline algorithms with our algorithms including the Thompson sampling-based variants introduced in Section I. Where NeuralLog-TS-1 and NeuralLog-TS-2 both choose the arm with best sampled reward where

$$\text{For NeuralLog-TS-1, } \tilde{r}_t(x) \sim \mathcal{N}(f(x; \theta_{t-1}), \nu_T^2 \sigma_{t-1}^2(x)),$$

$$\text{For NeuralLog-TS-2, } \tilde{r}'_t(x) \sim \mathcal{N}(g(x; \theta_0)^\top (\theta_{t-1} - \theta_0), \nu_T'^2 \sigma_{t-1}'^2(x)).$$

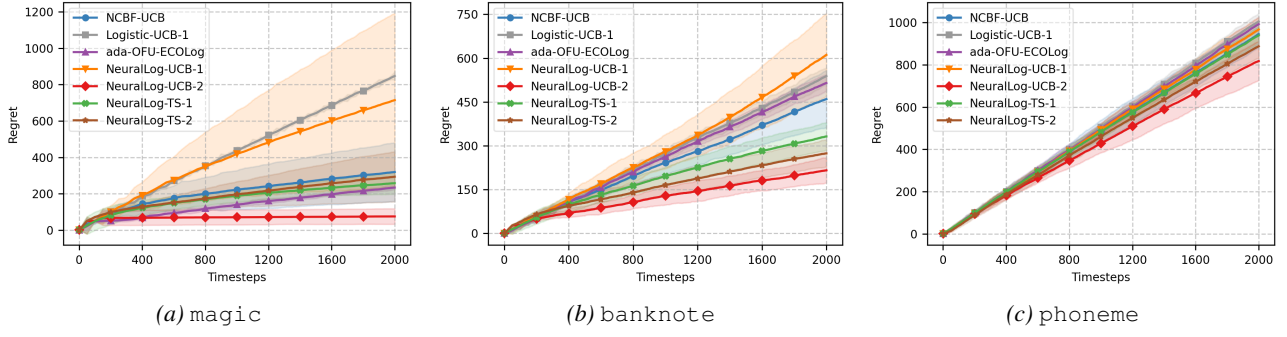
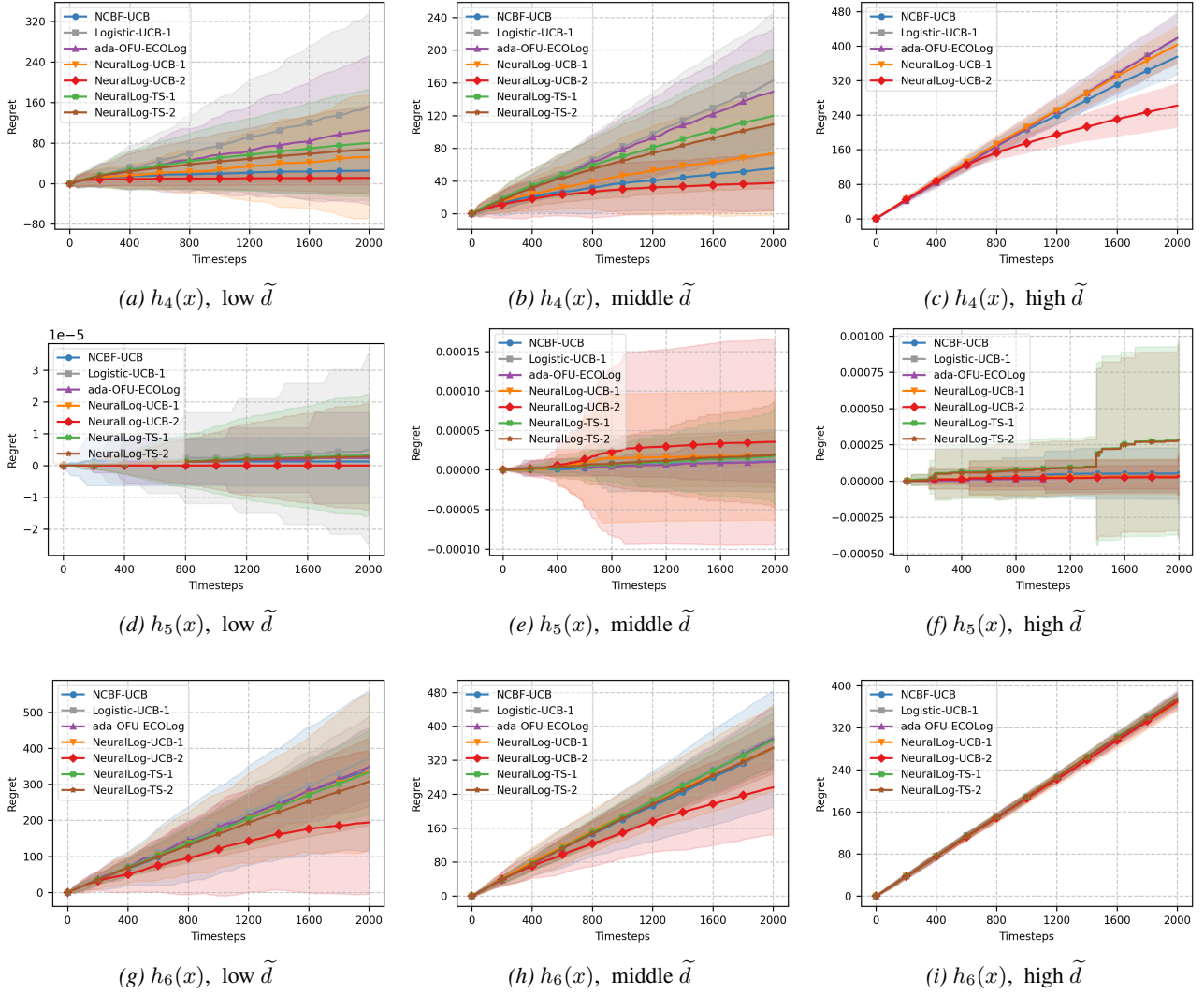


Figure 5. Comparison of cumulative regret of baseline algorithms for real-world dataset.


 Figure 6. Comparison of cumulative regret of baseline algorithms with varying effective dimension $\tilde{d}$.

We include the synthetic latent reward functions which are also used in Zhou et al. (2020): $h_4(x) = 10(x^\top \theta)^2$, $h_5(x) = x^\top \Theta^\top \Theta x$, $h_6(x) = \cos(3x^\top \theta)$. All other experimental parameters and details follow the same as described in Section 7.

Next, we include 3 more K -class classification tasks from (Dua & Graff, 2019): We reuse the same min-max normalization to $[-1, 1]$ as described in Section A. In the `magic` dataset (MAGIC Gamma Telescope), we convert all features to real-valued variables, impute any missing entries with 0, and then map the original class labels to a binary label by setting $y = 1$ for gamma ('g') events and $y = 0$ for hadron ('h') events. In the `banknote` dataset (UCI Banknote Authentication), we use the four real-valued attributes provided in the repository and keep the original binary labels $y \in \{0, 1\}$. For the `phoneme` dataset (Connectionist Bench (Nettalk Corpus)), we treat any categorical fields as numeric by casting them to an appropriate numeric type, and replace missing values with 0.

Figures 4 and 5 summarize the average cumulative regret of the five baseline algorithms together with our two Thompson sampling-based variants. Consistent with the results already observed in Figures 1 and 2, our NeuralLog-UCB-2 algorithm steadily achieves the best performance across the considered settings. Moreover, the two Thompson sampling-based variants also exhibit competitive performance compared to the baselines. Figure 6 demonstrates the influence of $\tilde{d}$ on data-adaptive algorithms by comparing cumulative regret across different values of $\tilde{d}$.

J.1. Open Bandit Dataset Replay Experiments

We further evaluate the algorithms on the Open Bandit Dataset (OBD) (Saito et al., 2020), a large-scale public logged bandit dataset collected from a fashion e-commerce platform. This experiment complements the classification-based real-world experiments in Sections A and 7. While the classification-to-bandit protocol gives access to the reward of every action in each round, logged bandit data provide a more realistic feature distribution and feedback structure. At the same time, as in most real-world bandit evaluations, the assumptions used in the theory are not guaranteed to hold exactly and the problem-dependent quantities such as S and κ cannot be identified tightly from the data. We therefore use OBD as a practical robustness test under several choices of the theory-motivated parameters.

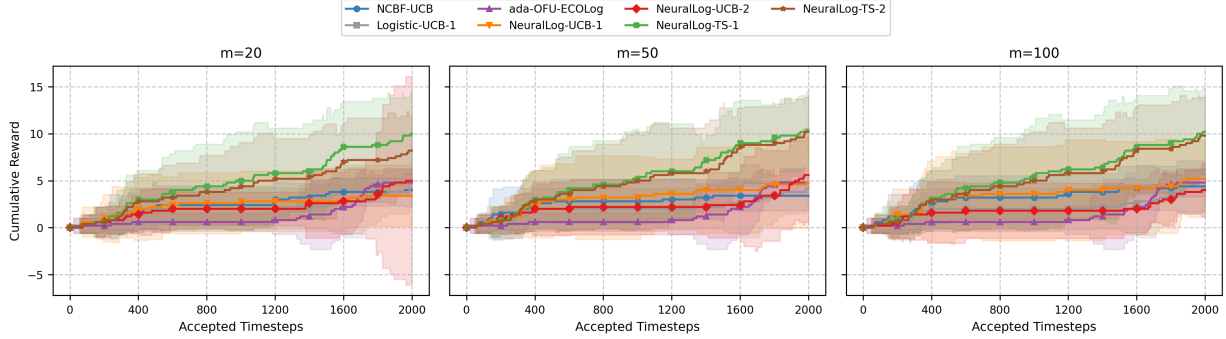


Figure 7. Open Bandit Dataset replay evaluation with varying neural network width m , while fixing $\kappa = 50$ and $S = 1$.

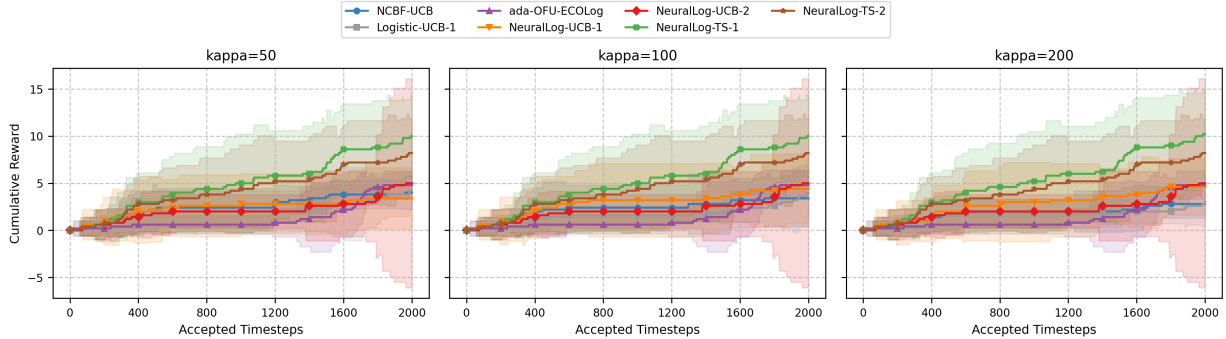


Figure 8. Open Bandit Dataset replay evaluation with varying κ , while fixing $m = 20$ and $S = 1$.

Open Bandit Dataset setup. We use the `random/all` split of OBD and run replay evaluation. In each logged event,

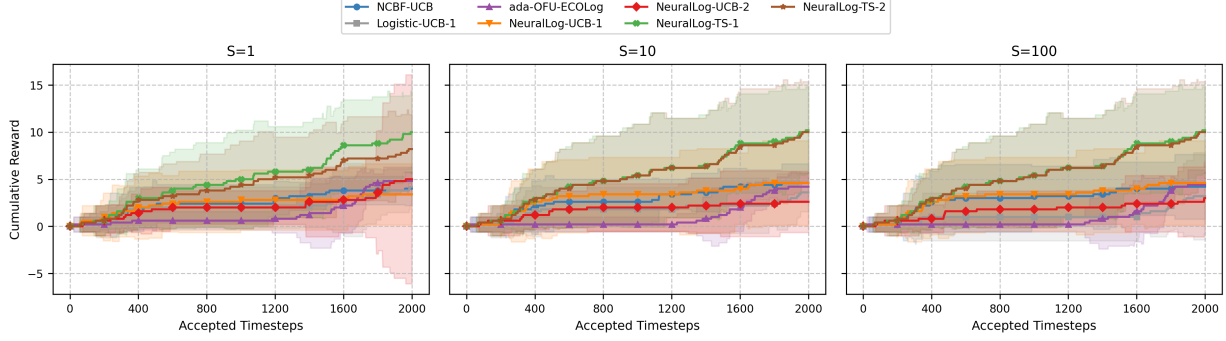


Figure 9. Open Bandit Dataset replay evaluation with varying S , while fixing $m = 20$ and $\kappa = 50$.

the agent receives a reduced action set consisting of the logged action and four alternative actions chosen by the highest user-item affinity scores, yielding a $K = 5$ -armed problem. Feedback is used only when the action selected by the agent matches the logged action; otherwise, the event is skipped. Thus the horizontal axis in Figures 7 to 9 denotes the number of accepted logged events. Since counterfactual rewards for unchosen actions are not available in OBD, we report cumulative reward and final click-through rate (CTR), rather than cumulative regret.

For the context-arm feature vector, we combine compressed user features, item features from `item_context.csv`, the display position, the action-specific user-item affinity score, and an interaction feature between user features and the affinity score. The categorical user and item features are one-hot encoded and compressed using truncated SVD, with user embedding dimension 16 and item embedding dimension 8. The affinity scores are log-transformed, scaled to $[-1, 1]$, and then used both as an action-specific scalar feature and as part of the interaction feature. We use $T = 2000$ accepted timesteps, batch size 50, and 5 repeated runs. The neural network is updated every 50 accepted events using 100 gradient descent steps with learning rate 0.01. For all algorithms, we use the same exploration and regularization parameters as in the synthetic and classification-based real-world experiments, except for the displayed choices of m , S , and κ .

Table 2. Final CTR at $T = 2000$ on the Open Bandit Dataset replay experiment. Higher is better. We boldface the top three methods in each row.

Parameter	Value	NCBF-UCB	Logistic-UCB-1	ada-OFU-ECOLOG	NeuralLog-UCB-1	NeuralLog-UCB-2	NeuralLog-TS-1	NeuralLog-TS-2
m	20	0.0020	0.0021	0.0024	0.0017	0.0025	0.0050	0.0041
	50	0.0017	0.0021	0.0024	0.0024	0.0028	0.0052	0.0051
	100	0.0022	0.0021	0.0024	0.0026	0.0020	0.0051	0.0049
κ	50	0.0020	0.0021	0.0024	0.0017	0.0025	0.0050	0.0041
	100	0.0017	0.0020	0.0024	0.0022	0.0025	0.0050	0.0041
	200	0.0014	0.0015	0.0024	0.0023	0.0025	0.0051	0.0041
S	1	0.0020	0.0021	0.0024	0.0017	0.0025	0.0050	0.0041
	10	0.0023	0.0018	0.0021	0.0023	0.0013	0.0051	0.0050
	100	0.0021	0.0016	0.0022	0.0023	0.0015	0.0051	0.0050

Results. Figures 7 to 9 and Table 2 summarize the OBD replay results. Across the considered values of m , the Thompson sampling-based variants, NeuralLog-TS-1 and NeuralLog-TS-2, consistently achieve the highest final CTR, and one of NeuralLog-UCB-1 or NeuralLog-UCB-2 is also among the top three methods. This indicates that the proposed neural logistic bandit algorithms remain competitive even when the network width is much smaller than the sufficient width required by the NTK-based theory. The same qualitative behavior is observed when varying κ : NeuralLog-TS-1, NeuralLog-TS-2, and NeuralLog-UCB-2 are the top three methods for all tested values. Finally, when varying S , the two Thompson sampling-based variants remain the strongest performers, while one of the UCB-based variants remains competitive. These results support the empirical robustness of our methods on real logged bandit data, where the model assumptions and the exact values of the theory-motivated parameters are necessarily misspecified.

K. Additional Future Directions

Although we successfully remove the direct dependence on p from the regret bound, a direct dependence on p reappears when we examine the per-round computational complexity. This is problematic in neural bandit settings where p scales with the horizon T , making the resulting algorithm computationally inefficient.

Let us briefly analyze the computational complexity of our algorithms. Since Algorithms 1 and 2 have the same order of complexity, we focus on Algorithm 1. For action selection, we must compute $f(x; \theta_{t-1})$ for K actions, which costs $\mathcal{O}(p)$ per action, and the quantity $\|g(x; \theta_0)/\sqrt{m}\|_{V_{t-1}^{-1}}$, which costs $\mathcal{O}(p^2)$ per action. Hence, the action-selection step has complexity $\mathcal{O}(Kp^2)$. The updates of the parameters λ_t , ι , and $\nu_t^{(1)}$ cost $\mathcal{O}(p^2)$ by their definitions. For neural network training, at round t we apply gradient steps over the full dataset of size t , which costs $\mathcal{O}(tp)$ per gradient step. Performing J_t iterations therefore costs $\mathcal{O}(J_t tp)$, where $J_t = \tilde{\mathcal{O}}(TL/\lambda_t)$. Finally, updating the design matrix V_t costs $\mathcal{O}(p^2)$. Altogether, the per-round computational complexity is $\mathcal{O}(J_t tp + Kp^2 + p^2)$. Moreover, Verma et al. (2025) can be seen to have essentially the same computational complexity, as their algorithm and training pipeline are close to ours.

In contrast, in the classical logistic bandit literature the algorithms operate directly in the feature space of dimension d , which is typically much smaller than p . For example, Filippi et al. (2010); Faury et al. (2020) obtain overall complexity on the order of $\mathcal{O}(d^2 K + d^2 T)$, and there has been significant recent progress on designing computationally efficient algorithms for logistic bandits: Abeille et al. (2021) achieve $\mathcal{O}(d^2 KT)$, and Faury et al. (2022) even propose an algorithm with complexity $\tilde{\mathcal{O}}(d^2 K)$. However, these favorable guarantees rely crucially on the strong assumption that the latent reward model is linear in the feature representation. From the perspective of practical applications, it is therefore important to develop neural bandit algorithms that retain the modeling flexibility of neural networks while achieving comparable computational efficiency, which we view as an important direction for future work.

As one illustrative example, in light of the connection between NTK-based neural bandits and kernelized bandits, one could consider importing techniques such as Nyström approximation as in Zenati et al. (2022) to reduce the effective computational cost in the neural bandit setting. Another approach is to adapt the method proposed in Xu et al. (2022) where an NTK-based neural bandit formulation is also used, but the neural network is trained so that its output is not the reward itself, but instead a new d -dimensional feature vector. The problem is then reduced to solving a linear bandit in this learned feature space with respect to an unknown parameter θ^* . This strategy can substantially reduce computational complexity and is attractive from an applied viewpoint, but it requires an additional Lipschitz-type assumption on the neural network on the theoretical side.